Chapter 1 - Internet and ICT Environment

Lesson 1.01 *What is ICT?*

40 minutes **CONCEPT**

Success Criteria - I can…

- Define ICT and identify its three components: Information, Communication, Technology
- Explain the input-process-output cycle using a real-world example
- Classify everyday devices as information, communication, or technology tools

Resources Needed:

- Working laptop or tablet for demonstration
- Labeled chart of ICT components
- 20 picture cards for sorting activity
- Exit ticket paper
- Chart paper and markers for group work

Key Vocabulary

ICT: Technology that handles information and helps people communicate together
Information: Facts, numbers, or pictures that give us useful knowledge daily
Communication: Sharing ideas or messages between people using different tools
Technology: Tools and machines that make our daily work easier

Engage - Retrieval & Hook (5 min)

TEACHER Bring a working laptop or tablet to class. Open it and ask: What is this? Is it just a machine, or something more? Let students touch the keyboard, move the mouse, watch the screen respond. Then ask: If I type a letter here and send it to someone in Bangalore, how does it travel? It does not go through the post office - so how? Guide them to think about invisible information flowing through wires and air. Write the word INFORMATION on the board. Circle it. Then write COMMUNICATION and TECHNOLOGY beside it. Ask: What happens when we put all three words together?

STUDENTS Gather around the device. Touch the keyboard and screen. Guess how messages travel through the internet. Suggest ideas: phone lines, satellites, magic, invisible wires. Watch as the three words combine into ICT - Information and Communication Technology.

Explore - Guided Discovery (10 min)

TEACHER Show a chart with three columns: Information, Communication, Technology. Under each, place picture cards. Information: book, calculator, newspaper, computer file. Communication: phone, letter, email, video call. Technology: fan, television, washing machine, robot. Ask students to work in pairs and sort 20 picture cards into the three columns. Walk around and challenge their choices: Is a telephone information or communication? Can it be both? After 10 minutes, ask each pair to explain one tricky card they disagreed on.

STUDENTS Work in pairs to sort picture cards into the three columns. Debate disagreements - for example, a smartphone carries information AND enables communication. Present their trickiest card to the class and explain their reasoning. Realize that many devices fit multiple categories.

Explain - Modelling & Concepts (10 min)

TEACHER Write the definition on the board: ICT is the set of tools that help us handle information and communicate with others. Break down each word: Information is data we use - names, numbers, pictures, sounds. Communication is sharing that information with other people. Technology is the physical tools we use to do this. Then draw a simple diagram: INPUT (data goes in) -> PROCESSING (computer works on it) -> OUTPUT (result comes out). Give a village example: a farmer tells the shopkeeper how much rice he has (INPUT), the computer calculates the total price (PROCESSING), and prints a receipt (OUTPUT). Ask students for their own examples.

STUDENTS Copy the definition and the input-process-output diagram into notebooks. Give their own examples: ATM - you insert your card (INPUT), the machine verifies your PIN and balance (PROCESSING), cash comes out (OUTPUT). Write three examples of ICT tools they use daily. Draw the IPO diagram for a mobile phone call.

Elaborate - Independent Application (10 min)

TEACHER Present this challenge: Your village school wants to go fully digital. The headmaster asks you to help. Make a list of 5 things the school currently does on paper, and for each one, explain how ICT could do it better. Walk around and prompt stuck students: What about attendance? Marks cards? Library records? Notices to parents? Exam schedules? Let students work in groups of three. After 15 minutes, ask each group to present their best idea.

STUDENTS Brainstorm school activities done on paper: attendance registers, marks cards, library register, notice boards, letter to parents. Suggest digital alternatives: attendance app, spreadsheet for marks, digital library catalog, SMS or app notifications. Draw or write their best idea on chart paper. Present to the class and explain why it would help.

Evaluate - Check & Close (5 min)

TEACHER Distribute exit tickets. Ask three questions: (1) Write the full form of ICT. (2) Give one example of input and one example of output from a computer. (3) Name one way ICT could help your village. Walk around as students write. Collect tickets as students leave. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Write answers individually. Question 1: Information and Communication Technology. Question 2: Input - typing on a keyboard; Output - text appears on screen. Question 3: Using a computer to maintain student records instead of paper registers.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 (KB, MB, GB) - place value concepts
Science: Electricity powers computer components - circuits and energy
Language: Labeling diagrams builds technical vocabulary

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part in one sentence
Advanced: Create a buying guide for a school computer lab - list parts, quantities, and estimated prices
SEND: Touch and identify real computer parts with teacher guidance, use simple one-word labels

Formative Assessment

Method: Exit ticket with three questions
CT Skill Assessed: Decomposition - breaking down a computer into input, processing, memory, and output components

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a labeled diagram of a computer from memory. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.02 *History of Computers*

40 minutes **CONCEPT**

Success Criteria - I can...

- List the five generations of computers in chronological order
- Describe how computers changed in size, speed, and cost across generations
- Compare a modern smartphone with ENIAC and explain the differences

Resources Needed:
- Photographs of ENIAC and modern smartphone
- Timeline cards for matching activity
- Blank paper and colored pencils for invention drawing
- Quiz paper
- Projector for photographs

Key Vocabulary

Generation: Stage showing how computers improved over many years gradually
Vacuum Tube: Glass switch used in earliest large room-sized computers
Transistor: Small switch that replaced tubes and made computers smaller
Microprocessor: Tiny chip containing the brain of a modern computer

Engage - Retrieval & Hook (5 min)

TEACHER Show a photograph of ENIAC - a computer that filled an entire room with wires, panels, and vacuum tubes, weighing 27 tonnes. Then show a photograph of a modern smartphone. Ask: "Both of these are computers. What differences do you see?" Write student observations on the board: size, wires, weight, cost, what they can do. Ask: "How did we go from THAT to THIS? What happened in between?" Tell them: "Today we will travel through time and see how computers evolved across five generations - each one smaller, faster, and cheaper than the last."

STUDENTS Compare the two photographs side by side. Notice the enormous size difference - ENIAC filled a room, a phone fits in a pocket. Guess how long the old computer took to do calculations. Express surprise that a phone is millions of times more powerful. Wonder what happened in between.

Explore - Guided Discovery (10 min)

TEACHER Draw a vertical timeline on the board with five sections. Place cards next to each: 1940s - Vacuum Tubes, 1950s - Transistors, 1960s - Integrated Circuits, 1970s - Microprocessors, 2000s - Artificial Intelligence. Under each, write one fact: filled a room, size of a suitcase, size of a fingernail, entire CPU on one chip, learns from data. Give each group additional fact cards (cost, speed, who used it, example machine). Ask them to match the facts to each generation. Walk around and ask: "Why did transistors replace vacuum tubes?"

STUDENTS Work in groups to match fact cards to each generation. Discuss: vacuum tubes were slow and hot, transistors were reliable and cool, integrated circuits were cheap, microprocessors enabled personal computers, AI enables smart devices. Present their matched timeline to the class.

Explain - Modelling & Concepts (10 min)

TEACHER Explain each generation with vivid analogies. First generation 1940s: Vacuum tubes - like light bulbs, big and hot, ENIAC had 18,000 tubes and could not be turned off without losing data. Second generation 1950s: Transistors - like tiny switches, much smaller and faster, UNIVAC was the size of a desk and used for business. Third generation 1960s: Integrated circuits - thousands of transistors on one silicon chip, size of a postage stamp, IBM System/360 could run many programs. Fourth generation 1970s: Microprocessors - entire CPU on one chip, led to personal computers, Apple II and IBM PC. Fifth generation present: AI and quantum computing - computers that learn from data, self-driving cars, voice assistants like Alexa.

STUDENTS Take notes on each generation with dates, key invention, and one example. Draw the comparison table: columns for generation, year, invention, size, speed, cost, who used it. Write one sentence about why each generation was better than the last. Understand the trend: smaller, faster, cheaper, more accessible.

Elaborate - Independent Application (10 min)

TEACHER Ask: "If computers keep getting smaller and faster, what will a computer look like in 2050?" Show examples of emerging technology: smartwatches that track health, smart speakers that answer questions, AR glasses that show information overlaid on the real world. Challenge: "Draw or describe your invention. What would it look like? What would it do? How would it help people in your village?"

STUDENTS Imagine future computers. Sketch or describe an invention: a pen that translates languages in real time, a mirror that shows the news and weather, a shirt that monitors your health and alerts a doctor. Present their invention to the class with a brief explanation.

Evaluate - Check & Close (5 min)

TEACHER Quick quiz: (1) List the five generations in order from oldest to newest. (2) What invention made personal computers possible? (3) Which generation are we in now? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Write the five generations in order: vacuum tubes, transistors, integrated circuits, microprocessors, AI. Answer: microprocessor. Answer: fifth generation with AI.

Cross-Curricular Connections

Mathematics: Timeline involves dates and ordering numbers chronologically
Science: Vacuum tubes and transistors involve electricity and semiconductors
Language: Describing inventions builds descriptive writing skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match each generation to its key invention using a word bank with pictures
On Level: Write one sentence describing each generation and draw a simple picture of the computer
Advanced: Research and present one specific computer from each generation with details about its capabilities
SEND: Point to pictures of old and new computers and describe the differences in simple words

Formative Assessment

Method: Quick quiz with five-generation ordering and identification questions
CT Skill Assessed: Pattern Recognition - identifying the trend of computers getting smaller, faster, and cheaper across generations

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a timeline of computer generations from 1940 to present. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.03 *Introduction to Internet* 40 minutes CONCEPT

Success Criteria - I can…

- Define the internet and explain how it connects computers worldwide
- Trace the path of data from a phone to a server and back
- Identify two advantages and two disadvantages of the internet

Resources Needed:
- String and labeled paper cups for network diagram
- Chart paper for group presentations
- Exit ticket paper
- Board markers for diagram drawing

Key Vocabulary

Internet: Worldwide network connecting millions of computers to share information
Server: Powerful computer that stores websites and sends data quickly
Packet: Small piece of data travelling separately across the network
ISP: Company that provides internet connection to homes and schools

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "How many of you watched a video on YouTube or talked to a relative on WhatsApp this week?" Show of hands. Then ask: "But have you ever wondered - where does that video come from? It is not stored on your phone. So where is it, and how does it reach you in seconds?" Write on the board: Where is YouTube? How does data travel? Tell them: "Today we will understand the invisible network that connects billions of devices across the world."

STUDENTS Raise hands to show internet use. Wonder where YouTube stores its videos. Guess: maybe on a big computer somewhere far away. Discuss how fast data seems to travel - faster than any postman could carry it.

Explore - Guided Discovery (10 min)

TEACHER Give each group a string and 6 small paper cups labeled: Your Phone, Home Router, Local Tower, ISP Office, Internet Cloud, YouTube Server. Ask them to connect the cups with string to show the path of a video request. Then ask: "Now the video needs to come back. Draw arrows showing the return path." After 8 minutes, each group presents their network diagram.

STUDENTS Connect the cups with string to show the data path: phone -> router -> tower -> ISP -> server. Draw return arrows for the video coming back. Present their diagram and explain the path. Realize that the internet is not a cloud in the sky - it is physical cables, towers, and servers in buildings.

Explain - Modelling & Concepts (10 min)

TEACHER Draw a detailed diagram on the board. Explain: Your device connects to a router in your home using Wi-Fi. The router connects to your Internet Service Provider (ISP) through a cable or mobile tower. The ISP connects to the global internet through undersea cables and satellites. When you request a video, your request breaks into small packets. Each packet finds its own path through the network. When all packets arrive at the server, they reassemble into the video. Use the letter analogy: you write a letter, the post office routes it through different cities, and the delivery person brings it to the address. Key terms: ISP, packets, protocol, bandwidth.

STUDENTS Copy the diagram showing phone -> router -> ISP -> internet -> server. Write definitions: ISP, packets, protocol, bandwidth. Trace the path of a Google search from start to finish. Draw arrows showing bidirectional data flow.

Elaborate - Independent Application (10 min)

TEACHER Present two scenarios. Scenario 1: The internet stops working for one day in your village. What happens? Scenario 2: A new fiber optic cable is laid near your village, giving everyone fast internet. What changes? Then discuss: "What are the dangers of the internet that we should be careful about?"

STUDENTS Discuss both scenarios in groups. For scenario 1: no online classes, no digital payments, no social media, bank ATMs might stop working. For scenario 2: faster learning, video classes, online shopping, telemedicine. For dangers: fake news, cyberbullying, addiction to games, privacy concerns.

Evaluate - Check & Close (5 min)

TEACHER Exit ticket with three questions: (1) In your own words, what is the internet? (2) What are packets and why does the internet use them? (3) Name one advantage and one disadvantage of the internet. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Write answers. Question 1: The internet is a global network of computers connected by cables and wireless signals. Question 2: Packets are small pieces of data that travel independently and reassemble at the destination. Question 3: Advantage - instant communication. Disadvantage - cyberbullying.

Cross-Curricular Connections

Mathematics: Bandwidth involves data measurement units (bits, bytes, kilobytes)
Science: Undersea cables and satellites use physics - light signals and electromagnetic waves
Language: Explaining how the internet works builds technical writing skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Trace the path of data by connecting labeled pictures in the correct order
On Level: Draw and label the data journey from phone to server with at least 4 steps
Advanced: Explain what happens when a packet is lost and how the internet fixes it
SEND: Point to pictures of internet devices and name them in simple words

Formative Assessment

Method: Exit ticket with definition, explanation, and opinion questions
CT Skill Assessed: Decomposition - breaking down the internet into components: devices, connections, protocols, and servers

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a diagram showing how data travels from your phone to YouTube and back. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.04 *Web Browsers*

40 minutes CONCEPT

Success Criteria - I can...

- Identify the main parts of a web browser interface: address bar, back, forward, refresh, tabs, bookmarks
- Navigate to websites by typing URLs in the address bar
- Use bookmarks to save and quickly access frequently visited websites

Resources Needed:
- Computer with projector for demonstration
- Printed Chrome diagram for labeling
- Student computers or tablets
- List of website addresses on the board

Key Vocabulary

Browser: Program that lets you open and view websites easily
Address Bar: Long box at top where you type website address
Tab: Small window letting you open many sites together
Bookmark: Saved link that helps you find favourite sites quickly

Engage - Retrieval & Hook (5 min)

TEACHER Open Google Chrome on the projector. Ask: "How many of you have seen this before? What do you call it?" Point to different parts: the long bar at the top, the small arrows on the left, the tiny star on the right. Ask: "What happens if I type something here and press Enter?" Type scratch.mit.edu and press Enter. The Scratch website loads with colorful blocks. Ask: "How did the computer know to go to that website? What does this bar at the top actually do?"

STUDENTS Recognize the browser from previous computer use. Name parts they know: the search bar, the back button. Watch the Scratch website load. Guess that the bar at the top is like a telephone number for websites.

Explore - Guided Discovery (10 min)

TEACHER Give each student a printed diagram of Chrome with blank labels. Parts to label: address bar, back button, forward button, refresh button, new tab button, bookmark star, tab strip. After labeling, give a practice task: (1) Go to google.com, (2) Click the back button, (3) Click forward, (4) Open a new tab and go to scratch.mit.edu, (5) Bookmark both sites.

STUDENTS Label the diagram correctly. Follow the practice steps one by one. Click back and forward to see the browsing history. Open a new tab using Ctrl+T or the plus button. Click the star to bookmark a site and choose a folder.

Explain - Modelling & Concepts (10 min)

TEACHER Explain each feature with demonstrations. Address bar: Type the website address here - the browser sends a request to the server at that address. Back button: Returns to the previous page you visited. Forward button: Goes forward if you pressed back. Refresh button: Reloads the page in case it did not load properly. Tabs: Let you have multiple websites open at once without opening new windows. Bookmark star: Saves a website address so you can find it quickly later.

STUDENTS Write notes on each feature. Practice using each one: open three tabs, navigate between them, bookmark a favorite site. Create a bookmark folder called "School" and add educational websites.

Elaborate - Independent Application (10 min)

TEACHER Challenge: "Navigate to five different websites by typing their addresses - no searching allowed." Give the addresses: wikipedia.org, scratch.mit.edu, youtube.com, kreis.karnataka.gov.in, code.org. Time how long each takes. Ask: "Which website loaded fastest? Why?"

STUDENTS Type each address carefully and press Enter. Explore each website briefly. Record loading times. Discuss why some sites loaded faster: Wikipedia loads fast because it is mostly text, YouTube loads slower because it has many videos.

Evaluate - Check & Close (5 min)

TEACHER Practical assessment: Ask each student to demonstrate five skills: (1) Type a URL and load a website, (2) Use the back button, (3) Use the forward button, (4) Open a new tab, (5) Bookmark a page. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Perform each skill when asked. Help a partner if they get stuck.

Cross-Curricular Connections

Mathematics: Loading time involves measurement and comparison of numbers
Science: Internet signals travel through cables and air at different speeds
Language: Typing URLs accurately requires spelling and attention to detail

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Label the browser diagram using a word bank with pictures
On Level: Navigate to three websites and bookmark two of them independently
Advanced: Clear browser history and cache, then explain what each action does
SEND: Point to and name browser parts with teacher guidance, practice clicking back and forward

Formative Assessment

Method: Practical skill demonstration - teacher observes each student performing five browser skills
CT Skill Assessed: Algorithm Design - following a sequence of steps to navigate the browser correctly

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a picture of a web browser window from memory. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.05 *Search Engines*

40 minutes **CONCEPT**

Success Criteria - I can...

- Explain how search engines find, index, and rank web pages
- Use three search techniques: exact phrases, exclusion, and site-specific search
- Evaluate whether a search result is trustworthy based on source and date

Resources Needed:
- Computer with projector for live search demonstration
- Search challenge worksheet for each group
- Quiz paper
- Board markers

Key Vocabulary

Search Engine: Tool that finds websites matching your typed words quickly
Keyword: Important word you type to find specific information online
Index: Huge list where search engine stores information from websites
Ranking: Ordering of search results from most useful to least

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "If I type school in the search box, what will Google show me?" Type it and show the results - millions of results from all over the world. Ask: "What if I only want schools in Karnataka?" Change the search to school Karnataka. Fewer and better results. Ask: "What if I want information from the government website specifically?" Type site:kreis.karnataka.gov.in school. Ask: "How did the results get better each time?"

STUDENTS Guess what Google will show for "school" - probably lots of different schools. Notice the millions of results. See how adding Karnataka narrows the results. Watch the site: search show only government websites. Discuss how search terms make a huge difference.

Explore - Guided Discovery (10 min)

TEACHER Give each group a worksheet with five search challenges. (1) Find the population of India using quotes for exact phrase. (2) Find information about cats but exclude the word pet. (3) Find information only from .edu websites about climate change. (4) Find the time in London. (5) Find a recipe for rice bath from a Karnataka website.

STUDENTS Work in groups to complete each challenge. Write their search queries: "population of India", cats -pet, climate change site:.edu, time London, rice bath recipe site:.in. Record results. Discuss which queries worked best.

Explain - Modelling & Concepts (10 min)

TEACHER Explain how search engines work in three steps. Step 1 - Crawling: crawlers visit every webpage and read content, following links from page to page. Step 2 - Indexing: the search engine stores information in a huge database called an index. Step 3 - Ranking: the algorithm looks at relevance, quality, freshness, and authority to show the best results first. Explain search tips: use specific keywords, put exact phrases in quotes, use minus to exclude words, use site: to search a specific website.

STUDENTS Take notes on crawling, indexing, ranking. Draw a diagram showing a spider visiting pages, storing information, showing results. Write five search tips. Practice creating better search queries.

Elaborate - Independent Application (10 min)

TEACHER Give a research question: "What is the average rainfall in Karnataka?" Students must search using three different queries and compare results. After 10 minutes, ask: "Which query worked best? How did you know the result was trustworthy?"

STUDENTS Search with three different queries. Record the top result for each. Compare quality. Discuss which source is most trustworthy and why: check the source (.gov, .edu), compare multiple sites, look for dates.

Evaluate - Check & Close (5 min)

TEACHER Quick quiz: (1) What are the three steps a search engine uses? (2) Write a search query to find information about tigers only from .gov websites. (3) How do you know if a search result is trustworthy? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Answer: crawling, indexing, ranking. Write: tigers site:.gov. Trustworthy signs: .gov or .edu source, recent date, multiple sources confirm.

Cross-Curricular Connections

Mathematics: Ranking involves comparing and ordering by multiple factors
Science: Crawlers follow links like scientists follow research citations
Language: Choosing search keywords requires understanding of synonyms

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Complete two search challenges with a word bank of suggested terms
On Level: Complete all five challenges and evaluate which result is most trustworthy
Advanced: Compare results from Google and another search engine, write about differences
SEND: Perform one simple search with teacher guidance and identify the top result

Formative Assessment

Method: Quick quiz on search engine steps and techniques
CT Skill Assessed: Algorithm Design - constructing effective search queries using specific rules

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write the three steps a search engine uses, in your own words. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.06 *URLs and Websites*

40 minutes **CONCEPT**

Success Criteria - I can…

- Identify the four main parts of a URL: protocol, domain name, top-level domain, and path
- Distinguish between secure HTTPS and non-secure HTTP websites
- Classify websites as educational, commercial, government, or organizational

Resources Needed:
- Large labeled URL diagram on chart paper
- Worksheet with 8 URLs
- 12 website cards for sorting
- Quiz paper
- Colored markers

Key Vocabulary

URL: Address that shows exact location of a website online
Protocol: Rule that controls how data travels between computers securely
Domain: Unique name identifying a website on the internet
HTTPS: Secure rule that encrypts data to keep information safe

Engage - Retrieval & Hook (5 min)

TEACHER Write two URLs on the board: http://example.com and https://kreis.karnataka.gov.in. Ask: "What do you notice is different between these two?" Students will point out the s in https. Ask: "What do you think that s means?" Then point to the rest: "What does .com mean? What about .gov.in?"

STUDENTS Compare the two URLs. Notice the s - guess it means secure. Notice .com vs .gov.in. Discuss what each part might mean, like how a postal address has house number, street, city, and state.

Explore - Guided Discovery (10 min)

TEACHER Show a large diagram of: https://www.kreis.karnataka.gov.in/schools/list. Label each part: protocol (https), subdomain (www), domain (kreis), second-level domain (karnataka), top-level domain (.gov.in), path (/schools/list). Give students a worksheet with 8 URLs to analyze.

STUDENTS Study the labeled diagram. Complete the worksheet: identify parts of URLs like google.com, scratch.mit.edu, kreis.karnataka.gov.in. Color-code the parts. Discuss why some start with http and others https.

Explain - Modelling & Concepts (10 min)

TEACHER Explain each part. Protocol: rules for data travel - http is older, https encrypts data (s = secure). Domain name: unique website name. Top-level domain: category - .com commercial, .edu educational, .gov government, .org organization, .in India. Path: specific page within the website. Show the lock icon for https sites. Explain: "Always look for the lock icon before entering personal information."

STUDENTS Write notes on each part. Draw a diagram labeling URL parts. Classify 10 websites by top-level domain: google.com (commercial), mit.edu (educational), kreis.karnataka.gov.in (government). Find the lock icon in the browser.

Elaborate - Independent Application (10 min)

TEACHER Present a safety scenario: "You need to enter your bank password. How do you know if it is safe?" Discuss: look for https, lock icon, correct domain. Then classify 12 website cards into categories.

STUDENTS Discuss safety: always check for https and lock icon. Complete classification: youtube.com (commercial), ncert.nic.in (educational), india.gov.in (government), redcross.org (organization). Create a poster of URL safety tips.

Evaluate - Check & Close (5 min)

TEACHER Quiz: (1) Label the parts of: https://www.school.edu.in/students. (2) What does the s in https mean? (3) Is http://free-gift-winner.com safe? Why? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Label: protocol https, subdomain www, domain school, second-level edu, top-level .in, path /students. Answer: s means secure. Answer: No - http not https, suspicious domain.

Cross-Curricular Connections

Mathematics: URL structure involves hierarchy and categorization
Science: Encryption uses mathematical algorithms to scramble data
Language: Domain names require correct spelling and punctuation

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match URLs to categories using a picture guide
On Level: Identify all parts of a URL and classify 8 websites
Advanced: Explain encryption and give examples of phishing URLs
SEND: Point to the lock icon and domain name on displayed URLs

Formative Assessment

Method: Quiz on URL parts, HTTPS safety, and website classification
CT Skill Assessed: Abstraction - filtering relevant information from a URL to determine its nature

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write down 6 URLs from websites you know. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.07 *Email Basics*

40 minutes **CONCEPT**

Success Criteria - I can...

- Identify the parts of an email address: username, @ symbol, domain
- Compose a formal email with subject line, greeting, body, and closing
- Use CC, BCC, and Reply All appropriately

Resources Needed:
- Printed example email for display
- Blank email template worksheet
- Jumbled email cards
- Board markers
- Student notebooks

Key Vocabulary

Email: Message sent instantly through the internet to another person
Inbox: Folder where all received emails are stored together
Subject: Short line telling what your email is about clearly
Attachment: File like photo or document sent together with email

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "How many of you have sent a letter by post? How long does it take?" Write answers: 3 days, 5 days, a week. Then: "What if you could send a letter that reaches someone in 3 seconds, anywhere in the world, for free?" Write an email address: student@kreis.karnataka.gov.in. Ask: "What do you notice? What does @ mean?"

STUDENTS Share experiences of sending letters. Notice it takes days. Compare with email which takes seconds. Examine the email address: @ separates the person's name from where they receive mail. @ means "at" - student at kreis.karnataka.gov.in.

Explore - Guided Discovery (10 min)

TEACHER Show a printed email on the projector. Point out: From, To, CC, BCC, Subject, Date, Greeting, Body, Closing, Signature. Give a worksheet with a blank email template. Students fill in: To address, Subject, greeting, 3-sentence body, closing.

STUDENTS Study the example email and label each part. Fill in the template: To: teacher@school.edu.in, Subject: Request for Library Book, Greeting: Dear Sir/Madam, Body: I would like to borrow a science book. May I collect it tomorrow? Closing: Thank you, Name.

Explain - Modelling & Concepts (10 min)

TEACHER Explain each part. Email address: username@domain. Subject line: short summary like a title. Greeting: Dear Sir/Madam (formal), Hi Name (informal). Body: main message - clear, short, polite. Closing: Yours sincerely (formal), Thanks, Best regards. CC: copy to someone who needs to know. BCC: secret copy - others cannot see. Reply All: reply to everyone.

STUDENTS Take notes. Write the format: To, Subject, Greeting, Body, Closing. Practice writing a formal email to the principal and an informal email to a friend. Compare tone and language differences.

Elaborate - Independent Application (10 min)

TEACHER Scenario: "The school science fair is next week. Email the principal for permission, the lab assistant to book the lab, and your friend to join your team." Walk around: "Use a clear subject, be polite, sign your name. What tone for the principal vs friend?"

STUDENTS Write three emails. Email 1 (formal): To principal, subject "Science Fair Permission", polite body. Email 2 (semi-formal): To lab assistant, subject "Lab Booking", clear body with date/time. Email 3 (informal): To friend, subject "Science Fair", casual body. Compare the three emails.

Evaluate - Check & Close (5 min)

TEACHER Give a jumbled email - put parts in correct order. Then: "What is the difference between CC and BCC?" Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Arrange jumbled email parts in correct order. Answer: CC shows all recipients; BCC hides the copied person from others.

Cross-Curricular Connections

Mathematics: Email timestamps involve time zones and date formats
Science: Email travels through servers and cables - physics of data transmission
Language: Writing emails practices formal and informal letter writing conventions

Digital Citizen Reminder: Never share your email password with anyone except your parents. Use a strong password with letters and numbers.

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Differentiation

Approaching: Fill in a pre-written email template using word banks
On Level: Write a complete formal email with correct format and tone
Advanced: Explain CC vs BCC with examples and write an email using both
SEND: Identify To, Subject, Body on a displayed example with picture labels

Formative Assessment

Method: Jumbled email reordering and CC/BCC explanation
CT Skill Assessed: Algorithm Design - following the correct sequence for composing an email

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three emails: (1) formal email to class teacher requesting leave, (2) email to a friend inviting them to study, (3) email to librarian asking to renew a book. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.08 *Online Safety*

40 minutes **CONCEPT**

Success Criteria - I can...

- Identify five common online dangers: phishing, cyberbullying, privacy leaks, inappropriate content, and addiction
- Apply the STOP rule: Stop, Think, Observe, Protect
- Create a personal online safety checklist

Key Vocabulary

Phishing: Fake message tricking you to share passwords or money
Cyberbullying: Using internet to hurt or scare another person repeatedly
Privacy: Keeping personal information safe from strangers online always
STOP Rule: Stop Think Observe Protect steps before clicking anything online

Engage - Retrieval & Hook (5 min)

TEACHER Read this scenario: "Ravi gets a message: 'Congratulations! You won a free mobile phone. Click here.' He clicks the link. What happens next?" Ask pairs to discuss. After 2 minutes, share predictions. Then reveal: "The link was fake. It stole Ravi's password and all his contacts." Ask: "How could Ravi have avoided this?"

STUDENTS Discuss the scenario in pairs. Predict: virus, stolen money, broken phone. Share ideas. Suggest: don't click unknown links, check if the message is real, ask an adult.

Explore - Guided Discovery (10 min)

TEACHER Give each group five scenario cards: (1) Stranger asks for your photo. (2) Classmate sends mean messages. (3) Website asks for bank details. (4) You see scary images while browsing. (5) You have been playing games for 4 hours. For each: identify the danger, decide what to do.

STUDENTS Read each scenario. For (1): stranger danger, block, tell adult. For (2): cyberbullying, don't respond, save evidence, tell teacher. For (3): phishing, close page, tell parent. For (4): inappropriate content, close, tell teacher. For (5): addiction, set timer, take breaks.

Explain - Modelling & Concepts (10 min)

TEACHER Explain the STOP rule. S - Stop: don't click, reply, or share. T - Think: is this real? Is this safe? O - Observe: look for warning signs - strange addresses, spelling mistakes, too-good-to-be-true offers. P - Protect: block, close, tell a trusted adult. Then explain five dangers: phishing, cyberbullying, privacy leaks, inappropriate content, addiction.

STUDENTS Write STOP rule with explanations. Write five dangers with examples. Create a mnemonic. Practice applying STOP to new scenarios.

Elaborate - Independent Application (10 min)

TEACHER Challenge: "Create a personal online safety checklist - 10 rules you will follow every day." Prompt: "What will you do before clicking a link? What if someone you do not know messages you?" After 12 minutes, combine best rules into a class safety charter.

STUDENTS Brainstorm 10 safety rules: never share passwords, don't click unknown links, tell an adult if something feels wrong, use a nickname, don't share photos without permission, take breaks, keep personal info private, be kind online, check website security, report bullying.

Evaluate - Check & Close (5 min)

TEACHER Scenario quiz: three new scenarios, write what you would do using STOP. (1) Email saying account will be deleted unless you click. (2) Classmate posts embarrassing photo. (3) Pop-up says you won a prize. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Write responses. (1) STOP - don't click. Think - is this real? Protect - delete, tell teacher. (2) STOP - don't respond angrily. Protect - save evidence, tell teacher, block. (3) STOP - don't enter info. Protect - close pop-up, tell adult.

Cross-Curricular Connections

Mathematics: Addiction involves time management - calculating hours online vs offline
Science: Screen time affects eyesight and sleep - biology of technology
Language: Writing safety rules practices imperative sentences and clear instructions

Digital Citizen Reminder: The STOP rule: Stop before clicking. Think about safety. Observe for warning signs. Protect yourself by telling an adult.

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Differentiation

Approaching: Match online dangers to safety actions using pictures
On Level: Write STOP rule and apply to three scenarios independently
Advanced: Research a real cyberbullying case and write a report
SEND: Sort safe/unsafe online actions with picture cards and teacher help

Formative Assessment

Method: Scenario-based quiz applying the STOP rule
CT Skill Assessed: Evaluation - assessing whether an online situation is safe and deciding the response

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write the STOP rule with explanations. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.09 *GPS and Digital Maps*

40 minutes **CONCEPT**

Success Criteria - I can...

- Explain how GPS uses satellites to determine a location on Earth
- Identify three real-world uses of GPS: navigation, agriculture, and emergency services
- Read a digital map to find directions between two points

Resources Needed:
- Scenario cards (5 per group)
- Chart paper for safety checklists
- Board markers
- Quiz paper
- STOP rule poster

Key Vocabulary

GPS: System using satellites to find exact location on Earth
Satellite: Machine orbiting Earth that sends signals to your phone
Coordinates: Numbers showing exact place of a location on map
Navigation: Finding best route from one place to another safely

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "How would you give directions from our school to the nearest bus stop? What landmarks would you mention?" Write directions: "Go straight, turn left at the temple, walk past the shop." Then: "What if someone from another city needed those directions? They would not know the temple. How do Google Maps or Uber drivers know exactly where to go?" Introduce GPS.

STUDENTS Give directions using local landmarks. Realize these only work for locals. Discuss how Google Maps works for anyone. Guess it uses satellites or some technology to know your location.

Explore - Guided Discovery (10 min)

TEACHER Give each group a printed map of the local area. Mark the school with a star. Find 5 locations: bus stand, hospital, market, post office, park. Draw routes from school to each. Estimate distance using map scale.

STUDENTS Study the map. Locate school and 5 destinations. Draw routes with rulers. Estimate distances. Discuss: which route is shortest? Safest? Present their route to the bus stand.

Explain - Modelling & Concepts (10 min)

TEACHER Explain GPS. GPS = Global Positioning System. 31 satellites orbit Earth. Your phone receives signals from at least 4 satellites. By comparing signal arrival times, your phone calculates its exact position - like triangulation. Accuracy: within 5 meters. Then explain digital maps: Google Maps, OpenStreetMap. Features: search, directions, traffic, distance.

STUDENTS Write notes: GPS uses 31 satellites, phone receives from 4+, calculates position by comparing signal times. Key terms: satellite, signal, triangulation, coordinates. Watch live demo: search for school, zoom in, get directions.

Elaborate - Independent Application (10 min)

TEACHER Challenge: "Your school is organizing a field trip. Use the digital map to plan the route. How far? How long will the bus take? Where should you stop for lunch?"

STUDENTS Study the route. Identify distance and time. Find landmarks. Choose lunch stop halfway. Present trip plan: "Leave at 8am, travel 45km, stop at 10am, arrive at 11am." Discuss how GPS helps drivers avoid getting lost.

Evaluate - Check & Close (5 min)

TEACHER Quick quiz: (1) What does GPS stand for? (2) How many satellites for location? (3) Two uses besides navigation? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Answer: Global Positioning System. Answer: At least 4. Uses: agriculture, emergency services, tracking, surveying.

Cross-Curricular Connections

Mathematics: Map scales involve ratios and distance measurement
Science: Satellites orbit Earth using gravity and physics
Language: Giving directions practices sequential and spatial language

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Find 3 locations by following arrow paths
On Level: Plan a route and estimate distance using map scale
Advanced: Compare two routes and explain which is better
SEND: Point to familiar locations on map, trace route with finger

Formative Assessment

Method: Quick quiz on GPS facts and digital map reading
CT Skill Assessed: Decomposition - breaking down a journey into steps: start, route, landmarks, destination

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a simple map of your dormitory area from memory. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.10 *Cloud Computing Basics*

40 minutes **CONCEPT**

Success Criteria - I can...

- Define cloud computing and explain how data is stored on remote servers
- Compare local storage (USB drive) with cloud storage (Google Drive)
- Identify three advantages of cloud storage: access anywhere, automatic backup, sharing

Resources Needed:

- Printed local maps or chart paper maps
- Rulers for distance measurement
- Computer with projector for Google Maps demo
- Quiz paper
- Map markers

Key Vocabulary

Cloud: Servers on internet that store your files safely always
Server: Powerful computer storing many files for many users together
Upload: Sending your file from device to cloud storage
Download: Bringing a file from cloud to your own device

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "If you write an essay on the school computer and forget to save it to a pen drive, what happens?" Students: it stays on that computer. Then: "What if there is a way to save your work and access it from any computer, anywhere, even your phone? And it backs up automatically?" Write: Where do your Google Docs live? Not on your phone. Not on this computer. So where?

STUDENTS Realize files on a school computer stay there unless copied. Think about the problem: forget pen drive, lose work. Wonder where Google Docs stores files - maybe on Google's computers. Discuss "the cloud" - someone else's computer you reach through the internet.

Explore - Guided Discovery (10 min)

TEACHER Show comparison: Local Storage vs Cloud Storage. Local: works offline, limited space, can be lost, one computer. Cloud: needs internet, usually free (15GB), backed up, any device. Scenario: "Your USB is broken but homework is due. How does cloud help?"

STUDENTS Study comparison and add rows. Discuss scenario: if homework is on Google Drive, access from any computer. Broken USB doesn't matter.

Explain - Modelling & Concepts (10 min)

TEACHER Explain cloud computing. "The cloud" is a building full of powerful computers called servers. Companies like Google, Microsoft, Amazon own them. When you save to Google Drive, it travels through the internet to their servers. Key concepts: Server, Upload, Download, Backup. Show upload/download process with arrows.

STUDENTS Write notes: Cloud = servers, files stored remotely, accessed through internet. Key terms: server, upload, download, backup. Draw the process. Write advantages and disadvantages.

Elaborate - Independent Application (10 min)

TEACHER Debate: "Should the school use USB drives or cloud storage for student records?" Consider: access, backup, security, cost, internet. Then discuss: "What if the internet goes down?"

STUDENTS Prepare arguments for/against cloud storage. Hold mini-debate. Conclude: use cloud but keep local backups. Discuss hybrid approach.

Evaluate - Check & Close (5 min)

TEACHER Quiz: (1) What is "the cloud"? (2) Two advantages, one disadvantage of cloud storage. (3) Where is a file stored when uploaded to Google Drive? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Answer: Network of servers storing data, accessed through internet. Advantages: access anywhere, backup. Disadvantage: needs internet. Answer: On Google's servers in a data center.

Cross-Curricular Connections

Mathematics: Storage units involve powers of 10: KB, MB, GB, TB
Science: Servers generate heat and need cooling - energy and environment
Language: Writing comparisons practices comparative and superlative language

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match cloud features to descriptions using word bank
On Level: Complete comparison chart and explain two advantages
Advanced: Explain cloud backup and discuss encryption briefly
SEND: Identify USB drive and cloud icon, name which stores files on internet

Formative Assessment

Method: Quick quiz on cloud concepts with opinion question
CT Skill Assessed: Abstraction - understanding "the cloud" as an abstraction for remote servers

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw two pictures: (1) USB drive with local files, (2) Cloud diagram with files on server accessed from phone and computer. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.11 *Digital Footprint*

40 minutes **CONCEPT**

Success Criteria - I can…

- Define digital footprint and distinguish between active and passive footprints
- Identify three examples of information that create a digital footprint
- Apply the Think-Before-You-Post rule to protect their online reputation

Resources Needed:
- Comparison chart on board
- Projector for cloud storage demo
- Quiz paper
- Board markers
- USB drive for demonstration

Key Vocabulary

Digital Footprint: Trail of data you leave whenever you use internet
Active Footprint: Data you share deliberately like posting photos online
Passive Footprint: Data collected secretly like cookies tracking your browsing
Reputation: What people think about you based on your online actions

Engage - Retrieval & Hook (5 min)

TEACHER Draw footprints on the board - physical footprints in sand. Ask: "When you walk on the beach, you leave footprints. Can others see where you walked?" Students: yes. Then: "Did you know you also leave footprints on the internet? Every photo you post, every comment you write - it all leaves a trail. This is called a digital footprint." Ask: "If someone searched your name online, what would they find?"

STUDENTS Look at the footprints. Understand physical footprints show where you've been. Apply to internet - online actions leave traces. Think about what they've posted online. Wonder what a stranger could find.

Explore - Guided Discovery (10 min)

TEACHER Give a worksheet with two columns: Active Digital Footprint and Passive Digital Footprint. Active: things you deliberately share - posting photos, comments, emails. Passive: collected without you knowing - tracking, cookies, search history. Sort 15 cards into columns.

STUDENTS Sort cards into active and passive. Discuss: liking a post - active (you clicked) or passive (platform records)? Present sorting. Conclude: passive is more dangerous because you don't know it's collected.

Explain - Modelling & Concepts (10 min)

TEACHER Explain digital footprint. Active: you choose to share - social media, comments, photos, emails. You control this. Passive: collected automatically - cookies, apps, ISPs, advertisers. You don't control this. Why it matters: employers, colleges can find your history. Once posted, almost impossible to delete. Think-Before-You-Post: Would I be comfortable if my teacher or future employer saw this? If not, don't post.

STUDENTS Write notes on active vs passive with 3 examples each. Write Think-Before-You-Post rule. Discuss real examples: student posted angry comment, teacher found it during job interview.

Elaborate - Independent Application (10 min)

TEACHER Challenge: "Audit your own digital footprint. List everything you posted online in the past month. For each, ask: Would I be comfortable if everyone could see this? Rate as green (fine), yellow (risky), red (should not have posted)."

STUDENTS List online activity from past month. Rate as green, yellow, red. Reflect on risky posts. Discuss patterns as a class.

Evaluate - Check & Close (5 min)

TEACHER Exit ticket: (1) What is a digital footprint? (2) One example active, one passive. (3) Write Think-Before-You-Post rule. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Answer: Trail of data left when using internet. Active: posting photo. Passive: websites tracking visits. Think-Before-You-Post: Ask "Would I be comfortable if my teacher saw this?"

Cross-Curricular Connections

Mathematics: Tracking involves numbers, dates, statistics
Science: Cookies use technology similar to sensors and data logging
Language: Writing about digital footprint practices persuasive and reflective writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match online actions to active/passive footprint using pictures
On Level: Complete sorting and write Think-Before-You-Post with example
Advanced: Research how cookies work, write explanation of website tracking
SEND: Sort safe/unsafe online actions with picture cards and teacher help

Formative Assessment

Method: Exit ticket on digital footprint definition, examples, and safety rule
CT Skill Assessed: Evaluation - assessing own online behavior and deciding what is safe to share

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write the definition of digital footprint. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.12 *ICT in Daily Life*

40 minutes **CONCEPT**

Success Criteria - I can...

- Identify how ICT is used in five sectors: education, health, agriculture, banking, communication
- Explain how ICT helps rural communities access services
- Design a simple ICT solution for a problem in their village

Resources Needed:
- Footprint drawings on board
- Sorting worksheet with 15 cards
- Audit worksheet
- Exit ticket paper
- Board markers

Key Vocabulary

AI: Machines mimicking human thinking to solve problems without explicit instructions always
Model: Trained system that makes predictions based on data it learned from examples
Data: Examples used to teach AI system patterns for making decisions later
Ethics: Rules ensuring fair and safe use of AI for everyone

Engage - Retrieval & Hook (5 min)

TEACHER Ask: "Think about everything you did from waking up to arriving at school. How many things involved technology?" Write a timeline: alarm clock, radio, phone, weather app, bus ticket, school register. Ask: "What would life be like without these?"

STUDENTS Think through morning routine. Identify technology uses. Imagine life without these: walking to get news, sending letters, no phones. Discuss how much easier ICT makes daily life.

Explore - Guided Discovery (10 min)

TEACHER Give each group a sector card: Education, Health, Agriculture, Banking, Communication. Brainstorm 4 ways ICT is used. After 10 minutes, present. Then: "Which sector benefits most from ICT in our village?"

STUDENTS Brainstorm ICT uses: Education - online classes, educational YouTube. Health - telemedicine, health apps. Agriculture - weather apps, market prices. Banking - ATMs, UPI. Communication - WhatsApp, video calls. Present. Discuss which is most important for their village.

Explain - Modelling & Concepts (10 min)

TEACHER Explain ICT impact with village examples. Education: student watches IIT lectures on YouTube. Health: farmer shows symptoms to doctor via video call. Agriculture: checks weather and market prices on phone. Banking:

sends money to son in city using UPI. Communication: video calls daughter in Bangalore. Discuss challenges: electricity, internet speed, digital literacy, language. Ask: "How can we overcome these?"

STUDENTS Take notes on each sector. Discuss challenges: power cuts, slow internet, older people don't know phones, apps in English. Suggest solutions: solar chargers, community Wi-Fi, digital literacy classes, local language apps.

Elaborate - Independent Application (10 min)

TEACHER Capstone challenge: "Design a simple ICT solution for a problem in your village." Identify one real problem, explain how ICT can solve it. Examples: telemedicine center, price SMS service, digital library. Present as 2-minute pitch.

STUDENTS Identify a real problem. Brainstorm ICT solutions. Design: what does it do? Who does it help? How does it work? Present as 2-minute pitch. Listen to other groups. Vote on most practical solution.

Evaluate - Check & Close (5 min)

TEACHER Chapter reflection: (1) Most useful ICT tool you learned? (2) How has ICT changed your village? (3) What ICT skill do you want to learn next? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Write reflection. Think about how everything connects: internet, browsers, search, email, safety, GPS, cloud, digital footprint - all parts of the ICT world.

Cross-Curricular Connections

Mathematics: Data about ICT usage involves statistics and graphs
Science: ICT in agriculture connects to biology, weather science, environmental studies
Language: Writing proposals practices persuasive and structured writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match ICT tools to sectors using picture matching
On Level: Identify ICT uses in 3 sectors and present one example each
Advanced: Design a complete ICT solution with diagram, requirements, and plan
SEND: Name ICT tools used at home, point to pictures of each

Formative Assessment

Method: Chapter reflection on learning, application, and future goals
CT Skill Assessed: Evaluation - assessing ICT impact on community and planning next steps

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a short essay (10 sentences) titled "How ICT Changed My Village." Describe three ways ICT is used, explain how each helps, and suggest one new ICT service. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 2 - Graphics in ICT

Lesson 2.01 *Introduction to Graphics*

40 minutes LAB

Success Criteria - I can...

- Define graphics and identify different types of computer images
- Distinguish between raster and vector graphics with real-world examples
- Explain how pixels form the building blocks of digital images

Resources Needed:

- Sector cards for group activity
- Chart paper for solution design
- Timer for presentations
- Reflection paper
- Board markers

Key Vocabulary

Graphics: Pictures or images created and edited on a computer
Canvas: Digital workspace where you draw or paint pictures freely
Tool: Option like brush or eraser that helps you create art
Palette: Set of colours you choose from to paint digitally

Engage - Retrieval & Hook (5 min)

TEACHER Show students two prints of the school annual day photograph - one clear, one heavily zoomed and blurry. Ask: 'Why does one look sharp and the other looks blocky even though both came from the same camera?'

STUDENTS Students examine both prints closely, pointing out the small squares visible in the blurry version. They shout out guesses about why the difference exists.

Explore - Guided Discovery (10 min)

TEACHER Give students magnifying glasses and ask them to examine newspaper photographs up close. Then open a simple image on the computer lab projector and zoom in step by step until individual pixels are visible.

STUDENTS Students sketch what they see at three zoom levels in their ICT journal: normal view, partially zoomed, and fully zoomed to pixel level. They label the tiny squares as 'pixels'.

Explain - Modelling & Concepts (10 min)

TEACHER Define graphics as visual representations created or manipulated by computers. Explain raster images as grids of colored pixels and vector images as math-defined shapes. Use the analogy of a rangoli made of colored rice grains (raster) versus a stencil outline (vector).

STUDENTS Students take notes on the differences between raster and vector graphics. They create a small comparison table in their notebooks listing properties like scalability, file size, and common uses.

Elaborate - Independent Application (10 min)

TEACHER Challenge students to find five examples of graphics around the school - the school logo on uniforms, the bus number plate, a textbook illustration, a poster in the corridor, and a photo on the notice board. Classify each as likely raster or vector and justify their reasoning.

STUDENTS Students walk around the school with their notebooks, sketching and classifying each graphic they find. They present their findings to a partner and discuss disagreements.

Evaluate - Check & Close (5 min)

TEACHER Give students a quick worksheet with three images: a scanned handwriting sample, a digital drawing of a circle, and a photograph of the school garden. They must label each as raster or vector and write one reason. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students complete the worksheet independently. Teacher collects and reviews for understanding of raster versus vector concepts.

Cross-Curricular Connections

Mathematics: Math: Counting pixels in a grid connects to understanding area and coordinate systems
Science: Science: How the human eye perceives color relates to how screens display pixel colors
Language: Language: Writing descriptive captions for images builds vocabulary and communication skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-labeled diagrams showing pixel grids with guided fill-in-the-blank notes
On Level: Students complete comparison tables and classify images at their own pace
Advanced: Research the difference between PNG, JPEG, and GIF formats and present findings to the class
SEND: One-on-one guided exploration of zooming into images with teacher support

Formative Assessment

Method: Worksheet classification of raster vs vector images with written justifications
CT Skill Assessed: Decomposition - breaking down a digital image into its fundamental pixel components

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Before bed, look at the poster in your dormitory and try to imagine it made of tiny colored squares. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.02 *Using Paint and Tux Paint*

40 minutes LAB

Success Criteria - I can…

- Navigate the Tux Paint interface and identify all major toolbars
- Use at least five different drawing tools to create a simple picture
- Save and reopen a drawing file in the correct format

Resources Needed:
- Newspaper clippings
- Magnifying glasses
- Computer lab projector
- Image viewer software
- ICT journal worksheets

Key Vocabulary

Paint: Simple program for drawing and colouring pictures digitally
Brush: Tool that draws lines with chosen colour and thickness
Eraser: Tool that removes mistakes from your drawing cleanly
Fill: Colouring a closed shape quickly with one click

Engage - Retrieval & Hook (5 min)

TEACHER Display a colorful drawing of the school campus created entirely in Tux Paint. Tell students: 'This was drawn using only the computer - no paintbrush, no crayons. Today you will learn to use the same tools.'

STUDENTS Students express excitement and curiosity. Some ask if it is really made on the computer. Teacher promises they will create their own by the end of the lesson.

Explore - Guided Discovery (10 min)

TEACHER Open Tux Paint on each computer and let students freely click on every tool, stamp, and selector for five minutes. Ask them to note which tools they like and what each one does.

STUDENTS Students experiment wildly - clicking brushes, stamps, magic effects, and colors. They call out to friends about funny effects they discover. Teacher circulates and asks guiding questions.

Explain - Modelling & Concepts (10 min)

TEACHER Walk through the Tux Paint interface systematically: the left toolbar for drawing tools, the bottom selector for tool options, the right panel for stamps and fonts, and the top buttons for save, open, and print. Demonstrate the paint brush, spray can, shapes, text, and eraser one by one.

STUDENTS Students follow along on their computers, replicating each tool demonstration. They take brief notes on what each tool does and its keyboard shortcut.

Elaborate - Independent Application (10 min)

TEACHER Give students a challenge: draw a picture of the school building using at least four different tools - the line tool for walls, the fill tool for coloring, the text tool for the school name, and the stamp tool for trees. They must save their work before the period ends.

STUDENTS Students work on their drawings with focus. They help each other remember tool locations and share discoveries about color combinations. Teacher checks that every student saves their file.

Evaluate - Check & Close (5 min)

TEACHER Ask students to open their saved file and demonstrate one tool to a neighbor. The neighbor must identify the tool by name. Teacher walks around listening to peer teaching. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students take turns demonstrating and observing. Teacher notes which students can confidently name and use the tools.

Cross-Curricular Connections

Mathematics: Math: Using the shape tools to draw geometric figures reinforces understanding of symmetry and angles
Science: Science: Drawing diagrams of plant parts or water cycle using Tux Paint supports science learning
Language: Language: Writing a short caption below the drawing using the text tool builds descriptive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a step-by-step picture card guide showing each tool with screenshots
On Level: Students follow the guided drawing task and save independently
Advanced: Create a complex scene using advanced stamps, magic effects, and multiple layers of detail
SEND: Sit beside the student and guide each mouse click through the drawing process

Formative Assessment

Method: Observation checklist of tool usage during drawing activity plus successful file save
CT Skill Assessed: Abstraction - simplifying a real-world scene into basic shapes and tools on screen

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of your favorite place in the school. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.03 *Color Theory for Digital Art* 40 minutes LAB

Success Criteria - I can...

- Identify primary, secondary, and complementary colors on the digital color wheel
- Use the color picker tool to select and mix exact shades
- Apply basic color harmony principles to create a visually pleasing digital drawing

Resources Needed:
- Computers with Tux Paint installed
- Projector for demonstrations
- Sample drawing handout
- Save location on shared drive

Key Vocabulary

Primary Colour: Red blue or yellow that makes all other colours
Secondary Colour: Colour made by mixing two primary colours together
Shade: Darker version of a colour made by adding black
Contrast: Strong difference between light and dark colours together

Engage - Retrieval & Hook (5 min)

TEACHER Show two versions of the same flower drawing - one with random clashing colors and one with harmonious colors. Ask: 'Which one looks better to you? Why?' Write student responses on the board.

STUDENTS Students immediately point to the harmonious version. They describe words like 'nice', 'matching', and 'beautiful' without knowing the formal color theory terms.

Explore - Guided Discovery (10 min)

TEACHER Give each student a printed color wheel and crayons. Ask them to mix colors on scrap paper and predict what happens when they combine red and yellow, blue and yellow, and red and blue. Then let them try the same mixing on the computer color picker.

STUDENTS Students mix physical colors first, then replicate the process digitally. They observe that the computer color picker shows them the exact RGB values. They fill in a color mixing chart in their journals.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the color wheel: primary colors (red, blue, yellow) cannot be made from other colors. Secondary colors (green, orange, purple) are made by mixing two primaries. Complementary colors sit opposite each other and create contrast. Show how the digital color picker uses RGB values from 0 to 255 for each channel.

STUDENTS Students label their color wheels with primary, secondary, and complementary categories. They experiment with the RGB sliders on the computer to create exactly ten different colors and record the RGB values.

Elaborate - Independent Application (10 min)

TEACHER Challenge students to create a digital drawing of a sunset using only warm colors (red, orange, yellow) and then recreate it using cool colors (blue, green, purple). They must explain which version feels different and why.

STUDENTS Students create two versions of the sunset, carefully choosing colors from the palette. They write a short reflection comparing the emotional feel of warm versus cool palettes.

Evaluate - Check & Close (5 min)

TEACHER Students present their warm and cool sunsets to a partner. The partner must identify which colors are primary, secondary, or complementary used in each drawing. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students explain their color choices and identify color categories in their partner's work. Teacher circulates and asks follow-up questions about color mixing.

Cross-Curricular Connections

Mathematics: Art: Color wheel concepts connect directly to traditional art and painting techniques
Science: Science: Light and color mixing through RGB relates to physics of light and wavelength
Language: Language: Using descriptive color adjectives in writing - crimson, azure, emerald - enriches vocabulary

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a pre-labeled color wheel and matching color name cards to sort
On Level: Complete the color mixing chart and digital drawing with some teacher guidance
Advanced: Experiment with advanced color schemes like triadic, split-complementary, and analogous
SEND: Use physical crayons to mix colors first, then match them on screen with one-on-one help

Formative Assessment

Method: Color mixing chart completion plus a two-version sunset drawing with written reflection
CT Skill Assessed: Pattern recognition - identifying how color combinations create visual patterns and harmony

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the colors in your dormitory at night - the bedsheets, walls, and light. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.04 *Drawing Tools and Techniques* 40 minutes LAB

Success Criteria - I can...

- Use the pencil, brush, spray, and line tools with control and precision
- Apply different brush sizes and opacity settings to create depth in a drawing
- Combine multiple drawing techniques to complete a structured composition

Resources Needed:
- Printed color wheel handouts
- Crayons and mixing paper
- Computers with Tux Paint
- RGB color picker reference sheet

Key Vocabulary

Pencil: Thin tool for drawing light outlines and sketches precisely
Brush: Thick tool for painting broad strokes with colour smoothly
Shape Tool: Quick way to draw perfect circles squares and lines
Colour Picker: Tool that copies any colour already on your canvas

Engage - Retrieval & Hook (5 min)

TEACHER Show a time-lapse video of a digital artist creating a landscape from scratch using basic tools. Pause at key moments and ask: 'Which tool do you think they used here? How can you tell?'

STUDENTS Students call out guesses like 'that must be the spray can for the trees' and 'the line tool for the buildings'. Their observation skills are activated before they touch the computer.

Explore - Guided Discovery (10 min)

TEACHER Set up five mini-stations on each computer: one for the pencil tool, one for the brush, one for the spray can, one for the line tool, and one for the shape tools. Students spend three minutes at each, creating a small sample on a practice canvas.

STUDENTS Students rotate through stations, making quick sketches. At the spray can station they create foggy effects. At the line tool they draw straight roads. They jot down which tool works best for what purpose.

Explain - Modelling & Concepts (10 min)

TEACHER Demonstrate each tool in detail on the projector: pencil for thin precise lines, brush for thick painterly strokes, spray can for textured areas, line tool for perfectly straight edges. Show how changing brush size affects the stroke and how opacity creates see-through layers.

STUDENTS Students replicate each technique on their own computers, creating a reference sheet with tool name, sample stroke, and best-use scenario. They practice controlling pressure and speed.

Elaborate - Independent Application (10 min)

TEACHER Give students the outline of the school campus map. They must complete the full drawing using at least four different tools: line tool for building edges, brush for trees and bushes, spray can for ground texture, and pencil for small details like windows and doors.

STUDENTS Students work carefully on the campus map, switching tools as needed. They compare techniques with neighbors and share tips about brush sizes that work well for different elements.

Evaluate - Check & Close (5 min)

TEACHER Teacher displays three student drawings side by side on the projector. Class discusses which tools were used in each and how the techniques differ. Students self-assess their own work using a rubric. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students evaluate the displayed drawings and rate their own work on tool variety, control, and composition. They identify one skill to improve.

Cross-Curricular Connections

Mathematics: Math: Measuring and drawing to scale on the campus map connects to ratio and proportion
Science: Science: Drawing scientific diagrams accurately using the line and shape tools
Language: Language: Writing tool descriptions in their own words strengthens technical vocabulary

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-drawn tracing outlines to practice tool control on familiar shapes
On Level: Complete the campus map using specified tools with moderate independence
Advanced: Add shading, depth, and texture techniques beyond the basic requirements
SEND: Use large brush sizes and simplified steps with teacher guiding each tool selection

Formative Assessment

Method: Completed campus map drawing evaluated with tool variety and technique rubric
CT Skill Assessed: Algorithmic thinking - planning a sequence of tool selections to achieve a desired drawing outcome

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Before sleeping, close your eyes and imagine drawing your bed using only straight lines. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.05 *Shapes and Patterns*

40 minutes LAB

Success Criteria - I can…

- Draw geometric shapes accurately using rectangle, circle, and polygon tools
- Create repeating patterns by duplicating and arranging shapes
- Identify mathematical properties of shapes while drawing digitally

Resources Needed:
- Time-lapse video of digital drawing
- Practice canvas worksheets
- School campus map outline
- Tool reference sheets

Key Vocabulary

Shape: Closed form like circle square or triangle with clear edges
Pattern: Design that repeats same shapes again and again regularly
Symmetry: Both sides of design looking exactly same like mirror
Repeat: Using same element many times to create pattern quickly

Engage - Retrieval & Hook (5 min)

TEACHER Bring a kolam pattern drawn by a grandmother and display it on the projector. Ask: 'How do you think Amma creates such perfect circles and symmetry? Can a computer do the same thing?'

STUDENTS Students share stories about watching their elders draw kolam. They are curious about how the computer can replicate these traditional patterns.

Explore - Guided Discovery (10 min)

TEACHER Give students a printed sheet with a half-finished kolam pattern and ask them to complete it by hand. Then open Tux Paint and ask them to create the same pattern using the shape tools instead of freehand drawing.

STUDENTS Students first struggle with the freehand version, then find the shape tools make perfect circles and squares easily. They compare the two results and discuss which is more precise.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the shape tools: rectangle (hold Shift for perfect squares), circle (hold Shift for perfect circles), and polygon (click vertices to create sides). Show how shapes can be filled, outlined, and sized precisely. Connect to math: a square has four equal sides, a circle has infinite symmetry.

STUDENTS Students practice drawing each shape type and recording how many clicks and drags are needed. They identify properties like equal sides and right angles in their digital shapes.

Elaborate - Independent Application (10 min)

TEACHER Challenge students to create a kolam-style pattern using at least three shape types arranged symmetrically around a center point. They must use the fill tool to color their pattern with at least four different colors.

STUDENTS Students design their patterns carefully, counting and measuring to ensure symmetry. They rotate their designs to check that all four sides look identical.

Evaluate - Check & Close (5 min)

TEACHER Students print or save their kolam patterns and present them to the class. Each student must explain which shapes they used and how they ensured symmetry. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their work, pointing out geometric properties. Class votes on the most symmetrical and most creative designs.

Cross-Curricular Connections

Mathematics: Math: Shape properties, symmetry, and angles are directly explored through digital drawing
Science: Art: Kolam and rangoli traditions connect cultural art forms to geometry
Language: Language: Describing geometric properties in sentences builds mathematical communication skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-drawn shape outlines to trace and arrange into patterns
On Level: Create a pattern using three shape types with some symmetry guidance
Advanced: Design a complex mandala-style pattern with rotational symmetry and detailed color schemes
SEND: Guide shape drawing step-by-step, placing each element together with the student

Formative Assessment

Method: Kolam pattern creation with shape identification and symmetry explanation
CT Skill Assessed: Decomposition - breaking a complex pattern into individual geometric shapes and their arrangement

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the tiles on the floor or the pattern on your bedsheet. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.06 *Symmetric Designs*

40 minutes LAB

Success Criteria - I can...

- Define bilateral and rotational symmetry in digital art context
- Use symmetry tools or manual techniques to create mirror-image designs
- Apply symmetry to create a butterfly, flower, or traditional motif

Resources Needed:
- Printed kolam pattern sheets
- Tux Paint software
- Color wheel reference
- Shape properties handout

Key Vocabulary

Symmetry: Balance where one half mirrors the other half perfectly
Mirror: Copying design exactly to opposite side of central line
Axis: Central line where folding gives two identical halves evenly
Mandala: Circular symmetric design made from repeated shapes beautifully

Engage - Retrieval & Hook (5 min)

TEACHER Fold a piece of paper in half, paint one side of a butterfly, and unfold it to reveal perfect symmetry. Ask: 'Can the computer do this automatically? What if you only had to draw half?'

STUDENTS Students marvel at the unfolded butterfly. They immediately want to try the technique and are excited to learn the computer can do it even better.

Explore - Guided Discovery (10 min)

TEACHER Give students a square piece of paper and ask them to fold it in half, then quarters, then cut a shape along the folded edge. Unfold to see the symmetric design. Then ask them to recreate the same design on the computer by drawing only half and copying-flipping.

STUDENTS Students create paper snowflakes first, then move to the computer. They discover that copying, flipping horizontally, and aligning the halves creates perfect symmetry faster than folding paper.

Explain - Modelling & Concepts (10 min)

TEACHER Define bilateral symmetry as mirror symmetry along one axis (like a human face). Define rotational symmetry as a design that looks the same after partial rotation (like a pinwheel). Show how to create symmetry on the computer: draw half, select, copy, paste, flip horizontal, and align. For rotational symmetry, copy the element and rotate by 90 or 60 degrees.

STUDENTS Students take notes on symmetry types and follow step-by-step instructions to create a symmetric butterfly. They label the line of symmetry on their screen using the text tool.

Elaborate - Independent Application (10 min)

TEACHER Students must create a symmetric flower with at least six petals arranged using rotational symmetry. They must use the rotate tool precisely and ensure all petals are evenly spaced around the center.

STUDENTS Students carefully rotate and position each petal, checking that the spacing is equal. They use a protractor on screen or estimate angles to achieve six-fold rotational symmetry.

Evaluate - Check & Close (5 min)

TEACHER Students submit their symmetric designs. Teacher checks each design for accurate symmetry by folding a printed copy or measuring distances from the center line digitally. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students observe their printed designs being tested for symmetry. They identify any imperfections and explain what caused them.

Cross-Curricular Connections

Mathematics: Math: Symmetry connects to geometry, angles, and rotational transformations
Science: Biology: Many living things show symmetry - butterflies, leaves, starfish - connecting to science
Language: Language: Writing a descriptive paragraph about their symmetric design practices English composition

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-drawn half-shapes to copy, flip, and complete into symmetric designs
On Level: Create a bilateral symmetric design following step-by-step instructions
Advanced: Design a complex rotational symmetry pattern with multiple nested elements
SEND: Guide each step of the copy-flip-align process with hands-on support

Formative Assessment

Method: Symmetric flower design checked for rotational symmetry accuracy with angle measurements
CT Skill Assessed: Algorithmic thinking - following a precise sequence of select-copy-flip-align steps to achieve symmetry

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at your own face in a mirror. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.07 *Image Editing Basics*

40 minutes LAB

Success Criteria - I can...

- Open, view, and perform basic edits on a digital photograph
- Use crop, resize, and brightness tools to improve an image
- Save edited images in different formats and understand format differences

Resources Needed:
- Paper and paint for butterfly activity
- Tux Paint with copy-paste-flip tools
- Protractor handouts
- Symmetry examples from nature

Key Vocabulary

Crop: Cutting unwanted parts to focus on important area only
Resize: Making picture bigger or smaller without losing clarity
Rotate: Turning image to change its direction or angle slightly
Filter: Effect that changes colours or mood of photo instantly

Engage - Retrieval & Hook (5 min)

TEACHER Show a dark, blurry photograph of the school playground. Ask: 'This photo is not very good. Can we fix it without taking the photo again?' Then reveal the same photo after editing - bright, cropped, and clear.

STUDENTS Students are amazed at the transformation. They immediately want to learn how to fix photos and ask if they can edit photos from their phones too.

Explore - Guided Discovery (10 min)

TEACHER Give students a USB drive with five pre-loaded photos of the school. Ask them to open each in an image viewer and describe what is wrong with each one - too dark, too zoomed out, crooked, too small, or has unwanted objects at the edges.

STUDENTS Students examine each photo and write descriptions of the problems. They predict what editing tools could fix each issue and share their predictions with a partner.

Explain - Modelling & Concepts (10 min)

TEACHER Demonstrate three essential editing tools: crop to remove unwanted edges and focus on the subject, resize to change image dimensions for different uses, and brightness/contrast to fix dark or washed-out photos. Show the file menu options for Save As and explain JPEG for photos and PNG for graphics with transparency.

STUDENTS Students follow along, editing a sample photo step by step. They record the steps in their journals: open, crop, resize, adjust brightness, save as JPEG.

Elaborate - Independent Application (10 min)

TEACHER Give students a poorly composed photo of the school garden. They must crop it to focus on the best flower, resize it to fit a newsletter column width, adjust brightness so details are visible, and save the final version as both JPEG and PNG to compare file sizes.

STUDENTS Students work through the editing steps systematically. They compare the JPEG and PNG file sizes and write a sentence explaining which format is better for photos and why.

Evaluate - Check & Close (5 min)

TEACHER Students display their before and after images side by side on the screen. Class discusses what changes were made and whether the edits improved the photo effectively. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their editing work, explaining each tool they used. Class provides feedback on whether the edits were appropriate and effective.

Cross-Curricular Connections

Mathematics: Math: Understanding image dimensions in pixels connects to area and measurement
Science: Science: Brightness and contrast relate to how light and sensors work in cameras
Language: Language: Writing a caption for the edited photo practices descriptive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide step-by-step picture cards showing each editing action with screenshots
On Level: Complete the editing task with written instructions and some teacher checkpoints
Advanced: Experiment with advanced edits like sharpening, color balance, and removing blemishes
SEND: One-on-one guided editing with teacher controlling mouse while student directs changes

Formative Assessment

Method: Before-after image comparison with written explanation of each editing tool used
CT Skill Assessed: Abstraction - deciding which parts of an image are essential and which can be removed through cropping

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could take a photo of one thing in the dormitory and make it look perfect, what would it be and how would you. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.08 *Layers Introduction*

40 minutes **LAB**

Success Criteria - I can...

- Explain what layers are and why they are useful in digital art
- Create, name, reorder, and toggle visibility of layers in a drawing application
- Build a simple scene using at least three separate layers for background, middle ground, and foreground

Resources Needed:
- USB drives with sample school photos

- Image editing software
- Before-after comparison template
- File format reference sheet

Key Vocabulary

Layer: Transparent sheet stacked to build complex picture step by step
Background: Bottom layer that holds main colour or photo behind everything
Foreground: Top layer that appears in front of all other layers
Opacity: How see-through or solid a layer appears on screen

Engage - Retrieval & Hook (5 min)

TEACHER Bring a stack of three transparent sheets - one with a sky painted, one with mountains, and one with a tree. Stack them on the projector to build a complete scene. Remove the middle sheet and ask: 'What happened? Why can I take away just the mountains?'

STUDENTS Students see that each sheet is independent. They understand that layers work the same way - you can change one part without affecting the others.

Explore - Guided Discovery (10 min)

TEACHER Give students three sheets of acetate and markers. Ask them to draw a background on one, buildings on the second, and people on the third. Stack them to create a scene. Then open Tux Paint or GIMP and find the layer panel to create the same effect digitally.

STUDENTS Students create physical layered scenes first, then discover the digital layer panel. They are excited to see they can hide and show each layer independently.

Explain - Modelling & Concepts (10 min)

TEACHER Define layers as transparent sheets stacked on top of each other. The bottom layer is the background, middle layers hold main elements, and the top layer has details. Show how to create a new layer, name it, draw on it, toggle visibility with the eye icon, and reorder by dragging. Explain that drawing on one layer does not affect others.

STUDENTS Students follow along creating three layers named Background, Middle, and Foreground. They draw a simple element on each layer and practice hiding and showing them.

Elaborate - Independent Application (10 min)

TEACHER Challenge students to create a layered landscape with five layers: sky background, mountains, trees, a river in the foreground, and animals on the very top. They must name each layer and be able to hide any single layer without affecting the others.

STUDENTS Students plan their scene on paper first, deciding which elements go on which layer. Then they build it digitally, carefully drawing on the correct layer and checking visibility toggles.

Evaluate - Check & Close (5 min)

TEACHER Students demonstrate their layered scene by toggling each layer on and off while a partner watches. The partner must identify what is on each layer and explain why layering is useful. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students take turns demonstrating and observing. Teacher checks that layers are properly named and that students can explain the purpose of each layer.

Cross-Curricular Connections

Mathematics: Math: Understanding stacking order connects to ordering and sequencing concepts
Science: Science: Layers in geology (soil layers, atmospheric layers) use the same concept of stacking
Language: Language: Writing a story about what happens when layers are removed practices cause-and-effect writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a pre-made layered file with layers already created for the student to explore and modify
On Level: Create a three-layer scene with clear instructions for each layer's content
Advanced: Create a five-layer scene with named layers and write a reflection on why layers help artists
SEND: Guide layer creation step-by-step, helping the student name and draw on each layer

Formative Assessment

Method: Layered scene demonstration with partner identification of layer contents
CT Skill Assessed: Decomposition - breaking a complex scene into independent, manageable layer components

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of your dormitory as layers - the floor is the bottom layer, beds are the middle layer, and you are the top layer. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.09 *Filters and Effects*

40 minutes LAB

Success Criteria - I can...

- Identify common image filters such as blur, sharpen, grayscale, and invert
- Apply at least three different filters to a photograph and observe the results
- Choose appropriate filters for specific artistic or practical purposes

Resources Needed:
- Transparent acetate sheets
- Markers
- Layer panel reference guide
- GIMP or Tux Paint with layer support

Key Vocabulary

Filter: Ready-made effect that changes photo look with one click
Blur: Softening image so details look less sharp and smooth
Brightness: How light or dark the whole picture appears to eyes
Effect: Special style applied to make picture more attractive quickly

Engage - Retrieval & Hook (5 min)

TEACHER Show the same photograph of the school gate with five different filters applied: grayscale, sepia, blur, sharpen, and invert. Number each version and ask students to match each filter name to the correct image.

STUDENTS Students match filters by discussing the visual characteristics of each version. They notice that grayscale makes it black and white, sepia makes it look old, and blur makes details fuzzy.

Explore - Guided Discovery (10 min)

TEACHER Give students a photograph of the school playground. Ask them to apply every filter they can find in the software and record what each one does. They should keep a filter journal with the filter name and a description of the change.

STUDENTS Students click through filters rapidly, laughing at the invert effect and admiring the sepia tone. They carefully write descriptions like 'blur makes everything fuzzy like looking through fog'.

Explain - Modelling & Concepts (10 min)

TEACHER Categorize filters into three groups: color filters (grayscale, sepia, invert, color balance) that change colors, quality filters (blur, sharpen, noise reduction) that change clarity, and artistic filters (oil painting, watercolor, emboss) that change the overall style. Explain when each type is useful - grayscale for printing on school newspapers, sharpen for fixing blurry photos.

STUDENTS Students categorize the filters they explored and add new ones from the demonstration. They create a reference card with filter categories and common uses.

Elaborate - Independent Application (10 min)

TEACHER Give students a portrait photo of the school headmaster. They must create four versions: one with sharpen for the school website, one with grayscale for the newspaper, one with sepia for a heritage display, and one artistic filter for a creative competition entry. Each must be saved with a descriptive filename.

STUDENTS Students carefully apply each filter and think about which version suits each purpose. They discuss with partners why different contexts need different visual styles.

Evaluate - Check & Close (5 min)

TEACHER Students display all four filtered versions and explain why they chose each filter for its specific purpose. Class discusses whether the choices were appropriate. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their filtered images with justifications. Teacher and peers ask questions about filter choices and suggest improvements.

Cross-Curricular Connections

Mathematics: Physics: Filters relate to how light wavelengths are blocked or passed through colored glass
Science: History: Sepia filters connect to understanding historical photographs and aging processes
Language: Language: Writing a persuasive paragraph about why a certain filter suits a purpose builds argumentative writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a matching worksheet with filter names and example outputs to connect
On Level: Apply three specified filters to a photo and describe each result in writing
Advanced: Experiment with filter combinations and write a creative story inspired by the filtered images
SEND: Select and apply filters together with the student, naming each one as it is used

Formative Assessment

Method: Four-version filtered photo set with written justifications for each filter choice
CT Skill Assessed: Evaluation - assessing which filter best serves a specific purpose and justifying the decision

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could make the dormitory look like it was from 100 years ago, what filter would you use? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.10 *Creating Digital Posters* 40 minutes **LAB**

Success Criteria - I can...

- Plan a poster layout including title, images, text, and color scheme
- Use a combination of drawing tools, text, and imported images to create a poster
- Apply design principles such as alignment, contrast, and hierarchy to a poster

Resources Needed:
- Photographs of school people and places
- Image editing software with filter options
- Filter reference cards
- Printed comparison sheets

Key Vocabulary

Poster: Large design combining text and images to share message
Layout: Arrangement of text and pictures to look balanced neatly
Heading: Big bold words at top that catch attention quickly
Alignment: Lining up text and images neatly along invisible lines

Engage - Retrieval & Hook (5 min)

TEACHER Display five posters from school events - some well-designed and some cluttered and hard to read. Ask students to vote for the best one and explain why. Write their criteria on the board: big text, clear images, not too crowded.

STUDENTS Students enthusiastically critique the posters. They identify what makes some posters effective and others confusing, building their design intuition before creating their own.

Explore - Guided Discovery (10 min)

TEACHER Give students three printed posters and a set of colored sticky notes. Ask them to rearrange the sticky notes on each poster to label the title, image, text, and background areas. Then discuss how the arrangement affects readability.

STUDENTS Students place sticky notes on posters, discovering that the title is usually at the top, images are central, and text is at the bottom. They discuss why this layout works.

Explain - Modelling & Concepts (10 min)

TEACHER Teach three design principles: hierarchy means the most important element (title) should be largest and most prominent, alignment means elements should line up neatly, and contrast means using light text on dark backgrounds or vice versa for readability. Show how to plan a poster on graph paper before creating it digitally.

STUDENTS Students sketch their poster plan on graph paper, marking where the title, image, text, and decorations will go. They apply hierarchy by making the title the largest element.

Elaborate - Independent Application (10 min)

TEACHER Students must create a digital poster for the school science fair with: a bold title, at least two images, a date and time, and three bullet points about the event. The poster must use at least three colors and follow the layout they planned on graph paper.

STUDENTS Students translate their paper plans to the digital canvas, carefully placing each element. They adjust colors and sizes to make the poster visually balanced and easy to read from a distance.

Evaluate - Check & Close (5 min)

TEACHER Print the posters and hang them on the classroom wall. Students walk around with sticky notes and place a star on the poster they think is most effective and a comment note explaining why. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students evaluate peer posters constructively, writing specific praise and suggestions. Teacher facilitates a discussion about common strengths and areas for improvement.

Cross-Curricular Connections

Mathematics: Math: Measuring poster dimensions and scaling images connects to ratio and proportion
Science: Social Science: Creating awareness posters about local issues connects to civic responsibility
Language: Language: Writing concise poster text practices persuasive and informative writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a poster template with pre-placed title and image boxes to fill in
On Level: Create a poster following the graph paper plan with moderate independence
Advanced: Design a poster from scratch with advanced layout, custom color scheme, and detailed text
SEND: Guide the student through each placement decision, helping with alignment and sizing

Formative Assessment

Method: Peer evaluation with star and comment notes, plus teacher rubric on design principles
CT Skill Assessed: Abstraction - simplifying event information into essential visual elements for effective communication

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of a poster for a dormitory cricket match. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.11 *Digital Art Projects*

40 minutes **LAB**

Success Criteria - I can…

- Plan and execute a complete digital art project from concept to final output
- Integrate multiple skills learned in previous lessons into a cohesive artwork
- Reflect on the creative process and evaluate their own artistic growth

Resources Needed:

- Sample event posters
- Graph paper for planning
- Colored sticky notes
- Computers with design software
- Printed poster samples

Key Vocabulary

Graphics: Pictures or images created and edited on a computer
Canvas: Digital workspace where you draw or paint pictures freely
Tool: Option like brush or eraser that helps you create art
Palette: Set of colours you choose from to paint digitally

Engage - Retrieval & Hook (5 min)

TEACHER Show a portfolio of digital art created by students from other residential schools - landscapes, portraits, abstract designs, and festival posters. Ask: 'Which one inspires you the most? What skills do you think the artist used to make it?'

STUDENTS Students examine the portfolio with admiration. They identify elements like symmetry, color harmony, layers, and text that they have learned about, connecting their skills to real artwork.

Explore - Guided Discovery (10 min)

TEACHER Give students a project planning sheet with sections for: theme, tools needed, layers plan, color palette, and steps. Ask them to brainstorm three possible project ideas and choose one. Examples: a festival celebration poster, a school map, or a nature scene.

STUDENTS Students brainstorm ideas enthusiastically. Some want to draw Diwali celebrations, others want to create a map of the school, and a few want to draw their village. They fill in the planning sheet with their chosen theme.

Explain - Modelling & Concepts (10 min)

TEACHER Review the project workflow: plan on paper, set up the canvas and layers, build the background first, add middle elements, add foreground details, apply color and filters, add text if needed, and save the final version. Discuss time management for the 40-minute period - spend 5 minutes planning, 25 minutes creating, and 10 minutes reviewing.

STUDENTS Students review their planning sheets and adjust them based on the workflow. They set up their digital canvas with named layers before starting to draw.

Elaborate - Independent Application (10 min)

TEACHER Students spend the full period working on their chosen project. Teacher circulates providing technical help and creative suggestions. Students must use at least three tools, two layers, and a thoughtful color palette.

STUDENTS Students work with focused energy on their projects. Some draw detailed village scenes, others create festival posters, and a few design abstract patterns. They help each other with tool questions.

Evaluate - Check & Close (5 min)

TEACHER At the end of the period, students save their work and write a reflection: what they created, which skills they used, what was challenging, and what they would do differently next time. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students write honest reflections about their creative process. They share one thing they are proud of and one thing they want to improve with a partner.

Cross-Curricular Connections

Mathematics: Art: The project integrates traditional art concepts with digital tools
Science: Social Science: Themes like festivals, village life, and school events connect to community studies
Language: Language: Writing the project reflection practices self-assessment and metacognitive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a structured template with pre-made background layers and guided next steps
On Level: Complete the project using the planning sheet with regular teacher checkpoints
Advanced: Create an original project that extends beyond the assignment with self-directed learning
SEND: Work alongside the student, making creative decisions together and providing constant support

Formative Assessment

Method: Self-reflection writing plus project evaluation using a rubric covering tools, layers, colors, and effort
CT Skill Assessed: Evaluation - assessing their own work against goals and identifying areas for improvement

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your favorite festival. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.12 *Graphics in Real Life*

40 minutes LAB

Success Criteria - I can...

- Identify real-world applications of computer graphics in their daily lives
- Explain how graphics are used in education, communication, and entertainment
- Create a graphic that solves a real problem in their school or community

Resources Needed:

- Student art portfolio samples
- Project planning sheets
- Computers with Tux Paint or GIMP
- Assessment rubrics
- Save folders for each student

Key Vocabulary

Graphics: Pictures or images created and edited on a computer
Canvas: Digital workspace where you draw or paint pictures freely
Tool: Option like brush or eraser that helps you create art
Palette: Set of colours you choose from to paint digitally

Engage - Retrieval & Hook (5 min)

TEACHER Bring a collection of items with graphics: a bus route map, a medicine packet with pictograms, a textbook cover, a 手机 wallpaper, and a school notice board poster. Ask: 'How many of these used a computer to be designed? How would life be different without graphics?'

STUDENTS Students examine each item and realize that nearly everything they see daily was designed using computer graphics. They discuss how maps help the bus driver and pictograms help people who cannot read.

Explore - Guided Discovery (10 min)

TEACHER Take students on a graphics walk around the school - photograph every sign, poster, logo, chart, and diagram they find. Back in the lab, organize the photos into categories: signs for directions, posters for information, logos for identity, charts for data.

STUDENTS Students photograph items eagerly, discovering graphics everywhere - the lunch menu chart, the class timetable, the attendance board, and the school logo on the gate. They categorize their photos on the computer.

Explain - Modelling & Concepts (10 min)

TEACHER Discuss three major real-world applications: education uses graphics in textbooks, presentations, and educational videos; communication uses graphics in social media posts, news graphics, and advertisements; entertainment uses graphics in movies, games, and cartoons. Show examples from each category and connect to how graphics make information easier to understand.

STUDENTS Students add examples from their own lives to each category. They discuss how their parents use graphics - a farmer reading a weather chart, a mother looking at a medicine label.

Elaborate - Independent Application (10 min)

TEACHER Students identify a real problem in their school or village that could be solved with a graphic. Examples: a better sign for the library, a map for new students, a poster about water conservation. They plan and begin creating this graphic using all skills learned in the chapter.

STUDENTS Students brainstorm real problems and choose the most impactful one. They sketch their solution on paper and begin the digital version, applying design principles, appropriate tools, and thoughtful colors.

Evaluate - Check & Close (5 min)

TEACHER Students present their problem-solving graphic to the class, explaining the problem, their design choices, and how the graphic would help in real life. Class votes on the most practical and impactful design. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present with confidence, connecting their design to real needs. They receive feedback on practicality and visual effectiveness from peers and teacher.

Cross-Curricular Connections

Mathematics: Social Science: Identifying community needs and designing solutions connects to civic engagement
Science: Math: Measuring sign dimensions and scaling for different sizes connects to practical measurement
Language: Language: Writing a proposal explaining the problem and solution practices persuasive and technical writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a list of school problems to choose from with picture examples of each solution
On Level: Select a problem and create a graphic solution with written explanation of design choices
Advanced: Research how professional graphic designers solve similar community problems and apply those

techniques
SEND: Choose a problem together with the student and co-create the graphic solution step by step

Formative Assessment

Method: Problem-solution graphic presentation with peer voting on practicality and impact
CT Skill Assessed: Application - using all chapter skills to solve a real-world problem through graphic design

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Before sleeping, think of one sign or poster that would make your dormitory life easier. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 3 - Audio & Visual Communication

Lesson 3.01 *Types of Communication* 40 minutes LAB

Success Criteria - I can...

- Classify communication into verbal, non-verbal, and digital categories
- Give examples of each communication type from daily school life
- Explain how digital communication differs from traditional methods

Resources Needed:
- Camera or phone for graphics walk
- Photo organization template
- Problem identification worksheet
- Design planning sheets
- Printing facilities

Key Vocabulary

Communication: Exchanging information between people using chosen medium effectively
Medium: Channel like phone or video used to send message
Broadcast: Sending same message to many people at once together
Feedback: Response showing whether receiver understood your message clearly

Engage - Retrieval & Hook (5 min)

TEACHER Play a game of Chinese Whispers with the class. Whisper a message from one student to the next. When the last student says it aloud, the message is usually wrong. Ask: Why did the message change? How can we make sure messages arrive correctly?

STUDENTS Students laugh at the distorted message and suggest solutions like writing it down or speaking more clearly. This naturally leads to discussing different communication methods.

Explore - Guided Discovery (10 min)

TEACHER Give groups a stack of cards with communication examples: talking face to face, sending a letter, making a phone call, using sign language, posting on social media, reading a newspaper. Ask them to sort the cards into categories they create themselves.

STUDENTS Groups discuss and create categories like with mouth, with hands, with machines, and with paper. Teacher guides them toward more formal categories: verbal, non-verbal, and digital.

Explain - Modelling & Concepts (10 min)

TEACHER Define verbal communication as using spoken words (face to face, phone, video call). Define non-verbal communication as using body language, gestures, facial expressions, and written symbols (letters, signs). Define digital communication as using electronic devices and internet (email, messaging, social media). Give examples from residential school life: morning assembly announcements (verbal), the school bell pattern (non-verbal), and the headmaster sending a message to all teachers (digital).

STUDENTS Students create a three-column chart in their notebooks and fill in examples from their own experience for each communication type. They discuss which type they use most at school.

Elaborate - Independent Application (10 min)

TEACHER Challenge groups to create a communication timeline: how did people in their village communicate 100 years ago, 50 years ago, 20 years ago, and today? They must include at least two examples for each time period and explain what technology made the change possible.

STUDENTS Groups research using their own family stories and general knowledge. They discover that grandparents sent letters, parents used landline phones, and they use smartphones and messaging apps.

Evaluate - Check & Close (5 min)

TEACHER Each group presents their timeline. Class identifies the pattern: communication has become faster, reaches farther, and uses more technology over time. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups present their timelines with explanations. Teacher asks probing questions about how speed of communication affects daily life.

Cross-Curricular Connections

Mathematics: History: Communication timeline connects to understanding historical changes in society
Science: Social Science: Different communication methods reflect cultural and economic development
Language: Language: Writing the timeline requires organizing information chronologically with clear sentences

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-sorted communication cards with pictures for matching to category labels
On Level: Sort communication examples into given categories and add two own examples
Advanced: Research and compare communication methods across different countries and cultures
SEND: Sort picture cards together with teacher support and discuss each one

Formative Assessment

Method: Three-column chart completion and group timeline presentation
CT Skill Assessed: Classification: categorizing different communication methods into logical groups

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about how you talk to your family at home versus how you talk to friends at school. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.02 *Audio Recording Basics*

40 minutes LAB

Success Criteria - I can...

- Explain how sound is captured and stored as digital audio
- Use a microphone to record a clear audio clip of at least 10 seconds
- Identify factors that affect recording quality such as background noise and distance

Resources Needed:
- Chinese Whispers game setup
- Communication example cards
- Chart paper for timelines
- Markers and colored pens

Key Vocabulary

Audio: Sound or music recorded and played through computer speakers
Record: Saving your voice or sound as a digital file
Playback: Listening again to recorded sound through speakers clearly
Microphone: Device that captures your voice and sends to computer

Engage - Retrieval & Hook (5 min)

TEACHER Record a student reading a poem while standing close to the microphone, then again while standing far away. Play both recordings and ask: Which sounds better? Why? Let students hear the difference immediately.

STUDENTS Students immediately notice the quality difference. They suggest reasons like the first one is louder and the second one has more echo, demonstrating their intuitive understanding of audio quality.

Explore - Guided Discovery (10 min)

TEACHER Give each group a school smartphone and ask them to record five short audio clips: whispering, speaking normally, shouting, clapping, and singing. They listen to each playback and rank them from clearest to noisiest.

STUDENTS Groups take turns recording and listening to playback. They notice that normal speech sounds clearest and shouting causes distortion. They discuss why the placement of the phone matters.

Explain - Modelling & Concepts (10 min)

TEACHER Explain that a microphone converts sound waves into electrical signals, which are then converted into digital data (numbers) that the computer stores. Key concepts: sample rate determines quality (higher is better), bit depth determines detail, and file formats like WAV (large, high quality) and MP3 (compressed, smaller) serve different purposes. Tips for good recording: speak clearly, stay 15 centimeters from the microphone, and reduce background noise.

STUDENTS Students take notes on the recording process. They practice speaking into a microphone at different distances and note which distance produces the clearest sound.

Elaborate - Independent Application (10 min)

TEACHER Students must record a 30-second audio message introducing themselves and their school to a student from another residential school. The recording must be clear, at a consistent volume, and free from background noise.

STUDENTS Students prepare their scripts, practice speaking, and take turns recording in a quiet corner of the lab. They listen to playback and re-record if needed, applying the tips they learned.

Evaluate - Check & Close (5 min)

TEACHER Students play their recordings for a partner. The partner rates the recording on three criteria: clarity, volume consistency, and background noise. They suggest one improvement for each recording. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students listen carefully to their partner recording and provide constructive feedback using the rating sheet. They re-record one final time incorporating the feedback.

Cross-Curricular Connections

Mathematics: Physics: Sound waves, frequency, and amplitude connect to the physics of sound
Science: Music: Understanding audio quality helps in recording music and understanding instruments
Language: Language: Speaking clearly and at a measured pace for recording practices public speaking skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Record single words or short phrases with teacher guiding microphone technique
On Level: Record a prepared introduction script with moderate independence
Advanced: Record and edit a multi-part audio message with transitions and background awareness
SEND: Record one sentence at a time with teacher helping with microphone distance and playback

Formative Assessment

Method: Partner rating of recording quality on clarity, volume, and noise criteria
CT Skill Assessed: Observation: identifying factors that affect audio quality through experimentation and comparison

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Whisper a story to your dormitory friend. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.03 *Audacity Introduction* 40 minutes LAB

Success Criteria - I can...

- Navigate the Audacity interface and identify the record, play, stop, and edit tools
- Record, play back, and trim an audio clip in Audacity
- Save an Audacity project and export the final audio as an MP3 file

Resources Needed:
- School smartphones or USB microphones
- Recording software like Audacity or Voice Recorder
- Quiet recording space
- Playback headphones
- Rating sheets

Key Vocabulary

Audacity: Free program for recording and editing sound files easily
Track: Single line holding one sound clip inside the project
Cut: Removing unwanted part of audio to make it shorter
Export: Saving edited sound as playable file like MP3

Engage - Retrieval & Hook (5 min)

TEACHER Play a short audio clip of a folk song that has a long silence at the beginning and an unwanted cough at the end. Ask: How can we remove the silence and the cough without cutting the song? Reveal that Audacity can do exactly this.

STUDENTS Students are intrigued by the idea of editing audio. They guess that some kind of scissors tool exists and are eager to find it.

Explore - Guided Discovery (10 min)

TEACHER Open Audacity on each computer and give students a guided exploration sheet with five tasks: find the record button, find the play button, find the zoom tool, find the selection tool, and find the export option. They have eight minutes to complete all five.

STUDENTS Students click through the Audacity interface, discovering buttons and menus. They call out to friends when they find something and help each other complete the exploration sheet.

Explain - Modelling & Concepts (10 min)

TEACHER Walk through Audacity interface: the transport toolbar (record, play, pause, stop, skip), the edit toolbar (selection, zoom, envelope, draw), the tracks panel, and the menu bar. Demonstrate recording a 10-second clip, playing it back, selecting the unwanted silence at the beginning, and cutting it with Edit and Remove Special. Show how to export as MP3 through File, Export, Export as MP3.

STUDENTS Students follow along step by step, replicating each action on their own computers. They take notes on the location and function of each important tool.

Elaborate - Independent Application (10 min)

TEACHER Give students a pre-recorded audio file that is 20 seconds long but has 5 seconds of silence at the start and 3 seconds of silence at the end. They must trim the silence, then record a 5-second spoken introduction and paste it at the beginning to create a complete 20-second intro clip.

STUDENTS Students open the provided file, use the selection tool to highlight and delete the silence, then record their own voice and paste it at the beginning. They listen to the final result.

Evaluate - Check & Close (5 min)

TEACHER Students export their completed intro clip as MP3 and play it for the class. Teacher and peers evaluate: is the silence removed, is the intro clear, and is the total length approximately 20 seconds? Hinge Q: [question with

3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their clips. Class provides feedback on quality and accuracy. Teacher checks that all students have successfully exported their files.

Cross-Curricular Connections

Mathematics: Music: Audacity skills connect to recording and editing music for school events
Science: English: Recording spoken introductions practices pronunciation and fluency
Language: Math: Understanding audio duration in seconds and minutes connects to time measurement

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a step-by-step screenshot guide for recording and trimming
On Level: Complete the intro clip task with written instructions and periodic teacher checks
Advanced: Add fade-in and fade-out effects to create a polished professional-sounding intro
SEND: Guide each click and action, helping the student navigate the interface

Formative Assessment

Method: Exported MP3 clip evaluated for trimmed silence, added intro, and correct duration
CT Skill Assessed: Algorithmic thinking: following a sequence of steps to transform raw audio into a finished product

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Hum your favorite song for 10 seconds. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.04 *Editing Audio* 40 minutes LAB

Success Criteria - I can…

- Perform cut, copy, paste, and delete operations on audio tracks
- Apply volume adjustments and fade effects to improve audio quality
- Combine two separate audio clips into one continuous track

Resources Needed:
- Computers with Audacity installed

- Pre-recorded audio clips on shared drive
- Audacity interface reference sheet
- Headphones for playback
- MP3 export guide

Key Vocabulary

Trim: Cutting extra silence from start or end of clip
Volume: How loud or soft the sound plays to listeners
Noise Reduction: Removing background hiss to make voice clearer instantly
Fade: Gradually increasing or decreasing sound at beginning or end

Engage - Retrieval & Hook (5 min)

TEACHER Play two recordings: one of a student speaking at varying volumes (too loud then too quiet) and one with an awkward pause in the middle. Ask: Can we fix these problems without recording again? Show the fixed versions after demonstrating the edits.

STUDENTS Students hear the problems clearly and are relieved to learn that editing can fix recording mistakes without re-recording everything.

Explore - Guided Discovery (10 min)

TEACHER Give students two short audio clips on their computers: a greeting and a farewell. Ask them to figure out how to swap the order so the farewell comes first and the greeting comes second. They must use only the tools in Audacity.

STUDENTS Students experiment with selection, cut, and paste tools. Some figure it out quickly and help others. They discover that selecting a section and dragging it is one method, while cut-and-paste is another.

Explain - Modelling & Concepts (10 min)

TEACHER Demonstrate four key editing operations: cut removes selected audio and copies it to clipboard, copy duplicates selected audio without removing it, paste inserts clipboard content at the cursor, and delete removes selected audio permanently. Then show volume adjustment using the Envelope Tool to make quiet parts louder and loud parts quieter. Finally, demonstrate the Fade In and Fade Out effects from the Effect menu.

STUDENTS Students practice each operation on a sample audio file, creating a reference recording of each technique. They record what each operation does in their journals.

Elaborate - Independent Application (10 min)

TEACHER Students receive a 30-second audio clip with three problems: the first 5 seconds are too quiet, there is an unwanted pause in the middle, and the last 5 seconds are too loud. They must fix all three issues using volume adjustment, cut, and fade out.

STUDENTS Students identify each problem area by listening carefully, then apply the appropriate editing technique. They play back their edited version and compare it to the original.

Evaluate - Check & Close (5 min)

TEACHER Students play their original and edited versions back to back for a partner. The partner rates whether each problem was fixed and suggests any remaining improvements. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present before and after versions. Partner provides detailed feedback on editing effectiveness.

Cross-Curricular Connections

Mathematics: Music: Editing audio skills apply to trimming songs for school performances
Science: English: Editing spoken recordings helps improve pronunciation practice materials
Language: Math: Measuring audio duration in seconds for precise editing connects to time calculations

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a guided worksheet with arrow keys showing exactly where to click for each operation
On Level: Complete the three-problem editing task with written step-by-step instructions
Advanced: Create a multi-track audio mix combining voice recording with background music
SEND: Guide each edit step by step, helping select the correct audio region and apply the operation

Formative Assessment

Method: Before and after audio comparison with partner evaluation of editing effectiveness
CT Skill Assessed: Decomposition: breaking an audio file into sections that can be independently modified and recombined

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of a conversation you had today. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.05 *Video Basics*

40 minutes LAB

Success Criteria - I can...

- Identify the main components of a video file: frames, resolution, and duration
- Use a simple video player to play, pause, and navigate through a video clip
- Record a short video clip using a smartphone or webcam with steady framing

Resources Needed:
- Audacity software
- Sample audio clips with known issues
- Before and after comparison template
- Editing operations quick reference card

Key Vocabulary

Video: Moving pictures made from many still images played quickly
Frame: Single still picture that is part of a longer video
Timeline: Bar showing order and length of clips in your project
Clip: Short piece of video you can move or trim separately

Engage - Retrieval & Hook (5 min)

TEACHER Show a short stop-motion animation made by flipping through hand-drawn pages rapidly. Ask: How many drawings do you think are in this 5-second animation? Then reveal that video is just many still images shown quickly, one after another.

STUDENTS Students are amazed to learn that video is a series of still images. They try flipping through their own notebook pages to create a simple animation effect.

Explore - Guided Discovery (10 min)

TEACHER Give students a video file on their computers and a worksheet. They must answer questions while watching: How long is the video? What is the main subject? Can you find the exact moment when something specific happens? What happens at the 10-second mark?

STUDENTS Students watch the video multiple times, pausing and rewinding to find specific moments. They realize that finding an exact moment in a video requires patience and careful scrubbing.

Explain - Modelling & Concepts (10 min)

TEACHER Explain that video is made of frames, which are individual images displayed in rapid succession (typically 24 to 30 frames per second). Resolution refers to the size of each frame in pixels (like 1920x1080 for full HD). Duration is the total length of time the video plays. Show these concepts using the video player information panel.

STUDENTS Students take notes on frames, resolution, and duration. They check the properties of the video file they watched and record its resolution, frame rate, and duration.

Elaborate - Independent Application (10 min)

TEACHER Students must use a smartphone to record a 15-second video of a classmate performing a simple task like solving a math problem on the board or demonstrating a science experiment step. The video must be steady, well-framed with the subject in the center, and have minimal background noise.

STUDENTS Students practice holding the phone steady and framing their subject before recording. They watch their recording and re-take if the framing is off or the video is shaky.

Evaluate - Check & Close (5 min)

TEACHER Students show their videos to a partner. The partner rates each video on steadiness, framing, and clarity. They suggest one improvement for each video. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students take turns watching and evaluating. Teacher circulates and provides feedback on camera technique.

Cross-Curricular Connections

Mathematics: Physics: Frames per second connects to understanding frequency and perception of motion
Science: Art: Framing and composition in video relate to visual art principles
Language: Language: Describing video content in written sentences practices descriptive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a video with pre-made timestamps and guided questions to answer
On Level: Record a 15-second video following a checklist of criteria
Advanced: Create a 30-second video with multiple shots and simple transitions using a video editor
SEND: Record a 10-second clip with teacher helping hold the phone and frame the subject

Formative Assessment

Method: Peer evaluation of recorded video on steadiness, framing, and clarity
CT Skill Assessed: Observation: analyzing video properties and recording technique through systematic evaluation

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Try to draw 10 frames of a bouncing ball on separate pages. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.06 *Video Conferencing* 40 minutes LAB

Success Criteria - I can...

- Explain how video conferencing works and its benefits for remote communication
- Navigate a video conferencing interface including mute, camera toggle, and chat features
- Practice proper video conferencing etiquette for educational settings

Resources Needed:

- Smartphones or webcams
- Sample video files on shared drive
- Video player software
- Recording space with good lighting
- Peer evaluation sheets

Key Vocabulary

Video Call: Talking face-to-face through camera and internet at the same time
Mute: Turning off microphone so others cannot hear you temporarily
Share Screen: Showing your computer screen to others during a call
Bandwidth: Amount of data internet can carry each second quickly

Engage - Retrieval & Hook (5 min)

TEACHER Tell students: Imagine you want to ask a question to a scientist in Bengaluru, but you cannot travel there. What if you could see them and talk to them on the screen at the same time? Then show a 30-second clip of a video conference between two classrooms.

STUDENTS Students are excited by the possibility of talking to people far away face to face. They ask questions about how it works and whether the internet connection is fast enough.

Explore - Guided Discovery (10 min)

TEACHER Give students a handout with screenshots of a video conferencing interface. Ask them to identify and label the buttons: camera, microphone, chat, share screen, end call, and participant list. Then let them explore a practice session in pairs on computers.

STUDENTS Students examine the screenshots and try to find the same buttons during their practice session. They experiment with muting and unmuting, turning the camera on and off, and sending chat messages while in a call.

Explain - Modelling & Concepts (10 min)

TEACHER Explain that video conferencing uses the internet to send both audio and video between devices in real time. Key features include: mute to silence your microphone, camera toggle to show or hide yourself, chat for typing messages during the call, screen share to show your computer screen to others, and recording to save the session. Etiquette rules: look at the camera not the screen, speak clearly, mute when not speaking, and sit in a well-lit quiet place.

STUDENTS Students role-play a video conference scenario in pairs. One student practices speaking to the camera while the other provides feedback on etiquette. They switch roles and try again.

Elaborate - Independent Application (10 min)

TEACHER Set up a mini video conference between two groups in different corners of the lab. Group A must explain the school schedule to Group B using video conferencing. Group B must take notes and ask three clarifying questions. Both groups must follow all etiquette rules.

STUDENTS Groups prepare their presentation and practice their delivery. During the conference, they take turns speaking, mute when listening, use the chat for non-urgent messages, and maintain good posture.

Evaluate - Check & Close (5 min)

TEACHER After the conference, each group rates the other on etiquette using a checklist: camera angle, speaking clarity, muting discipline, and professionalism. Teacher provides overall feedback. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups complete the etiquette checklist and discuss what went well and what they would improve. Teacher summarizes key etiquette points for future video calls.

Cross-Curricular Connections

Mathematics: Social Science: Video conferencing connects to understanding how technology bridges geographical distances
Science: English: Speaking clearly for video call practices formal presentation and public speaking skills
Language: Math: Understanding bandwidth and connection speed involves numerical concepts

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Practice mute and camera toggle with teacher guiding each button press
On Level: Participate in a short practice call with scripted questions and answers
Advanced: Lead a practice video call as moderator, managing mute and chat features
SEND: Join a call with teacher beside them, guiding each interaction step by step

Formative Assessment

Method: Etiquette checklist completed by peer groups after practice video conference
CT Skill Assessed: Evaluation: assessing communication effectiveness and etiquette during a real-time digital interaction

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could video call anyone in the world for 5 minutes, who would it be and what would you ask?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.07 *Presentation Tools*

40 minutes LAB

Success Criteria - I can...

- Identify the features of presentation software including slides, transitions, and animations
- Create a simple three-slide presentation with text, images, and a consistent theme
- Apply a slide transition and one animation effect to enhance a presentation

Resources Needed:
- Computers with video conferencing software
- Practice call setup between two groups
- Etiquette checklist handouts
- Quiet corners for separate group calls

Key Vocabulary

Slide: Single page inside presentation showing text and images together
Transition: Effect that moves you smoothly from one slide next
Animation: Movement added to text or pictures on a slide
Template: Ready-made design that keeps all slides looking consistent neatly

Engage - Retrieval & Hook (5 min)

TEACHER Show two presentations about the same topic: one is just text on white slides, the other has images, colors, and smooth transitions. Ask: Which one would you rather watch? Why does the second one hold your attention better?

STUDENTS Students immediately prefer the visually appealing version. They identify that images, colors, and movement make presentations more interesting and easier to follow.

Explore - Guided Discovery (10 min)

TEACHER Give students a printed handout with screenshots of LibreOffice Impress. Ask them to match each interface element to its function: slide panel, title bar, toolbar, design pane, notes area. Then open the software and find each element on screen.

STUDENTS Students compare the handout to the actual screen, pointing out differences and similarities. They click on each element to discover what it does and fill in their worksheet.

Explain - Modelling & Concepts (10 min)

TEACHER Introduce LibreOffice Impress as a tool for creating slide-based presentations. Key concepts: slides are individual pages, layouts define where text and images go, themes provide consistent colors and fonts, transitions are effects between slides, and animations are effects within a slide. Walk through creating a new presentation: choose a layout, add a title, add content, insert an image, and apply a transition.

STUDENTS Students follow along creating their first slide. They experiment with different layouts and themes, seeing how each changes the appearance of their content.

Elaborate - Independent Application (10 min)

TEACHER Students must create a three-slide presentation about their school: slide one is the school name and a photo, slide two lists three things they love about the school, and slide three is a thank you message. They must use a consistent theme and apply at least one transition between slides.

STUDENTS Students gather photos of the school from the shared drive and create their slides. They choose a theme that matches the school colors and practice adding transitions.

Evaluate - Check & Close (5 min)

TEACHER Students present their three slides to a partner. The partner evaluates: Is the theme consistent? Are the slides readable? Does the transition work smoothly? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present and receive feedback. They make one revision based on the feedback before the period ends.

Cross-Curricular Connections

Mathematics: Math: Slide dimensions and aspect ratios connect to understanding proportion and ratio
Science: Art: Choosing themes and layouts involves visual design and color theory
Language: Language: Writing slide content practices concise and clear writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a template with pre-made slides where the student fills in specific text boxes
On Level: Create a three-slide presentation following a step-by-step guide with screenshots
Advanced: Create a five-slide presentation with custom animations, speaker notes, and advanced transitions
SEND: Help the student select layouts and type text, guiding each step of the process

Formative Assessment

Method: Three-slide presentation evaluated on theme consistency, readability, and transition use
CT Skill Assessed: Abstraction: organizing information into discrete slides that each communicate one key idea

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you had to introduce your dormitory to a visitor using only three slides, what would each slide say?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.08 *Creating Presentations*

40 minutes LAB

Success Criteria - I can...

- Plan a presentation structure with an introduction, body, and conclusion
- Insert and format text, images, and shapes across multiple slides
- Deliver a presentation with clear narration and appropriate pacing

Resources Needed:

- Computers with LibreOffice Impress
- School photos on shared drive
- Presentation themes reference
- Slide layout guide
- Peer feedback form

Key Vocabulary

Slide: Single page inside presentation showing text and images together
Transition: Effect that moves you smoothly from one slide next
Animation: Movement added to text or pictures on a slide
Template: Ready-made design that keeps all slides looking consistent neatly

Engage - Retrieval & Hook (5 min)

TEACHER Ask a student volunteer to give a 1-minute talk about their favorite place with no preparation. Then ask another student to give the same talk but with three slides behind them. Compare the two experiences and ask the audience which was easier to follow.

STUDENTS Students observe that the prepared speaker with slides was more organized and easier to follow. They recognize that planning and visual aids make communication more effective.

Explore - Guided Discovery (10 min)

TEACHER Give students a topic card (My Village, My Favorite Food, A Festival We Celebrate) and a planning sheet with sections for: Title Slide, What I Want to Share, Key Details, and Closing Slide. They must plan five slides before opening the computer.

STUDENTS Students write bullet points for each slide on their planning sheet. They sketch where images should go and decide which text is most important for each slide.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the three-part structure: introduction slide grabs attention with a title and image, body slides share the main content with a mix of text and visuals, and conclusion slide summarizes and ends with a call to action or thank you. Demonstrate inserting images from files, formatting text with bold and color, and adding shapes as bullet point backgrounds.

STUDENTS Students review their planning sheets and adjust their content based on the three-part structure. They identify which of their bullet points go in introduction, body, and conclusion.

Elaborate - Independent Application (10 min)

TEACHER Students must create a five-slide presentation on their chosen topic. Requirements: title slide with name and image, three body slides with at least one image each, and a conclusion slide. All text must be readable from the back of the room (minimum 24 point font).

STUDENTS Students work on their presentations, carefully selecting images and writing concise text. They ask peers to check font size by standing at the back of the room.

Evaluate - Check & Close (5 min)

TEACHER Two or three students present to the class. The audience rates each presentation on structure, visual appeal, and speaking clarity. Teacher provides constructive feedback on each. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Student presenters speak clearly and advance slides at appropriate times. Audience members provide specific praise and suggestions using a rating form.

Cross-Curricular Connections

Mathematics: Social Science: Creating presentations about local communities builds awareness and pride
Science: English: Writing and delivering presentations practices both written and oral communication
Language: Math: Planning slide timing and pacing involves counting and time management

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a template with pre-written text that needs only image additions
On Level: Create a five-slide presentation using the planning sheet as a direct guide
Advanced: Design a presentation with custom layouts, animations, speaker notes, and a handout
SEND: Guide each step: choosing topics, writing text, placing images, and practicing delivery

Formative Assessment

Method: Class presentation with audience rating form and teacher rubric evaluation
CT Skill Assessed: Algorithmic thinking: planning and executing a sequence of steps from concept to delivered presentation

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Practice explaining what you had for lunch today in exactly 5 sentences. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.09 *Animation Basics*

40 minutes LAB

Success Criteria - I can…

- Explain how animation creates the illusion of movement through sequential images
- Create a simple flipbook or frame-by-frame digital animation
- Identify key animation concepts: frames, keyframes, and timing

Resources Needed:
- Topic cards and planning sheets
- Computers with LibreOffice Impress
- Image files on shared drive
- Presentation evaluation rubric
- Timer for pacing practice

Key Vocabulary

Animation: Creating movement by showing many pictures very quickly together
Frame: One picture in sequence that creates motion when played fast
Motion: Change of position of object from one place another
Timeline: Line showing order and timing of each frame in animation

Engage - Retrieval & Hook (5 min)

TEACHER Show a flipbook animation of a stick figure walking that the teacher made by drawing on sticky notes. Flip through it slowly, then quickly. Ask: Why does it look like the person is moving when I flip fast but not when I flip slow?

STUDENTS Students are fascinated by the flipbook. They immediately try to draw their own flipbooks on scrap paper, experimenting with flipping speeds.

Explore - Guided Discovery (10 min)

TEACHER Give each student a small pad of sticky notes and ask them to draw a circle at the bottom of page 1, then the same circle slightly higher on page 2, and continue for 10 pages. When they flip through, they should see the circle bounce upward.

STUDENTS Students draw their bouncing circles and flip through excitedly. Some make the circle go up and down, others make it go side to side. They share their creations with neighbors.

Explain - Modelling & Concepts (10 min)

TEACHER Define animation as the technique of displaying a sequence of images rapidly to create the illusion of movement. Each image is a frame. The speed of display determines how smooth the movement appears. In digital animation, keyframes are the important frames where changes happen, and the computer fills in the in-between frames (tweens). Show a simple Tux Paint animation or Scratch project to illustrate.

STUDENTS Students take notes on frames, keyframes, and timing. They compare their flipbook frame counts with classmates and discuss how more frames make smoother movement.

Elaborate - Independent Application (10 min)

TEACHER Students must create a 12-frame flipbook animation of any moving object: a ball bouncing, a bird flying, a flower growing, or a person waving. The animation must show clear progressive movement from frame to frame.

STUDENTS Students plan their animation by sketching all 12 frames on paper first before drawing on the sticky notes. They check that each frame shows a small change from the previous one.

Evaluate - Check & Close (5 min)

TEACHER Students show their flipbooks to the class. The class identifies what is moving and whether the movement looks smooth. Teacher measures frame consistency and progressive change. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their flipbooks and explain the movement they were trying to show. Classmates provide feedback on smoothness and clarity of the animation.

Cross-Curricular Connections

Mathematics: Physics: Understanding how persistence of vision creates the illusion of movement
Science: Art: Flipbook creation connects to traditional hand-drawn animation techniques
Language: Language: Writing a story about what happens in their animation practices narrative writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide pre-drawn first and last frames for the student to complete the middle frames
On Level: Create a 12-frame animation following a step-by-step frame guide
Advanced: Design a complex 20-frame animation with multiple moving elements and background changes
SEND: Draw each frame together with the teacher, with guidance on consistent spacing

Formative Assessment

Method: Flipbook presentation evaluated on frame progression, movement clarity, and consistency
CT Skill Assessed: Pattern recognition: identifying how small sequential changes create the perception of continuous motion

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a stick figure on the corner of your notebook pages. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.10 *Digital Storytelling* 40 minutes LAB

Success Criteria - I can…

- Structure a story with beginning, middle, and end using digital tools
- Combine text, images, and audio to create a digital story
- Select appropriate media elements to enhance narrative impact

Resources Needed:
- Sticky note pads
- Pencils and erasers
- Sample flipbook from teacher
- Animation frame reference sheet
- Tux Paint animation tools

Key Vocabulary

Story: Sequence of events told with words pictures and sound together
Narrative: Way you organise story with beginning middle and end
Scene: Single part of story showing one moment or place
Voiceover: Recorded speech added over pictures to explain story clearly

Engage - Retrieval & Hook (5 min)

TEACHER Play a 1-minute digital story about a student journey to school that combines photos, narration, and background music. Then ask: What made this story moving? Could the same story work as just text on paper?

STUDENTS Students discuss how the combination of pictures, voice, and music made them feel something. They realize that digital storytelling adds emotional layers that text alone cannot.

Explore - Guided Discovery (10 min)

TEACHER Give students three cards with different stories: a folk tale, a personal experience, and an imagined adventure. They must pick one and list five moments in the story that could be shown as images with text captions.

STUDENTS Students discuss which stories lend themselves to images and which are better as text. They sketch thumbnail ideas for each of their five moments.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the digital storytelling process: write a script (text for each slide), select or create images for each scene, record narration or add background music, and combine everything in a presentation tool. Emphasize the rule of three: each slide should have one image, a few words of text, and one audio element. Show how to align narration timing with slide transitions.

STUDENTS Students outline their story with five slides: opening scene, rising action, climax, falling action, and resolution. They assign an image and narration line to each slide.

Elaborate - Independent Application (10 min)

TEACHER Students must create a five-slide digital story about a moment from their school life. Each slide needs one image, text of no more than 20 words, and a 10-second narration. They record their narration in pairs, taking turns being the narrator.

STUDENTS Students write their scripts, select images from the shared drive, and record narration. They practice speaking clearly and at a measured pace for the recording.

Evaluate - Check & Close (5 min)

TEACHER Two or three digital stories are played for the class. Audience members rate each on story clarity, image quality, narration quality, and emotional impact. Teacher facilitates discussion. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students listen to each other stories and complete rating forms. They discuss which elements made each story effective and what they would do differently.

Cross-Curricular Connections

Mathematics: English: Digital storytelling integrates creative writing with multimedia presentation
Science: Social Science: Stories about village life and school experiences connect to community identity
Language: Music: Choosing background music that matches the mood of the story practices musical appreciation

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a pre-written story with images that needs only narration recording
On Level: Write and illustrate a five-slide story following a guided template
Advanced: Create an original digital story with custom images, recorded narration, and background music
SEND: Work with teacher to write the script, choose images, and record narration together

Formative Assessment

Method: Digital story presentation with audience rating on clarity, images, narration, and impact
CT Skill Assessed: Abstraction: selecting the most important moments from a story and representing them through multimedia

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of one memory from your first week at school. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.11 *Podcasting*

40 minutes LAB

Success Criteria - I can...

- Define a podcast and identify its key elements: topic, format, and episode structure
- Plan and record a short podcast episode on an educational topic
- Apply editing techniques to produce a clean, well-paced podcast segment

Resources Needed:
- Sample digital story files
- Image libraries on shared drive
- Recording software
- Story planning template
- Presentation software

Key Vocabulary

Podcast: Series of audio episodes you can listen anytime online
Episode: Single audio story that is part of larger series
Host: Person who speaks and guides the podcast for listeners
Subscribe: Following podcast to receive new episodes automatically when released

Engage - Retrieval & Hook (5 min)

TEACHER Play a 30-second clip from a children educational podcast about space. Ask: What makes this different from a teacher just talking in class? What elements make it interesting to listen to?

STUDENTS Students identify that the podcast uses multiple voices, sound effects, and a clear structure. They discuss how they would make their own interesting podcast.

Explore - Guided Discovery (10 min)

TEACHER Give students a podcast planning worksheet with sections for: Episode Title, Main Topic, Three Key Points, Opening Hook, and Closing Call to Action. Ask them to fill it in for a topic they know well (a recipe, a game, a local festival).

STUDENTS Students brainstorm topics they are passionate about. Some choose Diwali celebrations, others choose cricket rules, and a few choose their grandmother recipes. They fill in the worksheet with enthusiasm.

Explain - Modelling & Concepts (10 min)

TEACHER Explain podcast structure: introduction (welcome and topic overview), body (main content with three key points), and conclusion (summary and what to do next). Discuss recording tips: speak naturally, maintain consistent volume, and record in a quiet space. Show how to use Audacity to record, trim silence between sentences, and add a brief pause before the closing.

STUDENTS Students review their worksheets and refine their scripts for natural spoken language. They practice reading their scripts aloud, adjusting for conversational tone.

Elaborate - Independent Application (10 min)

TEACHER Students must record a 2-minute podcast episode in pairs. One student is the host and the other is the expert guest. They must include an introduction, three discussion points, and a conclusion. After recording, they must trim any long pauses and ensure volume is consistent throughout.

STUDENTS Students practice their dialogue before recording. During recording, they maintain eye contact for natural conversation. After recording, they use Audacity to clean up the audio.

Evaluate - Check & Close (5 min)

TEACHER Played podcasts are evaluated by the class on: content accuracy, speaking clarity, pacing, and engagement. Teacher provides a detailed rubric score and constructive feedback. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students listen to other pairs podcasts and fill in evaluation rubrics. They provide specific praise and one suggestion for improvement to each pair.

Cross-Curricular Connections

Mathematics: English: Writing podcast scripts practices conversational and persuasive writing
Science: Science: Educational podcasts about scientific topics reinforce content knowledge
Language: Social Science: Podcasts about local culture and traditions preserve and share community knowledge

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Read a pre-written podcast script with one question-and-answer section to record
On Level: Record a scripted podcast episode with moderate independence
Advanced: Create an unscripted podcast discussion with research notes and natural conversation
SEND: Record one section at a time with teacher helping with script and microphone technique

Formative Assessment

Method: Podcast evaluation rubric covering content, clarity, pacing, and engagement
CT Skill Assessed: Evaluation: assessing audio quality and content effectiveness through peer and self-evaluation

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Try explaining your favorite game to your dormitory friend in a 1-minute talk. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.12 *Media Production Project* 40 minutes LAB

Success Criteria - I can…

- Plan and execute a complete media project combining audio, visual, and presentation elements
- Collaborate effectively in a production team with defined roles
- Reflect on the entire media production process from concept to final output

Resources Needed:
- Sample podcast clips
- Podcast planning worksheets
- Audacity recording software
- USB microphones or smartphones
- Evaluation rubrics

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage - Retrieval & Hook (5 min)

TEACHER Show a 2-minute video documentary made by our school students about their school. It includes interviews, photos, narration, and text overlays. Ask: How many different skills did these students use? How long do you think it took to make?

STUDENTS Students count the skills: photography, writing, recording, editing, and presenting. They are inspired to create something similar about their own school.

Explore - Guided Discovery (10 min)

TEACHER Give groups a Production Planning Sheet with roles: Director (makes decisions), Scriptwriter (writes content), Camera Operator (captures images), Audio Engineer (records and edits sound), and Editor (combines everything). Groups must assign roles and choose a topic: a day at school, our village, or our favorite subject.

STUDENTS Groups negotiate roles based on individual strengths. Some students want to be directors, others prefer technical roles. They discuss and agree on their topic and structure.

Explain - Modelling & Concepts (10 min)

TEACHER Walk through the production workflow: Pre-production (planning, scripting, storyboarding), Production (recording, photographing, filming), and Post-production (editing, combining, reviewing). Each phase has specific tasks and time requirements. For a 40-minute period, spend 10 minutes on planning, 15 minutes on recording, and 15 minutes on editing.

STUDENTS Groups complete their planning sheet, create a simple storyboard with three scenes, and prepare their recording scripts. They discuss the technical requirements for each scene.

Elaborate - Independent Application (10 min)

TEACHER Groups must produce a 3-part media piece: a title slide with text, a 30-second recorded narration about their topic, and a closing slide with credits. They use the remaining time to execute their plan, with each member contributing their assigned role.

STUDENTS Groups work in parallel: some record audio, others prepare slides, and others select images. They reconvene to combine their work and review the complete piece.

Evaluate - Check & Close (5 min)

TEACHER Each group presents their final media piece. Class evaluates using a rubric covering planning, execution, teamwork, and final quality. Groups also submit a reflection on their production process. Hinge Q:

[question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups present their work proudly. They explain what went well, what was challenging, and what they learned about working as a production team.

Cross-Curricular Connections

Mathematics: English: Scriptwriting and narration practice both written and oral communication
Science: Social Science: Media production about community topics builds awareness and documentation skills
Language: Art: Visual composition, color choices, and layout design integrate artistic principles

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a structured template with pre-assigned roles and step-by-step task cards
On Level: Execute the production plan with the planning sheet guiding each phase
Advanced: Lead the production team, making creative and technical decisions independently
SEND: Participate in one specific role with teacher support for that role tasks

Formative Assessment

Method: Group presentation with production rubric plus individual reflection on teamwork
CT Skill Assessed: Application: integrating all media production skills into a collaborative project with real-world workflow

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about what makes a good story. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 4 - Data Representation & Processing

Lesson 4.01 *What is Data?*

40 minutes **CONCEPT**

Success Criteria - I can…

- Define data and distinguish between raw data and processed information
- Identify different types of data encountered in daily school life

- Explain why organizing data is important for making decisions

Resources Needed:
- Production planning sheets
- Storyboard templates
- Recording equipment
- Presentation software
- Peer evaluation rubrics
- Sample student media projects

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage - Retrieval & Hook (5 min)

TEACHER Bring a bag of mixed letters (A, B, C, numbers 1-9, and symbols) and dump them on a table. Ask: Is this useful? Now arrange them to spell the school name and class attendance numbers. Ask again: Is it useful now? What changed?

STUDENTS Students see that random symbols become meaningful when organized. They shout out that the arranged letters make words and the numbers could be attendance counts.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give each group a worksheet with a messy list of 20 items: student names, roll numbers, subjects, marks, and attendance days. Ask them to sort the data into categories and answer: How many students scored above 50? Which subject has the most students?

STUDENTS Groups struggle to answer questions from the messy list. They start organizing the data into a table and find the answers become easy. Teacher asks what they did differently.

Explain - Modelling & Concepts (10 min)

TEACHER Define data as raw facts and figures that have not been organized (like a list of numbers). Define information as organized data that tells us something useful (like a class report card). Give our school examples: daily attendance numbers are data, but a monthly attendance report is information. Student marks are data, but a class ranking is information. Data types include text (names), numbers (marks), and dates (admission date).

STUDENTS Students create their own examples of data versus information in their notebooks. They write three examples from school life: attendance data becomes a report, marks data becomes a result card, and inventory data becomes a stock list.

Elaborate - Independent Application (10 min)

TEACHER Students must find five examples of data being collected at their school (attendance register, library book list, exam marks, lunch count, bus route list) and explain what information each piece of data produces when organized.

STUDENTS Students walk through the school office, library, and classroom to observe data collection in action. They interview the librarian about how book data is organized and the clerk about attendance data.

Evaluate - Check & Close (5 min)

TEACHER Give students a scenario: the school principal wants to know which class has the best attendance this month. They must explain what data is needed, how to organize it, and what information it would produce. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students write their explanation on a worksheet. Teacher reviews for understanding of the data-information distinction and proper use of examples.

Cross-Curricular Connections

Mathematics: Math: Data and numbers connect directly to mathematical concepts of counting and statistics
Science: Social Science: Census data and population information use the same data-to-information process
Language: Language: Describing data in sentences practices clear and precise writing

Digital Citizen Reminder: Personal data like your name and marks is private. Never share someone else data without permission.

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Differentiation

Approaching: Provide a pre-sorted data set with picture labels for matching
On Level: Organize a given data set into a simple table with some guidance
Advanced: Collect original data from a school survey and organize it into meaningful information
SEND: Sort picture-based data together with teacher support and verbal explanations

Formative Assessment

Method: Scenario-based worksheet explaining what data is needed and how it becomes information
CT Skill Assessed: Abstraction: identifying which raw facts are relevant data for answering a specific question

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Count how many students sleep in your dormitory, how many beds are occupied, and how many are empty. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.02 *Number Systems*

40 minutes **CONCEPT**

Success Criteria - I can...

- Explain the decimal number system and why humans use base 10
- Describe the binary number system and its use in computers
- Convert simple numbers between decimal and binary

Resources Needed:
- Mixed letter bag
- Messy data worksheets
- Attendance register samples
- Office visit permission
- Data organization templates

Key Vocabulary

Number System: Method of writing numbers using specific set of digits
Decimal: Base ten system using digits zero to nine daily
Binary: Base two system using only zero and one inside computers
Place Value: Worth of a digit depending on its position in number

Engage - Retrieval & Hook (5 min)

TEACHER Ask students to count from 1 to 10 on their fingers. Then ask: How would you count to 100 with only 10 fingers? What if each finger could only be up or down (two states)? Show that binary uses this up-down logic.

STUDENTS Students try counting on fingers and realize that binary finger counting (up=1, down=0) lets them count much higher. They get excited when they can count to 1023 on both hands.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give each group ten coins. Ask them to represent the number 13 using the fewest coins possible (they will use place value: 1 ten and 3 ones). Then ask them to represent 13 using only two types of coins (1s and 2s) - they discover the binary approach.

STUDENTS Groups experiment with different coin arrangements. They realize that decimal uses ten symbols (0-9) while binary uses only two symbols (0 and 1). They practice writing numbers in both systems.

Explain - Modelling & Concepts (10 min)

TEACHER Explain that decimal (base 10) uses digits 0-9 and each position represents a power of 10 (ones, tens, hundreds). Binary (base 2) uses digits 0-1 and each position represents a power of 2 (1, 2, 4, 8, 16). Show conversion: decimal 13 = 8+4+1 = 1101 in binary. Explain that computers use binary because electronic circuits have two states: on and off.

STUDENTS Students practice converting five decimal numbers to binary and five binary numbers to decimal. They use a place value chart to keep track of their calculations.

Elaborate - Independent Application (10 min)

TEACHER Challenge students to write their birth date in binary. For example, 15 March = 15/03. Convert 15 to binary (1111) and 03 to binary (11). Then create a binary birthday card with the numbers decorated using 0s and 1s.

STUDENTS Students work through the conversions carefully, checking their work with a partner. They create creative binary birthday cards combining math and art.

Evaluate - Check & Close (5 min)

TEACHER Give students a quick quiz: convert three decimal numbers to binary and three binary numbers to decimal. Include one word problem about a computer receiving binary signals. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students complete the quiz independently. Teacher checks for understanding of place value in both systems and accuracy of conversions.

Cross-Curricular Connections

Mathematics: Math: Number systems and place value are core mathematical concepts
Science: Science: Electronic circuits use on-off states that binary represents, connecting to physics
Language: History: Different civilizations used different number systems (Roman, Mayan, Babylonian)

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a pre-made binary conversion table for reference during practice
On Level: Convert numbers up to 15 between decimal and binary with a place value chart
Advanced: Convert larger numbers (up to 100) and explain the relationship between binary and computer memory
SEND: Convert single-digit numbers with teacher guiding each step of the place value calculation

Formative Assessment

Method: Quiz on decimal-binary conversion including a word problem application
CT Skill Assessed: Decomposition: breaking a decimal number into powers of 2 to express it in binary

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Try counting to 20 using only your right hand where each finger position means 1 or 0. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.03 *Binary Basics*

40 minutes CONCEPT

Success Criteria - I can…

- Perform binary addition using the rules of base-2 arithmetic
- Explain how binary addition differs from decimal addition
- Apply binary addition to solve simple computer-related problems

Resources Needed:

- Ten coins per group
- Binary conversion worksheet
- Place value charts
- Binary birthday card template
- Binary flashlight demonstration

Key Vocabulary

Binary: Counting system using only zero and one for computers
Bit: Smallest unit of data holding either zero or one
Byte: Group of eight bits that stores one character like A
Conversion: Changing number from decimal to binary or opposite method

Engage - Retrieval & Hook (5 min)

TEACHER Write on the board: 1+1=10 and ask students if this is correct. They will say no, but explain that in binary, this is exactly right. Ask: Why does 1+1 not equal 2 in binary? What happens to the 10?

STUDENTS Students are confused and curious. They debate among themselves and some guess that binary has no digit 2, so it carries over. Teacher confirms their intuition.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give groups a set of binary addition problems on cards, ranging from simple (1+1) to medium (101+11). They must solve each problem using a table with place values 8, 4, 2, 1 and physical tokens to represent bits.

STUDENTS Groups place tokens on the place value table to represent each binary number, then combine them. When two tokens land on the same position, they carry one to the next position. Students discover the pattern of binary addition.

Explain - Modelling & Concepts (10 min)

TEACHER Teach binary addition rules: 0+0=0, 0+1=1, 1+0=1, 1+1=10 (write 0, carry 1), 1+1+1=11 (write 1, carry 1). Walk through examples step by step using the place value chart. Connect to computers: binary addition is how processors calculate everything from simple sums to complex graphics.

STUDENTS Students practice addition problems on their own, using the rules they learned. They check their work by converting both the binary numbers and their answer to decimal to verify.

Elaborate - Independent Application (10 min)

TEACHER Give students a challenge: a computer has 4 bits of memory (can store numbers up to 1111). Add the binary numbers 1011 and 1101. Does the result fit in 4 bits? What happens if it does not fit? Discuss overflow.

STUDENTS Students work through the addition and discover that the result (11000) needs 5 bits. They discuss what happens in a real computer when numbers overflow their storage capacity.

Evaluate - Check & Close (5 min)

TEACHER Students complete five binary addition problems of increasing difficulty. They must show their working using the place value chart and verify their answers by converting to decimal. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students work independently and self-check using a provided answer key. Teacher reviews for correct application of carry rules and understanding of overflow.

Cross-Curricular Connections

Mathematics: Math: Binary addition reinforces place value concepts and regrouping (carrying)
Science: Science: Binary arithmetic connects to how computer processors perform calculations
Language: Logic: Understanding binary rules develops systematic logical thinking

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a step-by-step worked example for each binary addition rule
On Level: Solve binary addition problems up to 4 bits with a place value chart
Advanced: Solve larger binary additions and explain the concept of overflow in computing
SEND: Add single-bit binary numbers with teacher guiding the carry process

Formative Assessment

Method: Five binary addition problems with working shown and decimal verification
CT Skill Assessed: Algorithmic thinking: applying a systematic set of rules to solve binary arithmetic problems

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write the number of stars you can see from your dormitory window in binary. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.04 *Data Storage*

40 minutes CONCEPT

Success Criteria - I can...

- Identify units of digital storage: bit, byte, kilobyte, megabyte, gigabyte
- Estimate the storage size needed for different types of files
- Explain why storage capacity matters for choosing devices and managing files

Key Vocabulary

Storage: Place where computer keeps data for later use safely
Memory: Temporary space where computer holds data while working actively
Hard Disk: Metal disk inside computer storing files permanently for long
Capacity: Amount of data a device can hold measured in GB

Engage - Retrieval & Hook (5 min)

TEACHER Show students two USB drives: a 1 GB drive and a 16 GB drive. Ask: If one photo is 3 MB, how many photos fit on each? Let them estimate, then calculate together.

STUDENTS Students estimate and are surprised by how many photos fit on even the small drive. They realize that storage capacity is measured in huge numbers.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give groups a storage estimation worksheet: a photo is 3 MB, a song is 4 MB, a video is 150 MB, a document is 50 KB. Ask them to calculate how many of each would fill a 1 GB drive (approximately 1000 MB).

STUDENTS Groups calculate and discover that one song fits about 250 times, but a video only fits about 6 times. They discuss why videos need so much more storage.

Explain - Modelling & Concepts (10 min)

TEACHER Define storage units: 1 bit is the smallest unit (0 or 1), 8 bits make 1 byte, 1000 bytes make 1 kilobyte (KB), 1000 KB make 1 megabyte (MB), 1000 MB make 1 gigabyte (GB), 1000 GB make 1 terabyte (TB). Explain that text files are small (KB), photos are medium (MB), and videos are large (MB to GB). Show a comparison chart of common file sizes.

STUDENTS Students create a storage unit reference chart in their notebooks with examples of what each size can hold: KB holds a page of text, MB holds a photo, GB holds a movie.

Elaborate - Independent Application (10 min)

TEACHER Students must plan storage for a school project: they need 30 photos (3 MB each), 5 songs (4 MB each), and a 3-minute video (150 MB). Calculate total storage needed. Which USB drive should they use: 1 GB, 4 GB, or 16 GB?

STUDENTS Students calculate total storage needed (90 + 20 + 150 = 260 MB), compare with available options, and recommend the most cost-effective drive. They justify their recommendation.

Evaluate - Check & Close (5 min)

TEACHER Give students a storage challenge: they have a 2 GB memory card and need to fit a mix of photos, songs, and one video. They must create a plan that fits within the storage limit. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students work through the storage calculations and create a fitting plan. Teacher checks their arithmetic and reasoning about storage choices.

Cross-Curricular Connections

Mathematics: Math: Unit conversion between KB, MB, GB connects to metric system and multiplication
Science: Science: Understanding data storage connects to how memory chips and hard drives work physically
Language: Economics: Storage capacity versus cost connects to budgeting and resource management

Digital Citizen Reminder: Regularly back up important data. Storage devices can fail, and data can be lost permanently.

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Differentiation

Approaching: Provide pre-calculated storage examples with visual fill-in-the-blank charts
On Level: Calculate storage needs for simple file collections with a reference chart
Advanced: Plan complex storage solutions considering file compression and different quality settings
SEND: Use visual storage bar charts to compare file sizes with teacher guidance

Formative Assessment

Method: Storage planning challenge with calculations and device recommendation
CT Skill Assessed: Application: using mathematical calculation to solve a real-world storage planning problem

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If your phone has 32 GB of storage and you have used 12 GB, how many more 3 MB photos can you take?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.05 *File Formats*

40 minutes CONCEPT

Success Criteria - I can…

- Identify common file formats and their extensions: .txt, .jpg, .png, .mp3, .mp4, .ods, .odp
- Match file formats to their appropriate uses and software
- Explain why choosing the right file format matters for quality and compatibility

Resources Needed:
- USB drives of different capacities
- Storage estimation worksheets
- File size reference chart
- Calculator
- Memory card examples

Resources Needed:
- Binary addition card sets
- Place value charts (8-4-2-1)
- Tokens or counters
- Binary addition worksheet
- Answer key for self-checking

Key Vocabulary

File: Single document or picture stored with a name on computer
Format: Type deciding how file stores data like JPG or MP3
Extension: Letters after dot showing file type like .jpg or .docx
Compression: Making file smaller to save space and transfer quickly

Engage - Retrieval & Hook (5 min)

TEACHER Show students three versions of the same image saved as .jpg, .png, and .bmp. Open each and zoom in. Ask: Why do they look different even though it is the same picture? What could cause the differences?

STUDENTS Students notice quality differences when zoomed in. They discuss how different formats store image information differently.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give students a matching activity with 14 cards: 7 file extensions (.txt, .jpg, .png, .mp3, .mp4, .ods, .odp) and 7 descriptions (text document, photo, photo with transparency, music, video, spreadsheet, presentation). They must match each extension to its description.

STUDENTS Students match cards and discuss their reasoning. Some match by familiarity (MP3 is music), others by elimination. Teacher guides discussions about why different formats exist.

Explain - Modelling & Concepts (10 min)

TEACHER Categorize file formats: text files (.txt) store plain text with no formatting, image files (.jpg for photos, .png for graphics with transparency), audio files (.mp3 for compressed music, .wav for high quality), video files (.mp4 for compressed video), and document files (.ods for spreadsheets, .odp for presentations). Explain that compression reduces file size but may lose quality (lossy like JPEG) or preserve it perfectly (lossless like PNG).

STUDENTS Students create a file format reference table in their notebooks with format name, extension, best use, and whether it is compressed or uncompressed.

Elaborate - Independent Application (10 min)

TEACHER Students must decide the best format for each scenario: saving a photo for a school poster (PNG for quality), emailing a document to a teacher (ODT for compatibility), recording a song for a podcast (MP3 for small size), and creating a presentation for class (ODP for editing). They must justify each choice.

STUDENTS Students discuss format choices in pairs and justify their decisions. They debate trade-offs between quality, file size, and compatibility.

Evaluate - Check & Close (5 min)

TEACHER Give students a scenario worksheet: someone saved a logo as JPEG and it looks blurry on the poster. They must explain why PNG would be better and identify three other format mistakes from given scenarios. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students analyze each scenario, identify the format problem, and suggest the correct format with justification.

Cross-Curricular Connections

Mathematics: Math: File size calculations connect to understanding compression ratios and percentages
Science: Art: Image format choice affects the quality of digital artwork and design projects
Language: Language: Writing format recommendations practices technical and persuasive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a completed file format reference chart for copying into notebooks
On Level: Match file extensions to descriptions with some hints from the reference chart
Advanced: Create a file format guide for the school computer lab recommending formats for common tasks
SEND: Match picture-based file icons to their names with teacher support

Formative Assessment

Method: Scenario-based worksheet identifying format problems and recommending correct formats
CT Skill Assessed: Evaluation: assessing which file format best serves a specific purpose based on quality, size, and compatibility

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could only keep 3 files on your phone forever, what format would each be in and why?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.06 *Spreadsheet Basics* 40 minutes CONCEPT

Success Criteria - I can...

- Navigate a spreadsheet interface and identify rows, columns, and cells
- Enter text and numerical data into specific cells in a spreadsheet
- Create a simple class marks sheet with headers and student data

Key Vocabulary

Spreadsheet: Grid of rows and columns for organising numbers and data
Cell: Single box formed by one row and one column intersection
Row: Horizontal line of cells running from left to right
Column: Vertical line of cells running from top to bottom

Engage - Retrieval & Hook (5 min)

TEACHER Show a handwritten class marks list on paper and then the same data in a spreadsheet on screen. Ask: Which one would you rather use to find out who scored highest in Math? Why is the spreadsheet faster?

STUDENTS Students notice that the spreadsheet makes it easy to sort and find information, while the paper list requires scanning every row.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Open LibreOffice Calc and give students a guided tour worksheet. Ask them to find: the name box, the formula bar, column letters (A, B, C), row numbers (1, 2, 3), and a specific cell (C5). They must click on each element and note what happens.

STUDENTS Students click through the spreadsheet interface, discovering that clicking on cell C5 highlights column C and row 5. They practice entering their name in cell A1 and a number in cell B1.

Explain - Modelling & Concepts (10 min)

TEACHER Define key spreadsheet terms: a workbook is the entire file, a worksheet is one page within it, rows are horizontal (numbered 1, 2, 3), columns are vertical (lettered A, B, C), and a cell is where a row and column meet (identified by column letter and row number like B3). Show how to enter data: click a cell, type, and press Enter to move down or Tab to move right.

STUDENTS Students practice entering data into specific cells, following a grid pattern. They enter their name in A1, class in A2, and five subject marks in B1 through F1.

Elaborate - Independent Application (10 min)

TEACHER Students must create a class marks sheet with: title in A1 (merged across A1 to F1), column headers in row 3 (Name, Kannada, English, Math, Science, Social), and five student entries with marks in rows 4-8. All data must be properly aligned.

STUDENTS Students merge cells for the title, type headers in bold, and enter five rows of student data. They check alignment and formatting with a partner.

Evaluate - Check & Close (5 min)

TEACHER Teacher displays a completed marks sheet on the projector. Students self-check their own work against the model, noting any differences in formatting, data entry accuracy, or alignment. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students compare their work to the model, identify discrepancies, and make corrections. Teacher circulates checking for proper cell references and data accuracy.

Cross-Curricular Connections

Mathematics: Math: Spreadsheet rows and columns connect to coordinate systems and data organization
Science: Social Science: Organizing class data connects to understanding demographics and records
Language: Language: Writing column headers practices clear and concise labeling

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a template spreadsheet with headers already entered for the student to fill in data
On Level: Create a marks sheet with step-by-step cell-by-cell instructions
Advanced: Create a marks sheet with formatting requirements including borders, colors, and merged title cells
SEND: Enter data into pre-formatted cells with teacher guiding cell selection and data entry

Formative Assessment

Method: Class marks sheet creation evaluated on data accuracy, formatting, and cell reference knowledge
CT Skill Assessed: Decomposition: breaking down a data organization task into individual cell entries and formatting steps

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Make a mental list of the five subjects you study and your marks in each. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.07 *Formulas and Functions* 40 minutes CONCEPT

Success Criteria - I can...

- Write basic spreadsheet formulas using the equals sign (=) for addition, subtraction, multiplication, and division
- Use the SUM function to calculate totals for a column of numbers
- Apply formulas to automatically calculate student total marks and percentages

Resources Needed:
- File extension matching cards
- Format reference chart
- Sample files in different formats
- File properties dialog screenshots
- Scenario worksheets

Key Vocabulary

Formula: Rule starting with equals that calculates answer from cell values
Function: Ready-made formula like SUM that does calculations quickly automatically

Reference: Address of a cell like A1 used inside formula for calculation
Result: Answer displayed after formula finishes its calculation correctly

Engage - Retrieval & Hook (5 min)

TEACHER Open the marks sheet from the previous lesson and ask: How long would it take to add up all the marks for 30 students using a calculator? Now show them one formula that calculates the total instantly. Ask: Which method would you prefer?

STUDENTS Students see the formula calculate instantly and are amazed. They want to learn how to make the spreadsheet do the math for them.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give students a marks sheet with five students and five subjects. Ask them to calculate each student total by hand using a calculator. Then show them the SUM formula and ask them to enter it for the first student. Compare the time taken.

STUDENTS Students laboriously add up marks with calculators, then enter one SUM formula and see it work instantly. They calculate how much time the formula saved and are convinced of its value.

Explain - Modelling & Concepts (10 min)

TEACHER Teach formula basics: every formula starts with =, use + for addition, - for subtraction, * for multiplication, and / for division. Cell references (like B4, C4) let formulas use values from other cells. The SUM function adds a range: =SUM(B4:F4) adds cells B4 through F4. Demonstrate calculating total, average, and percentage in the marks sheet.

STUDENTS Students practice writing formulas for each student total. They then calculate average marks per subject using =SUM(B4:B8)/5 and percentage using total/500*100.

Elaborate - Independent Application (10 min)

TEACHER Students must complete the marks sheet by adding: a Total column using SUM, a Percentage column using the formula total divided by maximum marks times 100, and an Average row at the bottom using SUM of each column divided by the number of students.

STUDENTS Students write formulas carefully, checking that cell references are correct. They verify their formulas by comparing the first result with a manual calculation.

Evaluate - Check & Close (5 min)

TEACHER Teacher gives a new data set (10 students, 4 subjects) and asks students to create formulas for total, percentage, and class average for each subject. They must enter all formulas within 10 minutes. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students work against the clock, entering formulas quickly and accurately. Teacher checks formula syntax and result accuracy.

Cross-Curricular Connections

Mathematics: Math: Spreadsheet formulas directly apply arithmetic operations and order of operations
Science: Science: Lab results often require calculating averages and percentages from data sets
Language: Economics: Budget calculations use the same addition, multiplication, and percentage formulas

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a template with SUM formulas already entered for the student to test and verify
On Level: Write SUM formulas for totals with a step-by-step formula building guide
Advanced: Write complex formulas including nested functions, absolute references, and conditional calculations
SEND: Enter =SUM formulas with teacher typing the formula while student selects the cells

Formative Assessment

Method: Timed formula entry challenge with accuracy check on totals, percentages, and averages
CT Skill Assessed: Algorithmic thinking: constructing a formula step by step to automate a mathematical calculation

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Add up the ages of everyone in your dormitory row. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.08 *Charts and Graphs*

40 minutes **CONCEPT**

Success Criteria - I can…

- Create a bar chart from spreadsheet data to visualize comparisons
- Create a pie chart to show proportions of a whole
- Choose the appropriate chart type for different data visualization needs

Resources Needed:
- Computers with LibreOffice Calc
- Guided tour worksheet
- Sample marks sheet for reference

- Spreadsheet terminology poster
- Cell reference practice grid

Key Vocabulary

Chart: Picture like bar or pie showing data visually for understanding
Axis: Line along side or bottom showing measurement scale clearly
Legend: Key explaining what each colour or symbol means in chart
Data Point: Single value plotted as dot or bar inside chart

Engage - Retrieval & Hook (5 min)

TEACHER Show the class marks table with numbers only, then show the same data as a colorful bar chart. Ask: Which one tells you faster which subject has the highest average? Why does the chart work better?

STUDENTS Students immediately see that the bar chart makes comparisons visual and instant. They can identify the tallest bar without reading any numbers.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give students a printed page with five different charts (bar, pie, line, scatter, pictograph) showing the same data. Ask them to match each chart type to its name and describe what each chart is good at showing.

STUDENTS Students examine the charts and discuss which ones are easiest to read for different purposes. They notice that bar charts compare categories, pie charts show percentages, and line charts show changes over time.

Explain - Modelling & Concepts (10 min)

TEACHER Teach two main chart types: bar charts compare categories (like subject averages) and pie charts show parts of a whole (like how students spend their day). Walk through creating each in LibreOffice Calc: select the data, go to Insert, choose Chart, select the type, add titles and labels. Explain chart elements: title, legend, axis labels, and data labels.

STUDENTS Students follow along creating a bar chart of subject averages and a pie chart of daily time allocation. They customize colors and add proper titles.

Elaborate - Independent Application (10 min)

TEACHER Students must create two charts from the marks sheet: a bar chart comparing average marks across all five subjects, and a pie chart showing the proportion of marks each student contributed to the class total. Both charts must have titles, axis labels, and legends.

STUDENTS Students select data ranges carefully, choosing the right cells for each chart. They customize chart appearance and verify that the visual matches the data.

Evaluate - Check & Close (5 min)

TEACHER Display student charts on the projector. Class evaluates each chart for: correct data representation, clear labels, appropriate chart type, and visual appeal. Students explain their design choices. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their charts and explain why they chose each chart type. Class votes on the clearest and most visually effective charts.

Cross-Curricular Connections

Mathematics: Math: Charts represent mathematical data visually, connecting numbers to geometric shapes
Science: Social Science: Census and survey data are commonly displayed using bar and pie charts
Language: Art: Chart design involves color choices, layout, and visual communication principles

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Provide a spreadsheet with pre-selected data and chart wizard steps to follow
On Level: Create one bar chart from provided data with step-by-step instructions
Advanced: Create multiple chart types from original data and write a reflection on which works best
SEND: Select data and choose chart type with teacher guiding each click

Formative Assessment

Method: Two charts created and presented with class evaluation using a chart quality rubric
CT Skill Assessed: Pattern recognition: identifying which chart type best reveals patterns and comparisons in data

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your study time, play time, and sleep time. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.09 *Data Entry Practice*

40 minutes CONCEPT

Success Criteria - I can...

- Enter data accurately and efficiently into spreadsheet cells

- Use keyboard shortcuts to speed up data entry (Tab, Enter, Arrow keys, Ctrl+Z)
- Validate entered data for correctness and consistency

Resources Needed:
- Marks sheet from previous lesson
- Formula reference card
- Calculator for verification
- New data set worksheet
- Spreadsheet function guide

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage - Retrieval & Hook (5 min)

TEACHER Time yourself entering 20 names into a spreadsheet using only the mouse versus using keyboard shortcuts. Show students the time difference on the clock. Ask: Which method was faster? How much time could be saved over a whole class list?

STUDENTS Students are impressed by the speed difference. They are eager to learn the keyboard shortcuts to become faster data entry specialists.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give students a practice sheet with 30 items to enter: 10 names, 10 numbers, and 10 dates. They must enter all data as fast and accurately as possible, first using only the mouse, then using keyboard shortcuts. Compare both times.

STUDENTS Students race against the clock, discovering that Tab moves right, Enter moves down, and arrow keys navigate freely. They record their times and see significant improvement.

Explain - Modelling & Concepts (10 min)

TEACHER Teach essential keyboard shortcuts: Tab moves to the next cell to the right, Enter moves down, arrow keys navigate in any direction, Ctrl+Z undoes the last action, Ctrl+C copies, Ctrl+V pastes. Demonstrate efficient data entry workflow: type data, press Tab, type next data, press Tab, and so on across a row, then press Enter to start the next row. Show how to use Fill Handle to copy formulas down a column.

STUDENTS Students practice each shortcut individually, then combine them into a fluid data entry workflow. They time themselves entering the same data set using different methods.

Elaborate - Independent Application (10 min)

TEACHER Students must enter a complete class register: 25 student names, 5 subject marks each (125 numbers total), and verify that all entries are correct by cross-checking with a printed source. They must complete the entry in under 10 minutes.

STUDENTS Students work efficiently using shortcuts, checking their work as they go. They use Ctrl+Z to fix mistakes without losing their rhythm. Teacher monitors their progress.

Evaluate - Check & Close (5 min)

TEACHER Compare student data entry speed and accuracy. The fastest and most accurate entries are displayed as examples. Students self-evaluate: How many errors did they make? Which shortcuts helped most? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students count their errors, calculate their accuracy rate, and identify which shortcuts were most helpful. Teacher recognizes the most improved students.

Cross-Curricular Connections

Mathematics: Math: Data entry accuracy connects to precision in mathematical calculations
Science: English: Typing skills and keyboard shortcuts improve general computer literacy
Language: Economics: Fast and accurate data entry is essential for business and office work

Digital Citizen Reminder: Accuracy matters more than speed. A wrong number in a data entry can cause incorrect calculations and bad decisions.

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Differentiation

Approaching: Enter 10 items using a simplified template with large cells and guided navigation
On Level: Enter 20 items using keyboard shortcuts with periodic accuracy checks
Advanced: Enter 30 items at competition speed with self-auditing and error correction
SEND: Enter 10 items with teacher guiding keyboard presses and explaining each shortcut

Formative Assessment

Method: Timed data entry challenge with speed and accuracy measurement
CT Skill Assessed: Algorithmic thinking: developing an efficient workflow sequence for accurate data entry

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Try writing all your classmates names as fast as you can. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.10 *Sorting and Filtering* 40 minutes **CONCEPT**

Success Criteria - I can...

- Sort spreadsheet data in ascending and descending order by one or more columns
- Apply filters to display only rows that meet specific criteria
- Use sorting and filtering to answer questions about a data set

Resources Needed:

- Marks spreadsheet with calculated averages
- Chart type comparison handout
- LibreOffice Calc chart tools
- Chart evaluation rubric
- Projector for presentations

Key Vocabulary

Sort: Arranging data in order like A to Z or smallest largest
Filter: Showing only rows that meet chosen condition and hiding others
Ascending: Order from smallest to largest or A to Z upward
Descending: Order from largest to smallest or Z to A downward

Engage - Retrieval & Hook (5 min)

TEACHER Show a marks sheet with 25 students in random order. Ask: Who scored the highest in English? Students struggle to find it by scanning. Then sort by English marks and ask again. Students instantly point to the top name.

STUDENTS Students see that sorting transforms a hard search into an instant answer. They want to learn how to sort data themselves.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give groups a marks sheet with 20 students. They must answer five questions: Who is first in Math? Who scored below 30 in Science? How many students scored above 80 in Kannada? Which subject has the highest class average? Who is ranked first overall?

STUDENTS Groups struggle with the unsorted data. Some start circling numbers, others write lists. After the teacher shows them sorting, they quickly answer all five questions.

Explain - Modelling & Concepts (10 min)

TEACHER Teach sorting: select the data range, go to Data, Sort, choose the column and ascending or descending order. Teach filtering: go to Data, AutoFilter, use dropdown arrows on headers to show only specific values (like all students with marks above 50). Demonstrate sorting by multiple columns: first by total marks descending, then by name ascending for ties.

STUDENTS Students practice sorting the marks sheet by different columns and observe how the order changes. They apply filters to show only students scoring above 75 in Math.

Elaborate - Independent Application (10 min)

TEACHER Students must sort the marks sheet by total marks to create a class ranking, then filter to show only students who need improvement (below 40 marks in any subject). They must explain which students need extra support and in which subjects.

STUDENTS Students sort and filter, then analyze the results. They identify struggling students and suggest which subjects need attention, connecting data analysis to real school decisions.

Evaluate - Check & Close (5 min)

TEACHER Give students a new data set (15 students, 4 subjects) and five questions that require sorting or filtering to answer. They must show which sort or filter they used for each question. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students answer questions and document their sorting/filtering steps. Teacher checks that they chose the correct operations for each question.

Cross-Curricular Connections

Mathematics: Math: Sorting involves comparing and ordering numbers, a fundamental math skill
Science: Social Science: Census data uses sorting and filtering to analyze population demographics
Language: Language: Writing explanations of data analysis practices clear technical communication

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Sort a simple list of 10 numbers in ascending order with button-by-button guidance
On Level: Sort and filter a marks sheet using written instructions with some independence
Advanced: Perform multi-column sorts and complex filter combinations to answer analytical questions
SEND: Apply a single sort with teacher clicking the menus while student names the operations

Formative Assessment

Method: Five-question data analysis challenge requiring appropriate sorting or filtering for each
CT Skill Assessed: Evaluation: selecting the most effective sort or filter operation to answer each data question

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you had to sort your dormitory friends by height, who would be first and who would be last?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.11 *Data Analysis*

40 minutes CONCEPT

Success Criteria - I can...

- Calculate basic statistics from spreadsheet data: mean, median, mode, minimum, and maximum
- Use spreadsheet functions to automate statistical calculations
- Interpret statistical results to draw conclusions about a data set

Resources Needed:

- Practice data sheets with 30 items
- Timer for speed tests
- Keyboard shortcut reference card
- Error checking guide
- Data entry competition certificates

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage - Retrieval & Hook (5 min)

TEACHER Show a class test score distribution on the board. Ask: Is this class doing well? How can we tell? What number best represents the whole class? Introduce the idea that different statistics answer different questions.

STUDENTS Students debate what well means. Some say average, others say most common score, others say the highest score. Teacher explains that different statistics give different perspectives.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Give groups a list of 15 test scores. Ask them to find the average by hand, the middle score, and the most common score. They must show all their working and explain what each number tells them about the class.

STUDENTS Groups calculate manually and discover that the average, median, and mode can give different values. They discuss what each reveals about the class performance.

Explain - Modelling & Concepts (10 min)

TEACHER Define and teach five statistical functions: AVERAGE (mean) adds all values and divides by count, MEDIAN finds the middle value when sorted, MODE finds the most frequent value, MIN finds the smallest, MAX finds the largest. Show how to use each function in LibreOffice Calc: =AVERAGE(B4:B28), =MEDIAN(B4:B28), =MODE(B4:B28), =MIN(B4:B28), =MAX(B4:B28).

STUDENTS Students practice using each function on the marks sheet. They record the results and compare them with their manual calculations from the explore activity.

Elaborate - Independent Application (10 min)

TEACHER Students must analyze the class marks data: calculate all five statistics for each subject, identify which subject has the highest average, which has the widest spread (difference between MIN and MAX), and which subject has the most consistent scores (smallest spread).

STUDENTS Students calculate statistics for all five subjects, create a comparison table, and write a paragraph explaining their findings. They identify patterns and anomalies in the data.

Evaluate - Check & Close (5 min)

TEACHER Students present their statistical analysis to the class. Each group must explain what their numbers mean in real terms: not just the average is 65, but the class is performing reasonably well with room for improvement. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups present their analysis with visual charts. Class asks questions about the findings. Teacher evaluates the depth of interpretation.

Cross-Curricular Connections

Mathematics: Math: Statistics functions are core mathematical tools for data analysis
Science: Science: Analyzing experiment results uses the same mean, median, and mode calculations
Language: Social Science: Census and survey data are analyzed using statistical methods

Digital Citizen Reminder: Statistics can be misleading. Always consider the full picture - average alone does not tell the whole story.

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Differentiation

Approaching: Calculate MIN and MAX from a small data set using the function wizard
On Level: Calculate all five statistics using functions and record results in a table
Advanced: Define a large data set with all statistics, create charts, and write an interpretation paragraph
SEND: Find MIN and MAX with teacher helping select the function and data range

Formative Assessment

Method: Statistical analysis report with calculations, comparison table, and written interpretation
CT Skill Assessed: Analysis: using multiple statistical measures to gain different perspectives on the same data set

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of the marks of five subjects you scored this term. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.12 *Data in Daily Life*

40 minutes **CONCEPT**

Success Criteria - I can…

- Recognize how data collection and analysis affect decisions in their daily lives
- Identify data-driven systems in their school, village, and state
- Create a data collection project that solves a real problem in their school community

Resources Needed:

- Marks sheet with 20 students
- Question cards requiring sort/filter operations
- Filtering practice worksheet
- Sorting steps reference card
- New data set for assessment

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage - Retrieval & Hook (5 min)

TEACHER Bring a bus timetable and ask: How does the school decide when the bus should leave? Show the attendance data that shows most students arrive by 8:15 AM. Ask: Was this decision based on data or guessing?

STUDENTS Students realize that the bus schedule was designed using arrival time data. They start noticing data-driven decisions around them.

Explore - Guided Discovery (10 min)

TEACHER Concrete (CPA): Use dried beans/grains and picture cards to physically sort and group data before moving to pictorial tables and abstract symbols. Take students on a data walk around the school: visit the attendance office (attendance data), the library (book borrowing data), the kitchen (meal count data), the garden (plant growth data), and the health room (height and weight data). Students note what data is collected and how it is used.

STUDENTS Students observe data collection in action at each stop. They interview staff about what decisions are made based on the data collected.

Explain - Modelling & Concepts (10 min)

TEACHER Discuss three real-world data systems: school attendance data helps the headmaster identify students who need support, agricultural data (rainfall, crop prices) helps farmers decide what to plant, and health data (weight, height) helps doctors monitor child growth. Emphasize that data collection, organization, analysis, and decision-making form a cycle that improves life.

STUDENTS Students connect each data system to their own lives. They discuss how their parents use agricultural data and how their own health records track their growth.

Elaborate - Independent Application (10 min)

TEACHER Students must identify one problem in their school (late arrivals, food waste, water usage, or library book losses) and design a data collection plan to address it. They must specify: what data to collect, how to collect it, how to organize it, what analysis to perform, and what decisions it would inform.

STUDENTS Students work in groups to design their data projects. They create data collection forms, plan their analysis approach, and present their proposals to the class.

Evaluate - Check & Close (5 min)

TEACHER Each group presents their data project proposal. The class votes on the most practical and impactful project. Teacher selects the best proposal to actually implement in the school. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups present with confidence, connecting their data project to real school needs. Class provides constructive feedback on feasibility and potential impact.

Cross-Curricular Connections

Mathematics: Social Science: Data collection and analysis are tools for understanding and improving communities
Science: Math: Real-world data applications make statistics relevant and meaningful
Language: Science: Scientific observation and data collection follow the same systematic process

Digital Citizen Reminder: Data about real people has real consequences. Always collect, store, and share data responsibly and ethically.

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Differentiation

Approaching: Identify one data collection example from the school data walk and describe its purpose
On Level: Design a simple data collection form for one school problem with teacher guidance
Advanced: Design a complete data project with collection plan, analysis method, and decision framework
SEND: Participate in the data walk and describe one observation with teacher support

Formative Assessment

Method: Data project proposal presentation with peer voting on practicality and impact
CT Skill Assessed: Application: using all data skills to design a real-world data collection and analysis project

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Count how many hours you sleep each night this week. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 5 - Programming

Lesson 5.01 *What is Programming?* 40 minutes DEBUG

Success Criteria - I can...

- Define programming and explain why computers need instructions
- Write a simple set of step-by-step instructions for a everyday task
- Identify how programming is used in devices around them

Resources Needed:
- Test score data sets
- Statistical function reference
- LibreOffice Calc functions guide
- Analysis template
- Comparison table template

Key Vocabulary

Programming: Writing instructions so computer performs tasks step by step
Code: Written instructions that computer can understand and follow exactly
Instruction: Single command telling computer to do one specific action
Bug: Mistake in code that makes program behave incorrectly or stop

Engage - Retrieval & Hook (5 min)

TEACHER Ask a student volunteer to follow instructions to make a cup of chai, but give them instructions one word at a time: Boil. Water. Add. Tea. Leaves. Ask: Was that easy to follow? What was missing? Programming is the same - the instructions must be clear and complete.

STUDENTS Students laugh at the vague instructions and suggest better ones. They realize that computers need just as much clarity and detail as a confused chai maker.

Explore - Guided Discovery (10 min)

TEACHER Give each pair a Blind Drawing Challenge: one student has a simple picture (house, flower, star) and must describe it step by step so their partner can draw it without seeing the original. Compare the drawings.

STUDENTS Students discover that vague instructions produce wildly different drawings. They learn that precise language and ordered steps are essential for accurate results.

Explain - Modelling & Concepts (10 min)

TEACHER Define programming as writing instructions that a computer can follow to complete a task. Explain that computers are very fast but not smart - they do exactly what you tell them, nothing more and nothing less. Give our school examples: the school bell timer runs a program, the attendance system uses a program, and Tux Paint follows your program commands.

STUDENTS Students list five programs they use daily: Tux Paint, LibreOffice, the school management system, games on phones, and YouTube. They discuss what instructions each program must be following.

Elaborate - Independent Application (10 min)

TEACHER Students must write a complete program (set of instructions) for a classmate to follow that will result in drawing a specific shape on the board. The program must have at least 8 steps, use only simple commands, and be tested by a partner. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students write their instruction sets, then swap with a partner who follows them literally. When the drawing does not match, they identify which instructions were unclear and revise them.

Evaluate - Check & Close (5 min)

TEACHER Each pair presents their best instruction set and the drawing it produced. Class evaluates which instructions were clearest and most effective. Teacher connects the activity to real programming concepts. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Pairs present their instruction sets and drawings. Class discusses which instructions worked well and which needed improvement.

Cross-Curricular Connections

Mathematics: Math: Following step-by-step instructions connects to mathematical procedures and algorithms
Science: English: Writing clear instructions practices imperative writing and precise language
Language: Social Science: Many daily tasks follow programmed sequences - cooking recipes, farming steps, cleaning routines

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Write 5-step instructions for a simple task like folding a paper boat
On Level: Write 8-step instructions for a drawing task with clear and precise language
Advanced: Write a 12-step program that includes conditional instructions
SEND: Write 5 simple instructions with teacher helping clarify each step

Formative Assessment

Method: Instruction set evaluated by partner who tests it and rates clarity and completeness
CT Skill Assessed: Decomposition: breaking a complex task into clear, ordered, step-by-step instructions

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write instructions for making your bed perfectly. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.02 *Scratch Interface* 40 minutes **DEBUG**

Success Criteria - I can...

- Navigate the Scratch interface and identify the stage, sprite panel, block palette, and script area
- Create a new project and add a sprite to the stage
- Run a simple script by connecting motion blocks

Resources Needed:
- School data collection forms
- Bus timetable and attendance records
- Library borrowing logs
- Kitchen meal count sheets
- Project proposal templates

Key Vocabulary

Scratch: Visual programming tool using blocks to create stories and games
Stage: Area where your characters move and show their actions live
Sprite: Character or object that moves and acts on the stage
Block: Colourful puzzle piece representing one programming instruction visually

Engage - Retrieval & Hook (5 min)

TEACHER Open a finished Scratch project: a cat that dances to music across the stage. Ask: How do you think this was made? What parts do you see moving? Can we build something like this today?

STUDENTS Students are excited by the dancing cat. They want to make their own characters move and are eager to explore the Scratch interface.

Explore - Guided Discovery (10 min)

TEACHER Open Scratch on each computer and give students 5 minutes to click on everything: the stage area, the sprite list, the block categories, and the script area. Ask them to note what each area does.

STUDENTS Students click wildly, discovering that clicking a motion block makes the cat move. They call out to friends about exciting discoveries and help each other find the green flag.

Explain - Modelling & Concepts (10 min)

TEACHER Walk through the Scratch interface: the stage is where the action happens (480 by 360 pixels), sprites are characters on the stage, the block palette on the left has categories (Motion, Looks, Sound, Events, Control), and the script area in the center is where you build programs by snapping blocks together.

STUDENTS Students follow along, exploring each area. They learn to switch between block categories and drag three blocks to make the cat move: move 10 steps, turn 15 degrees, and say Hello for 2 seconds.

Elaborate - Independent Application (10 min)

TEACHER Students must create a project with the cat sprite that moves forward, turns, and says their name when the green flag is clicked. They must also change the stage background to a school setting. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students experiment with different block combinations to achieve their desired effect. They change backgrounds and test their scripts multiple times.

Evaluate - Check & Close (5 min)

TEACHER Students demonstrate their projects to a partner. The partner must identify which blocks were used and what each one does. Teacher circulates and checks for understanding. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students explain their blocks to partners and answer questions about what each block does. Teacher notes which students can confidently navigate the interface.

Cross-Curricular Connections

Mathematics: Math: The stage uses a coordinate system (x from -240 to 240, y from -180 to 180) connecting to number lines
Science: Art: Choosing sprites and backgrounds involves creative design decisions
Language: English: Naming sprites and adding dialogue practices descriptive language

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Follow a step-by-step screenshot guide to drag three blocks and run them
On Level: Create a simple script using Motion and Looks blocks with written instructions
Advanced: Design a project with multiple sprites, backgrounds, and custom block combinations
SEND: Drag blocks with teacher help, seeing what each one does as it runs

Formative Assessment

Method: Project demonstration with partner identification of blocks and their functions
CT Skill Assessed: Abstraction: understanding how the Scratch interface organizes programming concepts into visual blocks

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could make any character dance on screen, what character would you choose and how would it move?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.03 *Motion Blocks*

40 minutes **DEBUG**

Success Criteria - I can...

- Use motion blocks to move sprites in specific directions and distances
- Apply glide and turn blocks to create smooth movement paths
- Combine multiple motion blocks to navigate a sprite through a maze

Resources Needed:

- Blind drawing pictures
- Paper and pencils for instruction sets
- Sample simple programs
- Devices showing programmed instructions

Key Vocabulary

Motion: Movement of sprite from one place to another on stage
Steps: Number of small moves sprite takes in chosen direction forward
Direction: Angle showing which way sprite will face and move toward
Coordinate: Pair of numbers showing exact position on stage like x y

Engage - Retrieval & Hook (5 min)

TEACHER Place a toy car on a grid drawn on the board. Ask: How do I tell the car to move 3 squares right, then 2 squares up? What happens if I say move 3 right but forget to turn? Show that direction and distance both matter.

STUDENTS Students direct the car and see that forgetting direction or distance produces wrong results. They understand that motion requires both how far and which way.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch maze challenge: a maze drawn on the stage with a sprite at the start and a star at the end. They must use only motion blocks to navigate the sprite through the maze.

STUDENTS Students experiment with move, turn, and glide blocks to navigate the maze. They discover that small adjustments to step size and rotation angles are needed to stay on the path.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the key motion blocks: move 10 steps, turn 15 degrees, go to x y, glide 1 secs to x y, point in direction, set x to and change y by. Show how to combine these to create complex paths.

STUDENTS Students practice each motion block individually, observing what happens on the stage. They record the effect of each block in their journals.

Elaborate - Independent Application (10 min)

TEACHER Students must make their sprite trace a square using exactly four move blocks and four turn blocks. They must then modify it to trace a triangle and a hexagon, adjusting the turn angles each time. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students calculate the correct turn angles: 90 degrees for a square, 120 degrees for a triangle, 60 degrees for a hexagon. They test and refine their scripts until the shapes are accurate.

Evaluate - Check & Close (5 min)

TEACHER Students display their shape-tracing scripts. Teacher verifies that each shape is drawn correctly. Students explain how they calculated the turn angles. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their scripts and explain their angle calculations. Class discusses the pattern: the turn angle times the number of sides equals 360 degrees.

Cross-Curricular Connections

Mathematics: Math: Shape tracing connects to geometry, angles, and the rule that exterior angles sum to 360 degrees
Science: Physics: Motion in different directions relates to vectors and displacement
Language: Geography: Coordinates on the stage connect to latitude and longitude on maps

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Move a sprite forward and backward using only the move block with step size changes
On Level: Trace a square using move and turn blocks following a step-by-step guide
Advanced: Trace multiple shapes calculating angles independently and explaining the 360-degree rule
SEND: Move a sprite forward with teacher help adjusting the step size

Formative Assessment

Method: Shape tracing scripts verified for accuracy with explanation of angle calculations
CT Skill Assessed: Algorithmic thinking: planning a precise sequence of motion commands to trace geometric shapes

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Walk from your bed to the door. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.04 *Looks and Sound* 40 minutes DEBUG

Success Criteria - I can...

- Use Looks blocks to change sprite costumes, size, and display speech bubbles
- Use Sound blocks to play sounds and change volume
- Create a short animation combining looks and sound blocks

Resources Needed:
- Computers with Scratch installed or web access
- Sample Scratch projects
- Scratch interface guide
- Sprite library
- Background image collection

Key Vocabulary

Costume: Different appearance of sprite that you can switch quickly for animation
Backdrop: Picture behind sprites that sets scene like classroom or forest
Sound: Music or voice played when program runs for effect
Effect: Change like ghost or colour that alters how sprite looks

Engage - Retrieval & Hook (5 min)

TEACHER Show a Scratch project where the cat says something, changes costume to look surprised, and plays a sound effect. Ask: How many different things happened at once? How do you make a character express emotions on screen?

STUDENTS Students notice the combination of text, costume change, and sound. They want to learn how to make their sprites express feelings and reactions.

Explore - Guided Discovery (10 min)

TEACHER Give students a sprite with multiple costumes. Ask them to try every Looks block: say, think, show, hide, switch costume, next costume, change size. Note what each one does.

STUDENTS Students cycle through costumes, making the sprite grow and shrink, appear and disappear. They discover the say block creates speech bubbles and the think block creates thought bubbles.

Explain - Modelling & Concepts (10 min)

TEACHER Teach Looks blocks: say Hello for 2 seconds, think Hmm for 2 seconds, switch costume to, next costume, change size by 10, show and hide. Teach Sound blocks: play sound until done, start sound, change volume by, set volume to. Show how to combine them for expressive animation.

STUDENTS Students create a simple animation: sprite says Good morning, switches to a smiling costume, and plays a cheerful sound. They practice combining looks and sound for emotional effect.

Elaborate - Independent Application (10 min)

TEACHER Students must create a 10-second animated scene: a sprite enters from the left, says a greeting, changes costume to show an emotion, plays a sound effect, and exits to the right. The animation must include at least three different looks blocks and two sound blocks. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students plan their animation on paper first, then build it in Scratch. They test and adjust timing, costume changes, and sound effects until the animation flows smoothly.

Evaluate - Check & Close (5 min)

TEACHER Students show their animations to the class. Class evaluates: Does the animation tell a clear story? Are looks and sound well synchronized? Is the timing appropriate? Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their animations and answer questions about their design choices. Class provides specific feedback on the effectiveness of the looks and sound combination.

Cross-Curricular Connections

Mathematics: English: Creating animated dialogues practices creative writing and character development
Science: Music: Choosing sound effects and understanding volume connects to audio production
Language: Art: Costume design and visual expression connect to character art and storytelling

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Make a sprite say Hello and change size using two Looks blocks
On Level: Create a short animation following a storyboard with specific looks and sound blocks
Advanced: Design a 15-second story with multiple costume changes, sound effects, and synchronized timing
SEND: Add a speech bubble and one costume change with teacher selecting the blocks

Formative Assessment

Method: Animated scene evaluated on story clarity, looks-sound synchronization, and timing
CT Skill Assessed: Pattern recognition: identifying how combining visual and audio blocks creates expressive digital storytelling

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If your pillow could talk, what would it say? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.05 *Events and Control*

40 minutes **DEBUG**

Success Criteria - I can…

- Use event blocks to trigger scripts when specific actions occur
- Apply control blocks like repeat, wait, and if-then to manage program flow
- Create a program that responds to keyboard input and runs multiple scripts simultaneously

Resources Needed:
- Scratch with maze backgrounds
- Grid drawing on board
- Toy car and grid
- Motion blocks reference card
- Shape angle calculation worksheet

Key Vocabulary

Event: Action like clicking flag that starts part of program running
Control: Block that decides order or repetition of steps in program
Trigger: Signal that causes sprite to begin doing something immediately afterwards
Loop: Repeating same steps many times without writing them again

Engage - Retrieval & Hook (5 min)

TEACHER Play a game of Simon Says with the class. Ask: What is the rule that starts every action? In Scratch, every script needs a starting rule too. What happens if Simon does not say Simon Says?

STUDENTS Students understand that actions need a trigger to start. They connect the Simon Says rule to event blocks that start Scratch scripts.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch project with four scripts, each triggered by a different event: green flag clicked, space key pressed, sprite clicked, and when this sprite switches costume. They must trigger each event and observe what happens.

STUDENTS Students click the green flag, press space, click the sprite, and watch costume-triggered scripts. They discover that events are the triggers that start programs.

Explain - Modelling & Concepts (10 min)

TEACHER Teach event blocks: when green flag clicked, when space key pressed, when this sprite clicked, when backdrop switches. Teach control blocks: wait 1 secs, repeat 10, forever, if-then. Show how events and control blocks work together to create interactive programs.

STUDENTS Students build a script that: when space is pressed, the sprite moves, waits 1 second, says something, and repeats the sequence three times.

Elaborate - Independent Application (10 min)

TEACHER Students must create an interactive project: pressing the up arrow makes the sprite jump, pressing the down arrow makes it duck, pressing left makes it move left, and pressing right makes it move right. Each action must use a different event block and include at least one wait block. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build four separate scripts, each with its own event trigger. They test all four directions and adjust the wait times so the movement feels natural.

Evaluate - Check & Close (5 min)

TEACHER Students demonstrate their interactive projects. Teacher tests each keyboard trigger and evaluates: Does each key produce the correct action? Are the movements smooth? Is the code organized? Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their projects and explain how each event triggers a different action. Teacher and peers test the projects and provide feedback.

Cross-Curricular Connections

Mathematics: Math: If-then conditions connect to logical propositions and mathematical inequalities
Science: Science: Event-driven programming models real-world cause and effect relationships
Language: Games: Understanding how game controls work connects to game design principles

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Use only the green flag event to trigger a simple animation sequence
On Level: Create two scripts with different event triggers and one control block
Advanced: Build a four-direction interactive project with events, waits, and loops
SEND: Trigger one script with green flag and one key press with teacher guiding block placement

Formative Assessment

Method: Interactive project demonstration with teacher testing all event triggers and control flow
CT Skill Assessed: Algorithmic thinking: designing multiple event-driven scripts that work together in one project

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of your morning routine. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.06 *Variables*

40 minutes **DEBUG**

Success Criteria - I can...

- Define variables as containers that store and change values during a program
- Create and use variables to track score, count, or position in a Scratch project
- Apply variables to build a simple score-tracking system

Resources Needed:
- Scratch with costume editor
- Sound library in Scratch
- Animation planning worksheet
- Character emotion cards
- Timer for animation length

Key Vocabulary

Variable: Named box storing a value that can change during program
Value: Number or word stored inside a variable for later use
Assign: Putting a value into a variable using equals sign
Update: Changing value already stored inside a variable to new value

Engage - Retrieval & Hook (5 min)

TEACHER Give each student an empty box and ask them to put a number on a sticky note inside. Then ask them to change the number. Ask: What stayed the same (the box) and what changed (the number inside)? Variables are like labeled boxes that hold changing information.

STUDENTS Students understand that the box is the variable (it has a name and holds something) and the number is the value (it can change). This concrete analogy makes the abstract concept tangible.

Explore - Guided Discovery (10 min)

TEACHER Open Scratch and create a variable called Score. Show students that they can set it to 0, change it by 1, and display it on the stage. Ask them to create a variable called Counter and make it count from 0 to 10 when the green flag is clicked.

STUDENTS Students create variables and watch them appear on the stage. They experiment with set, change, and show/hide blocks. They make the counter increase rapidly and see the number climb.

Explain - Modelling & Concepts (10 min)

TEACHER Define a variable as a named container that stores a value that can change during the program. In Scratch, create variables with the Variables palette: Make a Variable, name it, and choose for all sprites or this sprite only. Use set variable to value to initialize, change variable by to update, and the variable reporter block to read the value.

STUDENTS Students create three variables (Score, Lives, Level) and build scripts that demonstrate setting, changing, and reading each one. They display all three on the stage.

Elaborate - Independent Application (10 min)

TEACHER Students must create a click-counting game: a sprite appears on the stage and the player must click it as many times as possible in 10 seconds. A Score variable counts each click, and a Timer variable counts down from 10 to 0. When the timer reaches 0, the game stops and displays the final score. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build the timer using a forever loop with wait and change blocks, and the score using a when this sprite clicked event. They test and adjust the timing.

Evaluate - Check & Close (5 min)

TEACHER Students play each other games and report the highest scores. Teacher verifies that the score variable increments correctly and the timer counts down accurately in each project. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students test peer projects, checking score accuracy and timer functionality. They report any bugs they find and suggest fixes.

Cross-Curricular Connections

Mathematics: Math: Variables connect to algebraic thinking where letters represent unknown or changing values
Science: Science: Experiments use variables (independent, dependent, controlled) in the same way
Language: Economics: Money in a wallet works like a variable - it has a name (balance) and changes value

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Create one variable and use set to assign it a value, displaying it on stage
On Level: Create a counter variable that increases with a key press and decreases with another
Advanced: Build a complete click game with score, timer, and final result display
SEND: Create a variable with teacher help and watch it display on the stage

Formative Assessment

Method: Click-counting game verified for correct score tracking and accurate timer countdown
CT Skill Assessed: Abstraction: representing a changing quantity (score) with a named variable that the program can read and modify

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: How many steps do you walk from the dormitory to the classroom? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.07 *Conditions*

40 minutes **DEBUG**

Success Criteria - I can...

- Explain what a condition is and how it controls program behavior
- Use if-then and if-then-else blocks to make decisions in Scratch programs
- Create a quiz program that gives different responses based on user answers

Resources Needed:

- Scratch with keyboard event examples
- Simon Says game setup
- Interactive project templates
- Event blocks reference card
- Control flow diagram

Key Vocabulary

Condition: Question with true or false answer deciding next step
If: Block that runs code only when condition is true otherwise
Else: Alternative path taken when if condition is false instead
Boolean: Value that is either true or false for decision making

Engage - Retrieval & Hook (5 min)

TEACHER Play a guessing game: I am thinking of a number between 1 and 10. Students guess and the teacher responds Higher or Lower. Ask: What am I doing with each guess? I am checking a condition and deciding what to say next.

STUDENTS Students recognize that the teacher is making decisions based on whether the guess is correct, too high, or too low. This is exactly what if-then-else does in programming.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch project with an if-then block that checks: is the score greater than 10? They must change the condition and observe what happens in different scenarios.

STUDENTS Students modify conditions and test with different score values. They discover that the block checks the condition and only runs the code inside if it is true.

Explain - Modelling & Concepts (10 min)

TEACHER Define a condition as a statement that is either true or false, and the program decides what to do based on the result. Teach if-then: runs the code inside only if the condition is true. Teach if-then-else: runs one set of code if true, another if false. Show comparison operators: =, , , and, or.

STUDENTS Students create a temperature classifier and test it with different values. They write three new conditions in their notebooks and predict what the program would do for each.

Elaborate - Independent Application (10 min)

TEACHER Students must create a math quiz: the program displays a question, the player types an answer, and the program checks if it is correct. If correct, display Correct and add 1 to score. If wrong, display Try Again. The quiz must have at least three questions. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build the quiz using ask-and-wait blocks for input, if-then-else for checking answers, and variables for score tracking. They test all three questions thoroughly.

Evaluate - Check & Close (5 min)

TEACHER Students play each other quizzes and verify: Does it accept correct answers? Does it reject wrong answers? Does the score track accurately? Does it handle all three questions? Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students test peer projects using a testing checklist. They identify any conditions that do not work correctly and suggest fixes.

Cross-Curricular Connections

Mathematics: Math: Conditions use comparison operators that connect to mathematical inequalities
Science: Science: Hypothesis testing follows if-then logic: if the hypothesis is correct, then the prediction should happen
Language: Daily Life: We make if-then decisions constantly: if it rains, take umbrella; if hungry, eat food

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Use a single if-then block to check one condition with a simple true/false test
On Level: Build an if-then-else quiz with one question and correct/wrong feedback

Advanced: Create a three-question quiz with score tracking and varied feedback for each answer
SEND: Test a pre-built if-then block by changing the condition value with teacher guidance

Formative Assessment

Method: Math quiz project verified for correct condition checking across all questions
CT Skill Assessed: Algorithmic thinking: designing condition logic that correctly evaluates user input and produces appropriate responses

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three if-then rules for your dormitory: If it is lights out time, then stop talking. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.08 *Loops*

40 minutes **DEBUG**

Success Criteria - I can...

- Explain why loops are essential for repeating actions efficiently in programs
- Use repeat-until and forever loops to control repetition in Scratch
- Replace multiple repeated blocks with a single loop to simplify code

Resources Needed:
- Scratch with Variables palette
- Empty boxes and sticky notes for analogy
- Variable creation guide
- Click game template
- Score tracking examples

Key Vocabulary

Loop: Command that repeats steps without writing them again multiple times
Repeat: Doing same action fixed number of times automatically for efficiency
Infinite Loop: Endless repetition that never stops without break condition needed
Counter: Variable that counts how many times loop has repeated so far

Engage - Retrieval & Hook (5 min)

TEACHER Ask students to clap 20 times. Then ask: Was that boring after a while? What if you had to write a Scratch script with 20 move blocks? Is there a better way? Show the repeat block doing the same thing with just 2 blocks instead of 20.

STUDENTS Students are relieved to discover loops. They realize that repetitive tasks are perfect for loops and that loops save enormous amounts of effort.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch script with 10 say blocks that make the cat count from 1 to 10. Ask them to simplify it using as few blocks as possible. They discover the repeat block can replace all 10 say blocks.

STUDENTS Students try different approaches: some use repeat 10 with a variable, others experiment with forever. They compare the number of blocks in their solutions.

Explain - Modelling & Concepts (10 min)

TEACHER Teach three loop types: repeat (runs a set number of times), repeat-until (runs until a condition becomes true), and forever (runs continuously). Show how to use a variable counter inside a repeat loop to count. Demonstrate the difference between repeat (finite) and forever (infinite).

STUDENTS Students build a counting loop that counts from 1 to 20, then modify it to count backwards from 20 to 1. They experiment with forever loops and learn how to stop them.

Elaborate - Independent Application (10 min)

TEACHER Students must create a dance animation: the sprite repeats a sequence of moves 8 times to complete a full circle. They must then modify it to use repeat-until so the dance stops when the sprite returns to its starting position. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build the dance loop, testing it multiple times. They discover that repeat-until requires a condition and adjust until the dance completes perfectly.

Evaluate - Check & Close (5 min)

TEACHER Students compare two versions of the same program: one with repeated blocks and one with a loop. Class evaluates which is easier to read, modify, and debug. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present both versions and explain why the loop version is better. Class discusses when loops are useful and when they are not appropriate.

Cross-Curricular Connections

Mathematics: Math: Loops connect to multiplication as repeated addition and patterns in sequences
Science: Music: Musical scales and rhythms are loops of notes that repeat in patterns
Language: Agriculture: Planting and harvesting follow seasonal loops that repeat each year

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Replace 5 repeated blocks with a repeat loop following a step-by-step guide
On Level: Create a counting loop that counts from 1 to 20 using a variable
Advanced: Design a complex animation using nested loops and repeat-until conditions
SEND: Change a repeat count from 10 to 5 with teacher explaining the loop mechanics

Formative Assessment

Method: Loop comparison showing simplified code with explanation of why loops are more efficient
CT Skill Assessed: Decomposition: identifying repeated patterns in code and replacing them with a single loop structure

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: What do you do every morning in the same order? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.09 *Drawing with Code*

40 minutes **DEBUG**

Success Criteria - I can…

- Use the pen extension to draw shapes and patterns programmatically
- Control pen color, size, and position to create artistic designs
- Apply loops and pen blocks to generate geometric patterns automatically

Resources Needed:
- Scratch with sensing blocks
- Guessing game setup
- Math quiz worksheet
- Condition examples from daily life
- Comparison operator reference card

Key Vocabulary

Pencil: Thin tool for drawing light outlines and sketches precisely
Brush: Thick tool for painting broad strokes with colour smoothly
Shape Tool: Quick way to draw perfect circles squares and lines
Colour Picker: Tool that copies any colour already on your canvas

Engage - Retrieval & Hook (5 min)

TEACHER Show a complex geometric pattern drawn entirely by Scratch pen blocks. Ask: How long do you think this took to program? Reveal that it was created with just 5 blocks and a loop. Ask: What pattern do you see that could be repeated?

STUDENTS Students study the pattern and identify repeating elements. They are amazed that complex art can come from simple repeated instructions.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch project with the pen extension enabled. Ask them to try every pen block: pen down, pen up, set pen color, set pen size, clear, and stamp. They must create at least three different marks on the stage.

STUDENTS Students experiment with pen blocks, drawing lines, changing colors, and stamping shapes. They discover that pen down followed by movement draws lines.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the pen extension blocks: pen down (starts drawing), pen up (stops drawing), set pen color to, set pen size to, clear, stamp. Show how to combine pen blocks with motion and loops: to draw a square, use pen down, repeat 4 times (move 100, turn 90), pen up.

STUDENTS Students build a square drawing script, then modify it to draw a triangle, a pentagon, and a star. They record which turn angles produce which shapes.

Elaborate - Independent Application (10 min)

TEACHER Students must create a spirograph pattern: use a forever loop to move forward, turn 1 degree, and change pen color. The sprite will draw thousands of tiny lines that form a complex spiral pattern. They must adjust the turn angle and step size to create different designs. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students experiment with different turn angles and step sizes, discovering that small changes produce dramatically different patterns. They save their favorite designs.

Evaluate - Check & Close (5 min)

TEACHER Students display their pen drawings on the projector. Class identifies which shapes used which turn angles. Students explain how changing one block parameter affects the entire pattern. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their designs and explain the code behind each one. Class discusses the relationship between code parameters and visual output.

Cross-Curricular Connections

Mathematics: Math: Pen drawing connects to geometry, angles, symmetry, and coordinate systems
Science: Art: Programmed art (generative art) is a growing field that combines creativity with code
Language: Science: Fractal patterns in nature can be generated using recursive pen drawings

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Draw a simple line using pen down, move, pen up following step-by-step instructions
On Level: Draw a square using pen blocks and a repeat loop with guided angle calculation
Advanced: Create complex spirograph patterns with independent turn angle and color experiments
SEND: Draw a single shape with teacher helping with pen blocks and turn angles

Formative Assessment

Method: Pen drawing evaluated for geometric accuracy, color use, and explanation of code-pattern relationship
CT Skill Assessed: Pattern recognition: understanding how code parameters (angle, step, color) create visual patterns

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a star on paper. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.10 *Storytelling in Scratch*

40 minutes DEBUG

Success Criteria - I can...

- Plan a digital story with characters, setting, plot, and dialogue
- Use Scratch blocks to create an interactive narrative with multiple scenes
- Combine motion, looks, sound, and event blocks to bring a story to life

Resources Needed:
- Scratch with loop blocks
- Sample repeated-block script
- Dance animation template
- Loop comparison worksheet
- Pattern cards for loop inspiration

Key Vocabulary

Story: Sequence of events told with words pictures and sound together
Narrative: Way you organise story with beginning middle and end
Scene: Single part of story showing one moment or place
Voiceover: Recorded speech added over pictures to explain story clearly

Engage - Retrieval & Hook (5 min)

TEACHER Read a short folk tale aloud, then show a Scratch version of the same tale with animated characters, speech bubbles, and sound effects. Ask: Which version was more engaging? What did the Scratch version add that the reading did not have?

STUDENTS Students compare the two versions and identify that animation, voice, and interaction made the Scratch story more immersive and memorable.

Explore - Guided Discovery (10 min)

TEACHER Give students three story cards with different folk tales from Karnataka. They must pick one and identify: the main character, the setting, three key events, and the ending. Then plan how each element could be shown in Scratch.

STUDENTS Students discuss which tales work best for Scratch and sketch rough storyboards showing which sprites and backgrounds would be needed for each scene.

Explain - Modelling & Concepts (10 min)

TEACHER Teach the storytelling process in Scratch: create a storyboard with scenes, assign sprites to characters, write dialogue using say blocks, add backdrops for each scene, use events to transition between scenes, and add sound effects for atmosphere. Show how to use the backdrop switching event to trigger scene changes automatically.

STUDENTS Students create a detailed storyboard with at least four scenes, each with a backdrop, character sprites, dialogue, and a transition trigger.

Elaborate - Independent Application (10 min)

TEACHER Students must create a four-scene Scratch story with: at least two character sprites, four different backdrops, speech bubbles for dialogue, at least one sound effect, and automatic scene transitions. The story must have a clear beginning, middle, and end. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build their stories scene by scene, testing each transition before moving to the next. They coordinate multiple sprites and backdrops to create a flowing narrative.

Evaluate - Check & Close (5 min)

TEACHER Each group performs their Scratch story for the class. Audience evaluates: story clarity, character movement, dialogue quality, scene transitions, and overall engagement. Teacher provides detailed feedback.

Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Groups present their stories and answer questions about their creative choices. Class votes on the most engaging and well-constructed story.

Cross-Curricular Connections

Mathematics: English: Digital storytelling integrates creative writing with visual and auditory presentation
Science: Social Science: Karnataka folk tales connect to local culture and heritage preservation
Language: Drama: Character animation and dialogue delivery connect to theatrical performance skills

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Follow a provided storyboard to build a two-scene story with pre-made sprites
On Level: Create a four-scene story using the planning template with moderate independence
Advanced: Design an interactive story where the viewer chooses what happens next using conditions
SEND: Build one scene of a story with teacher helping place sprites and write dialogue

Formative Assessment

Method: Story performance with audience evaluation rubric and creative choice reflection
CT Skill Assessed: Abstraction: planning a narrative structure and translating it into sequential Scratch commands

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Tell your dormitory friend a story about your village. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.11 *Game Design Basics* 40 minutes DEBUG

Success Criteria - I can...

- Identify the key elements of a game: objective, rules, obstacles, and feedback
- Create a simple Scratch game with a player-controlled sprite and a goal
- Apply score tracking, timing, and win conditions to a complete game

Resources Needed:

- Scratch with pen extension
- Geometric pattern samples
- Pen blocks reference card
- Pattern inspiration gallery
- Calculator for angle measurements

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage - Retrieval & Hook (5 min)

TEACHER Play a simple game of catching a ball thrown in the air. Ask: What is the goal? What are the rules? What happens when you succeed? What happens when you fail? Map each answer to a game design concept.

STUDENTS Students identify the game elements naturally: the goal is to catch, the rule is to use hands, success means the ball does not hit the ground, and failure means it drops. They see how these map to programming concepts.

Explore - Guided Discovery (10 min)

TEACHER Give students three Scratch games to play: a catching game, a maze game, and a quiz game. After playing each, they must identify: the objective, the player controls, the obstacles, the scoring system, and the win condition.

STUDENTS Students play enthusiastically and fill in a game analysis worksheet. They discover that all games share common elements even though they look different.

Explain - Modelling & Concepts (10 min)

TEACHER Teach game design elements: the objective tells the player what to achieve, rules define what actions are allowed, obstacles create challenge, feedback tells the player how they are doing (score, sounds, messages), and the win condition ends the game. Show how to build each element in Scratch: objective using a say block at the start, rules using if-then conditions, obstacles using sprites that move randomly, feedback using score variables and sound effects.

STUDENTS Students plan their own game by filling in a design document: game name, objective, player controls, obstacles, scoring, and win condition.

Elaborate - Independent Application (10 min)

TEACHER Students must create a complete game with: a player sprite controlled by arrow keys, at least one obstacle sprite that moves, a goal sprite to collect, a score variable that increases on collection, and a win message when the score reaches 5. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students build their games step by step: first the player movement, then the obstacle, then the goal collection, then the score, and finally the win condition. They test and balance the difficulty.

Evaluate - Check & Close (5 min)

TEACHER Students play each other games and complete a game review form rating: fun factor, difficulty, clarity of objective, and quality of feedback. Teacher evaluates the technical implementation. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students play peer games and provide constructive feedback. They identify what makes each game fun and suggest improvements.

Cross-Curricular Connections

Mathematics: Math: Score systems and win conditions use numerical comparison and counting
Science: Physics: Game physics (gravity, bouncing, speed) connects to real-world motion concepts
Language: Psychology: Game design involves understanding what motivates players and creates engagement

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Modify a pre-made game by changing the score target and obstacle speed
On Level: Create a simple catching game following a step-by-step tutorial
Advanced: Design an original game with multiple obstacles, power-ups, and a scoring system
SEND: Build player movement with teacher help and test a pre-made game collection system

Formative Assessment

Method: Complete game played by peers with game review form ratings and teacher technical evaluation
CT Skill Assessed: Application: combining all programming skills (motion, events, variables, conditions, loops) into a functional game

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of a game you play outside. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.12 *Debugging* 40 minutes **DEBUG**

Success Criteria - I can...

- Define debugging and explain why finding and fixing errors is a normal part of programming
- Identify common programming errors: syntax, logic, and runtime errors
- Use systematic testing strategies to find and fix bugs in Scratch projects

Resources Needed:
- Folk tale story cards
- Storyboard planning templates
- Scratch with backdrop library
- Sound effects collection
- Story evaluation rubric

Key Vocabulary

Bug: Error that makes program give wrong result or stop unexpectedly
Debug: Process of finding and fixing errors in code step by step
Test: Running program to check if it works as expected correctly
Trace: Following code line by line to find where mistake happens

Engage - Retrieval & Hook (5 min)

TEACHER Show a Scratch project that is supposed to make the cat dance but instead it goes off the stage and disappears. Ask: What went wrong? How would you figure out the problem? Reveal that finding and fixing bugs is called debugging.

STUDENTS Students laugh at the broken project and start suggesting what might be wrong. They are motivated to learn how to find and fix problems systematically.

Explore - Guided Discovery (10 min)

TEACHER Give students a Scratch project with three hidden bugs: the sprite moves too fast, the score never increases, and the game never ends. They must play the project, find all three bugs, and describe what is wrong with each one.

STUDENTS Students play the buggy project and write down each problem they notice. They discuss with partners what might be causing each bug and suggest possible fixes.

Explain - Modelling & Concepts (10 min)

TEACHER Define debugging as the process of finding and fixing errors in a program. Categorize three types of errors: syntax errors prevent the program from running (wrong block shape), logic errors make the program run but produce wrong results (score never increases because the variable is never changed), and runtime errors cause the program to crash or behave unexpectedly (sprite goes off stage). Teach debugging strategies: read the code carefully, add say blocks to show variable values, test one section at a time, and use the step-through feature.

STUDENTS Students practice debugging by adding say blocks to display variable values in a project. They discover that seeing the actual values helps identify logic errors.

Elaborate - Independent Application (10 min)

TEACHER Students receive a project with five bugs of different types. They must use a debugging checklist to find and fix each one: identify the symptom, read the relevant code, hypothesize the cause, test the fix, and verify the solution works. Modify: change one parameter/block and predict outcome before running.

STUDENTS Students work through their debugging checklist systematically. They document each bug they find: what was the symptom, what was the cause, and how they fixed it.

Evaluate - Check & Close (5 min)

TEACHER Students swap their fixed projects with a partner. The partner must verify that all five bugs are fixed and rate the debugging documentation. Teacher reviews the debugging process and the quality of fixes. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their debugging process to the class. Class discusses which debugging strategies were most effective and why.

Cross-Curricular Connections

Mathematics: Science: Debugging follows the scientific method: observe, hypothesize, experiment, and conclude
Science: Math: Finding errors in calculations follows a similar logical debugging process
Language: English: Writing clear bug reports practices precise technical communication

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Find and fix one obvious bug using a hint card that tells them where to look
On Level: Find and fix three bugs using a systematic checklist with some guidance
Advanced: Find and fix five bugs independently and write a detailed bug report for each
SEND: Identify one bug with teacher help and watch the debugging process

Formative Assessment

Method: Debugging documentation reviewed for systematic process and correct bug fixes
CT Skill Assessed: Evaluation: systematically testing, analyzing, and fixing errors in a program using logical reasoning

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of something that went wrong today. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 6 - Software Applications

Lesson 6.01 *What is Software?* 40 minutes DEBUG

Success Criteria - I can...

- Define software and distinguish it from hardware
- Identify different types of software: system software and application software
- Give examples of software they use daily and explain what each one does

Resources Needed:
- Sample Scratch games for analysis
- Game design document template
- Game review forms
- Scratch game tutorials
- Score and win condition examples

Key Vocabulary

Software: Set of programs that tells hardware what to do each time
Application: Program designed for specific task like writing or drawing pictures
System Software: Programs managing computer hardware and basic operations silently behind scenes
License: Permission describing how you may legally use a program without breaking rules

Engage - Retrieval & Hook (5 min)

TEACHER Hold up a keyboard and ask: Can this make a document by itself? Now point to the computer screen with LibreOffice Writer open. Ask: Which one is the tool and which one is the instructions? Explain that hardware is the body and software is the mind.

STUDENTS Students understand that the keyboard is just a tool without software to interpret the key presses. They see that software gives hardware its purpose.

Explore - Guided Discovery (10 min)

TEACHER Give students a sorting activity with 16 cards: 8 hardware items (keyboard, mouse, monitor, CPU, printer, speaker, webcam, USB drive) and 8 software items (LibreOffice, Tux Paint, Scratch, Firefox, Ubuntu, Audacity, GIMP, VLC). They must sort them into two groups and name each group.

STUDENTS Students sort the cards into hardware and software groups. Some struggle with items like Ubuntu (is it hardware or software?). Teacher guides discussions about why operating systems are software.

Explain - Modelling & Concepts (10 min)

TEACHER Define software as a set of instructions that tell hardware what to do. Categorize software into two types: system software (operating systems like Ubuntu that manage the computer) and application software (programs like LibreOffice Writer that do specific tasks). Give our school examples: Ubuntu is the system software on school computers, Tux Paint is application software for drawing, and the school management system is custom application software.

STUDENTS Students create a two-column chart in their notebooks listing system software and application software examples. They draw arrows connecting each software to the hardware it runs on.

Elaborate - Independent Application (10 min)

TEACHER Students must create a Software Identification Poster: find five software programs on the school computers, identify whether each is system or application software, write what each one does, and draw a picture of its main screen.

STUDENTS Students explore the school computers, opening different programs and noting their names, types, and purposes. They create informative posters with accurate descriptions.

Evaluate - Check & Close (5 min)

TEACHER Students present their posters. Teacher and classmates ask questions: Is this system or application software? What would happen if we removed it? Could another software do the same job? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their findings and answer questions about each software program. Class discusses which software is essential and which is optional.

Cross-Curricular Connections

Mathematics: Science: Software-hardware relationship is like the mind-body relationship in biology
Science: Social Science: Software development is a major industry that creates jobs worldwide
Language: Language: Writing software descriptions practices technical and descriptive writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match hardware pictures to software names using a pre-made matching sheet
On Level: Identify three software programs and classify them as system or application
Advanced: Research and present on five software programs including free and paid options
SEND: Sort picture cards into hardware and software groups with teacher support

Formative Assessment

Method: Software identification poster evaluated for accuracy of classification and quality of descriptions
CT Skill Assessed: Classification: categorizing software into system and application types based on function

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If your phone is hardware, what is the software that makes it work? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.02 *Operating Systems*

40 minutes **DEBUG**

Success Criteria - I can...

- Explain what an operating system does and why computers need one
- Identify the main functions of an operating system: managing files, programs, and hardware
- Compare at least two different operating systems and their features

Resources Needed:
- Pre-buggy Scratch projects
- Debugging checklist
- Common error reference guide
- Say block debugging technique examples
- Bug report template

Key Vocabulary

Operating System: Main software managing all hardware and programs on computer together
Kernel: Core part of operating system controlling hardware directly without visible interface
Interface: Screen and buttons letting you communicate with computer easily every time
Process: Single task currently running on the computer like playing music

Engage - Retrieval & Hook (5 min)

TEACHER Ask: What would happen if you had a kitchen with all the tools but no one to tell the knife to cut, the stove to heat, and the pot to boil? The operating system is like the head chef who organizes everything. Show the Ubuntu desktop and ask students what they see.

STUDENTS Students look at the Ubuntu desktop and identify familiar elements: the taskbar, desktop icons, the menu button. They realize these are managed by the operating system.

Explore - Guided Discovery (10 min)

TEACHER Give students a worksheet with six operating system tasks: opening a file, connecting to the internet, managing the display, organizing folders, installing programs, and managing sound. They must match each task to what the operating system does for the user.

STUDENTS Students match tasks and discuss how the operating system handles each one behind the scenes. They realize they never think about these tasks because the operating system manages them.

Explain - Modelling & Concepts (10 min)

TEACHER Define an operating system (OS) as the main software that manages all other software and hardware on a computer. Key functions: file management (organizing folders and files), program management (opening and switching between programs), hardware management (making the keyboard, mouse, and screen work together), memory management (deciding what goes in RAM), and security (protecting against viruses and unauthorized access). Show Ubuntu as an example of an open-source operating system used in residential schools.

STUDENTS Students take notes on OS functions and create a diagram showing the operating system in the center with arrows pointing to all the hardware and software it manages.

Elaborate - Independent Application (10 min)

TEACHER Students must compare Ubuntu with Windows or Android: create a comparison chart showing the name, type (desktop or mobile), cost (free or paid), who made it, and three features of each. Explain why our school chose Ubuntu for school computers.

STUDENTS Students research the two operating systems using information provided by the teacher. They create detailed comparison charts and write a paragraph explaining why free software is important for schools.

Evaluate - Check & Close (5 min)

TEACHER Students present their comparison charts and explain the advantages of each operating system. Class votes on which operating system they would choose for the school and why. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their findings and answer questions. Class discusses the benefits of open-source software for education.

Cross-Curricular Connections

Mathematics: Social Science: Operating systems are developed by companies and communities around the world
Science: Economics: Free and open-source software saves money for schools and governments
Language: Language: Writing comparison charts practices organizing information for clear presentation

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Label the parts of the Ubuntu desktop from a labeled screenshot
On Level: List three functions of an operating system with examples from Ubuntu
Advanced: Create a detailed comparison of two operating systems with feature analysis
SEND: Identify the desktop, taskbar, and menu with teacher pointing them out

Formative Assessment

Method: Operating system comparison chart with written explanation of why Ubuntu is suitable for schools
CT Skill Assessed: Evaluation: comparing two operating systems and justifying which is better suited for a specific purpose

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: What operating system is on your phone? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.03 *Word Processors*

40 minutes **DEBUG**

Success Criteria - I can...

- Identify the main features of a word processor: typing, formatting, and saving
- Create a formatted document with title, headings, and body text
- Apply basic formatting: bold, italic, font size, and alignment

Resources Needed:
- Hardware and software sorting cards
- Computer lab for exploration
- Poster materials
- Software identification worksheet
- Sample software screenshots

Key Vocabulary

Word Processor: Program for typing and formatting text documents beautifully and neatly
Formatting: Changing font size colour or style to improve appearance quickly
Header: Text appearing at top of every page like title automatically
Spell Check: Tool that finds and suggests fixes for spelling mistakes instantly

Engage - Retrieval & Hook (5 min)

TEACHER Show two versions of the same letter: one handwritten on plain paper and one typed in a word processor with proper formatting. Ask: Which one looks more professional? Which one would you rather receive? Why does formatting matter?

STUDENTS Students identify that the typed letter looks more organized and easier to read. They notice the title is bigger, the paragraphs are aligned, and the text is clear.

Explore - Guided Discovery (10 min)

TEACHER Open LibreOffice Writer and give students 5 minutes to type anything they want. Then ask them to try: changing the font size, making text bold, centering text, and saving the file. They must report what they discovered.

STUDENTS Students type freely and experiment with formatting. They discover the toolbar buttons and call out to friends about what each one does.

Explain - Modelling & Concepts (10 min)

TEACHER Introduce LibreOffice Writer as a word processor. Key features: the typing area where you write, the toolbar for formatting (bold B, italic I, underline U, font size, alignment), the ruler for margins, and the file menu for save, open, and print. Demonstrate creating a document: type a title, press Enter, type a paragraph, select the title and make it bold and 24pt, center it, select the paragraph and make it 12pt, and save as .odt.

STUDENTS Students follow along, creating a document about their school. They practice each formatting tool as the teacher demonstrates it.

Elaborate - Independent Application (10 min)

TEACHER Students must create a one-page document about their village with: a centered title (bold, 24pt), an underlined subtitle (14pt), three paragraphs about the village (12pt, justified alignment), and a final sentence in italics. The document must be saved as village_report.odt.

STUDENTS Students write about their village, applying each formatting requirement. They check their document against the checklist before saving.

Evaluate - Check & Close (5 min)

TEACHER Students open their saved documents and display them on the projector. Class evaluates: Is the title formatted correctly? Are the paragraphs aligned? Is the font size consistent? Are all formatting requirements met? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their documents. Class provides specific feedback on formatting accuracy and document appearance.

Cross-Curricular Connections

Mathematics: English: Writing about their village practices descriptive and informative writing
Science: Social Science: Documenting village information builds awareness of local community
Language: Art: Choosing fonts and formatting involves visual design principles

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Type a simple paragraph with help choosing the font and size
On Level: Create a formatted document following a step-by-step formatting guide
Advanced: Design a professional document with custom margins, headers, and multiple formatting styles
SEND: Type their name and one sentence with teacher helping with formatting tools

Formative Assessment

Method: One-page village report evaluated against formatting checklist for all requirements
CT Skill Assessed: Algorithmic thinking: following a sequence of formatting steps to transform plain text into a professional document

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three sentences about your home village in your notebook. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.04 *Spreadsheets*

40 minutes **DEBUG**

Success Criteria - I can...

- Explain how spreadsheets organize data in rows and columns
- Enter data into a spreadsheet and use basic formulas for calculations
- Create a simple chart from spreadsheet data

Resources Needed:
- Ubuntu desktop screenshots
- Windows screenshots
- Android screenshots
- Comparison chart template
- OS function cards
- our school computer lab

Key Vocabulary

Spreadsheet: Grid program for calculations budgets and charts with numbers efficiently
Formula: Equation that calculates result from numbers in other cells automatically
Chart: Visual picture showing numbers as bars or pies for easy understanding
Cell Range: Group of cells selected together like A1 to B5 for calculation

Engage - Retrieval & Hook (5 min)

TEACHER Show a handwritten monthly expense list on paper and then the same data in a spreadsheet with automatic totals and a chart. Ask: Which one tells you faster where the most money is spent? Why is the spreadsheet more useful?

STUDENTS Students see that the spreadsheet total and chart make the data instantly understandable, while the paper list requires manual calculation.

Explore - Guided Discovery (10 min)

TEACHER Give students a printed expense list for a hypothetical family (rent, food, school fees, transport, savings) with amounts. Ask them to add up the total by hand, then open the same data in LibreOffice Calc and use the SUM function to calculate the total automatically.

STUDENTS Students add manually and discover the SUM function saves time and reduces errors. They compare their manual total with the formula result.

Explain - Modelling & Concepts (10 min)

TEACHER Review spreadsheet basics: rows (numbered), columns (lettered), cells (intersection). Introduce formulas: = sign starts a formula, + for addition, - for subtraction, * for multiplication, / for division, and =SUM for adding ranges. Show how to create a bar chart: select data, Insert, Chart, choose Bar chart. Explain that spreadsheets are used for budgets, marks sheets, and data analysis in real life.

STUDENTS Students practice writing formulas for the expense data and create a bar chart showing the proportion of each expense category.

Elaborate - Independent Application (10 min)

TEACHER Students must create a personal monthly budget spreadsheet with: categories (food, transport, school, savings, other), estimated amounts, a total formula, and a pie chart showing the percentage breakdown of each category.

STUDENTS Students fill in realistic amounts for their household expenses, write the total formula, and create a pie chart. They verify the pie chart percentages add up to 100 percent.

Evaluate - Check & Close (5 min)

TEACHER Students present their budget spreadsheets. Teacher verifies formula accuracy and chart correctness. Class discusses: Which category takes the most money? How could a family reduce expenses? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their budgets and explain their data choices. Class discusses real-world financial planning using spreadsheet tools.

Cross-Curricular Connections

Mathematics: Math: Spreadsheet formulas apply arithmetic operations and percentages
Science: Social Science: Budgeting connects to understanding household economics and financial planning
Language: Life Skills: Creating a budget is a practical skill for managing money

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Fill in a pre-made budget template with given amounts and test the total formula
On Level: Create a budget from scratch with four categories and a SUM formula
Advanced: Design a detailed budget with formulas for percentages, a pie chart, and expense analysis
SEND: Enter amounts into pre-labeled cells with teacher helping write the SUM formula

Formative Assessment

Method: Monthly budget spreadsheet evaluated for formula accuracy, chart correctness, and realistic data
CT Skill Assessed: Application: using spreadsheet formulas and charts to create a functional household budget

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you had 100 rupees to spend in a week, how would you divide it? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.05 *Presentations*

40 minutes DEBUG

Success Criteria - I can...

- Identify the features of presentation software and when to use it
- Create a multi-slide presentation with consistent themes and transitions
- Deliver a presentation with clear speaking and appropriate pacing

Resources Needed:

- Computers with LibreOffice Writer
- Village information prompts
- Formatting checklist
- Document template example
- Save location guide

Key Vocabulary

Slide: Single page of presentation holding text images and animations together nicely
Theme: Set of colours and fonts giving uniform look to all slides
Transition: Effect showing how one slide moves to the next smoothly
Presenter: Person who explains slides and talks to audience during show

Engage - Retrieval & Hook (5 min)

TEACHER Show a science project report as a written document and then the same information as an engaging presentation with images, animations, and speaker notes. Ask: If you had to present to the class, which would you choose? Why?

STUDENTS Students prefer the presentation because it is visual, organized, and easier to follow during a live talk. They understand that different formats serve different purposes.

Explore - Guided Discovery (10 min)

TEACHER Give students a topic card (My Favorite Book, My Dream Job, A Place I Want to Visit) and a planning sheet with five slides: title, three content slides, and conclusion. They must plan each slide content before opening the computer.

STUDENTS Students write bullet points for each slide, deciding what images to use and what text to include. They discuss which information is essential for each slide.

Explain - Modelling & Concepts (10 min)

TEACHER Review LibreOffice Impress features: slide layouts, themes for consistent design, transitions between slides, animations within slides, and speaker notes for the presenter. Walk through the workflow: choose a theme, create slides with the three-part structure (intro, body, conclusion), add images, apply transitions, write speaker notes, and practice delivery.

STUDENTS Students follow along creating their presentation, choosing a theme that matches their topic and building each slide with the correct layout.

Elaborate - Independent Application (10 min)

TEACHER Students must create a five-slide presentation on their chosen topic with: a title slide with their name and topic, three content slides with images and text, a conclusion slide with a summary, a consistent theme throughout, at least one transition, and speaker notes for each slide.

STUDENTS Students build their presentations carefully, checking theme consistency and practicing what they will say for each slide.

Evaluate - Check & Close (5 min)

TEACHER Three students present to the class. Audience rates each on: slide design, content clarity, speaking confidence, and use of transitions. Teacher provides detailed feedback on presentation skills. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present with clear speaking, advance slides at the right time, and answer one question from the audience. Class provides constructive feedback.

Cross-Curricular Connections

Mathematics: English: Writing presentation content practices concise and persuasive writing
Science: Social Science: Presentations about community topics build public speaking and advocacy skills
Language: Art: Slide design involves visual composition and color theory

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Fill in a template presentation with pre-made slides and text boxes
On Level: Create a five-slide presentation using the planning sheet with step-by-step guidance
Advanced: Design a presentation with custom animations, speaker notes, and advanced transitions
SEND: Add text to pre-made slides with teacher helping with theme and image selection

Formative Assessment

Method: Five-slide presentation with audience rating on design, content, speaking, and transitions
CT Skill Assessed: Abstraction: organizing information into a logical slide structure that supports effective oral presentation

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you had to present your dormitory to a new student, what five things would you show them?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.06 *Web Browsers Revisited*

40 minutes DEBUG

Success Criteria - I can...

- Identify advanced web browser features: bookmarks, tabs, history, and downloads
- Use a web browser safely and efficiently to find and save educational resources
- Explain how web browsers work to display websites from the internet

Resources Needed:

- LibreOffice Calc
- Expense list worksheets
- Budget template
- Chart creation guide
- Calculator for verification

Key Vocabulary

Browser: Program that lets you open and view websites easily
Address Bar: Long box at top where you type website address
Tab: Small window letting you open many sites together
Bookmark: Saved link that helps you find favourite sites quickly

Engage - Retrieval & Hook (5 min)

TEACHER Open five browser tabs with different educational websites and switch between them quickly. Ask: How can one program show so many different websites? What is happening behind the scenes when I click a link?

STUDENTS Students are curious about how the browser loads different websites. They realize that the browser is fetching information from computers around the world.

Explore - Guided Discovery (10 min)

TEACHER Give students a browser exploration challenge: open three new tabs, bookmark two websites, use the back button, clear the browsing history, and download a picture from a website. They must complete all tasks and record each step.

STUDENTS Students navigate the browser independently, discovering features they have never used before. They are excited to learn about bookmarks and tabs.

Explain - Modelling & Concepts (10 min)

TEACHER Explain how web browsers work: you type a URL or click a link, the browser sends a request to a web server (a computer that stores the website), the server sends back the website files (HTML, CSS, images), and the browser assembles and displays the page. Teach advanced features: bookmarks save favorite websites for quick access, tabs let you have multiple websites open at once, history records where you have been, and downloads save files from the internet to your computer.

STUDENTS Students take notes on how browsers work and practice using each feature. They bookmark five educational websites and organize them into folders.

Elaborate - Independent Application (10 min)

TEACHER Students must complete a Web Research Challenge: find and bookmark three educational websites about a topic assigned by the teacher, download one useful image from each, and create a list of the URLs with a brief description of each site.

STUDENTS Students search carefully using educational search terms, evaluate websites for reliability, bookmark the best ones, and download appropriate images.

Evaluate - Check & Close (5 min)

TEACHER Students share their bookmarked websites and downloaded images. Class evaluates: Are the websites educational and reliable? Are the images appropriate? Are the descriptions accurate? Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their research findings. Class discusses how to evaluate website reliability and choose trustworthy educational resources.

Cross-Curricular Connections

Mathematics: Science: Understanding how the internet works connects to networking and data transmission
Science: Social Science: Evaluating website reliability is essential for research and informed citizenship
Language: Language: Writing website descriptions practices summarizing and technical writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Open a browser and navigate to a specific URL with teacher guidance
On Level: Bookmark three websites and use tabs to switch between them
Advanced: Complete a web research challenge with independent searching and evaluation
SEND: Type a URL and click a bookmark with teacher guiding mouse and keyboard

Formative Assessment

Method: Web research challenge evaluated on bookmarked sites, downloaded images, and description accuracy
CT Skill Assessed: Evaluation: assessing website reliability and selecting appropriate educational resources from the internet

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could bookmark one website to show your parents, what would it be and why?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.07 *Educational Software*

40 minutes **DEBUG**

Success Criteria - I can…

- Identify different types of educational software and their learning purposes
- Evaluate educational software based on content quality, ease of use, and learning value
- Compare traditional learning methods with software-assisted learning

Resources Needed:
- Topic cards and planning sheets
- LibreOffice Impress
- Theme gallery
- Image collection
- Presentation evaluation rubric
- Timer for practice

Key Vocabulary

Educational App: Program designed to teach a specific subject interactively and engagingly for learning
Simulation: Model mimicking real life to practise safely without danger or cost
Quiz: Short test checking understanding of topic just learned recently in class
Tutorial: Step-by-step guide teaching you how to use a new tool

Engage - Retrieval & Hook (5 min)

TEACHER Show two ways to learn multiplication: flashcards on paper and an interactive multiplication game on the computer. Ask students to try both for 3 minutes. Then ask: Which one was more fun? Which one helped you learn faster? Why?

STUDENTS Students enjoy the game more and notice they practiced more problems in less time. They discuss why the interactive element made learning engaging.

Explore - Guided Discovery (10 min)

TEACHER Give students access to four educational programs: a math quiz game, a typing tutor, a science simulation, and a language learning app. They must spend 3 minutes on each and rate each on a worksheet: fun level (1-5), learning level (1-5), and ease of use (1-5).

STUDENTS Students rotate through the programs, eagerly trying each one. They debate with friends about which is most fun and which teaches the most.

Explain - Modelling & Concepts (10 min)

TEACHER Categorize educational software: drill and practice (repetitive exercises like math quizzes), tutorials (step-by-step teaching like typing tutors), simulations (virtual experiments like science labs), and educational games (learning through play). Discuss what makes good educational software: accurate content, clear instructions, appropriate difficulty, feedback on answers, and engaging design.

STUDENTS Students classify the four programs they tried into categories and discuss what features made each one effective or ineffective.

Elaborate - Independent Application (10 min)

TEACHER Students must create an Educational Software Review Card for their favorite program: name, type, what it teaches, how it teaches, what is good about it, what could be improved, and a rating out of 5 stars.

STUDENTS Students write detailed reviews based on their experience. They include specific examples of what they learned and suggestions for improvement.

Evaluate - Check & Close (5 min)

TEACHER Students present their review cards. Class votes on the most useful educational software. Teacher discusses how software can supplement but not replace traditional teaching. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their reviews and answer questions. Class debates the role of technology in education.

Cross-Curricular Connections

Mathematics: Psychology: Educational software design connects to how the brain learns best
Science: Math: Drill and practice software provides data on student progress and areas needing improvement
Language: Language: Writing software reviews practices critical and evaluative writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Use one educational program and rate it with smiley faces (happy, okay, sad)
On Level: Rate three programs on a 1-5 scale and write one sentence about each
Advanced: Write a detailed review comparing traditional and software-assisted learning
SEND: Try one program with teacher help and describe what they learned

Formative Assessment

Method: Software review card evaluated for accuracy, completeness, and critical evaluation
CT Skill Assessed: Evaluation: critically assessing educational software based on learning effectiveness and design quality

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you could design an educational game, what subject would it teach and how would it work?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.08 *Mobile Apps*

40 minutes **DEBUG**

Success Criteria - I can...

- Identify the difference between mobile apps and desktop software
- Evaluate mobile apps for educational value and safety
- Understand how apps access device features like camera, location, and contacts

Resources Needed:
- Web browsers with internet access
- Educational website list
- Bookmark folder template
- Download folder
- Website evaluation checklist

Key Vocabulary

App: Small program installed on phone to perform specific task quickly
Install: Downloading and setting up app so it appears on your phone
Permission: Allowing app to use camera or location only when you agree
Update: New version fixing bugs or adding features to existing app

Engage - Retrieval & Hook (5 min)

TEACHER Ask: How many of you have a smartphone? What apps do you use most? Now ask: Did you know that the calculator app on your phone and the calculator on the computer do the same thing but are different software? Explain that apps are software designed specifically for mobile devices.

STUDENTS Students share their favorite apps and realize that apps are specialized versions of software for touchscreens and portable devices.

Explore - Guided Discovery (10 min)

TEACHER Give students a printed list of 10 app categories: games, social media, education, productivity, health, photography, music, maps, shopping, and news. For each category, they must name one app they know and rate its educational value from 1 to 5.

STUDENTS Students brainstorm apps they know and discuss which ones are educational versus purely for entertainment. They are surprised that some apps they use daily have educational features.

Explain - Modelling & Concepts (10 min)

TEACHER Explain that mobile apps are software designed for smartphones and tablets. Key differences from desktop software: apps are smaller, designed for touch input, can access device features (camera, GPS, microphone, contacts), and are distributed through app stores. Discuss app permissions: why apps ask for access to camera (for photo features), location (for maps), and contacts (for messaging). Emphasize that not all permissions are necessary for all apps.

STUDENTS Students take notes on app features and permissions. They discuss which permissions each app category typically needs and which permissions might be unnecessary.

Elaborate - Independent Application (10 min)

TEACHER Students must create an App Safety Checklist: identify five apps they use, list what permissions each asks for, decide if each permission is necessary, and rate each app on safety (safe, somewhat safe, risky). They must explain their safety ratings.

STUDENTS Students analyze their own apps, questioning why each permission is needed. They identify apps that ask for unnecessary permissions and rate them as risky.

Evaluate - Check & Close (5 min)

TEACHER Students share their safety checklists. Class discusses which apps are safe and which need caution. Teacher summarizes app safety rules: read permissions before installing, avoid apps that ask for too much access, and keep apps updated. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present their app analyses. Class creates a class list of recommended safe educational apps.

Cross-Curricular Connections

Mathematics: Social Science: Apps connect people globally but also raise privacy concerns
Science: Science: Apps use device sensors (GPS, accelerometer, camera) that work on scientific principles
Language: Math: Some apps collect data about usage patterns that are analyzed using statistics

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Name three apps they use and identify if each is for games, learning, or communication
On Level: List five apps with their permissions and rate each on a simple safety scale

Advanced: Define app permissions in detail, explaining why each permission is necessary or unnecessary
SEND: Name two apps with teacher help discussing what each permission means

Formative Assessment

Method: App safety checklist evaluated for thoroughness of permission analysis and accuracy of safety ratings
CT Skill Assessed: Evaluation: critically assessing mobile apps based on their permissions, educational value, and safety

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the apps on a family phone. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.09 *Software Updates*

40 minutes **DEBUG**

Success Criteria - I can...

- Explain why software needs regular updates
- Identify the three types of updates: security patches, bug fixes, and feature additions
- Understand the consequences of not updating software

Resources Needed:
- Educational software installed on computers
- Review card template
- Rating worksheets
- Comparison chart
- Traditional vs digital learning examples

Key Vocabulary

Update: New version fixing problems or adding features to software automatically for safety
Patch: Small fix correcting security hole or bug quickly without full reinstall
Version: Number showing which release of software you are using currently on device
Upgrade: Moving to better version with more powerful features than before

Engage - Retrieval & Hook (5 min)

TEACHER Show a phone notification that says Software Update Available and ask students if they usually click Update or Ignore. Ask: What do you think happens if you never update? Show a news article about a computer virus that attacked computers that had not been updated.

STUDENTS Students share their update habits. Some always update, others ignore updates. The virus article motivates them to understand why updates matter.

Explore - Guided Discovery (10 min)

TEACHER Give students three scenarios: a) A school computer has not been updated for 2 years and suddenly stops working during a class, b) A phone gets a virus because the security update was ignored, c) A new app feature is added in an update that makes typing easier. Ask them to match each scenario to an update type.

STUDENTS Students match scenarios and discuss which update type prevented which problem. They realize that updates serve different important purposes.

Explain - Modelling & Concepts (10 min)

TEACHER Define software updates as changes made by the developer to improve the software. Three types: security updates fix vulnerabilities that hackers could exploit, bug fixes correct errors that cause crashes or wrong behavior, and feature updates add new capabilities. Explain the update process: the developer creates the update, distributes it through the internet, the user installs it, and the software runs better. Warn about risks of not updating: viruses, crashes, missing features, and compatibility problems.

STUDENTS Students create a diagram showing the update cycle: problem found, update created, update distributed, update installed, problem solved.

Elaborate - Independent Application (10 min)

TEACHER Students must create a Software Update Poster for the school: explain what updates are, why they matter, what happens without them, and when to install them. The poster must include at least three real examples of problems caused by not updating.

STUDENTS Students research update-related incidents and create informative posters with clear explanations and eye-catching design.

Evaluate - Check & Close (5 min)

TEACHER Students display their posters and present their key messages. Class discusses: Should we always update immediately? What are the risks of updating too quickly? Teacher explains the balance between updating promptly and waiting for stable versions. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present and answer questions about update strategies. Class creates a school update policy recommendation.

Cross-Curricular Connections

Mathematics: Science: Software updates connect to the concept of evolution and improvement over time
Science: Social Science: Cybersecurity and updates are important for national digital infrastructure
Language: Math: Understanding version numbers (1.0, 2.1, 3.0) connects to number systems

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match three update scenarios to their update types using a picture card activity
On Level: Explain three types of updates with one example of each in their own words
Advanced: Research a specific software update and present its security improvements to the class
SEND: Identify one reason to update with teacher explaining the other types

Formative Assessment

Method: Software Update Poster evaluated for accuracy of update types, real examples, and design quality
CT Skill Assessed: Decomposition: breaking down the concept of software updates into types, purposes, and consequences

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Does your family phone have any pending updates? Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.10 *Installing Software*

40 minutes **DEBUG**

Success Criteria - I can…

- Explain the process of installing software on a computer
- Identify safe sources for downloading software
- Understand the difference between installing and uninstalling software

Resources Needed:
- Smartphone with common apps installed
- App permission examples
- Safety checklist template
- Educational app recommendations
- App store screenshots

Key Vocabulary

Install: Copying program files to computer so program runs correctly afterwards for use
Installer: Helper program that guides you through installation steps simply one by one
Wizard: Series of screens helping you set up software without needing expert knowledge
Requirement: Minimum hardware or software needed before installation succeeds properly every time

Engage - Retrieval & Hook (5 min)

TEACHER Show a computer with LibreOffice Writer already installed and another with only the operating system. Ask: Can you write a document on the second computer? Why not? Explain that software must be installed before it can be used, just like you must plant a seed before it grows.

STUDENTS Students see that the empty computer cannot do useful tasks without installed software. They understand that installation puts the software onto the computer.

Explore - Guided Discovery (10 min)

TEACHER Give students a step-by-step handout showing software installation: downloading from a website, opening the installer, clicking Next through the wizard, choosing installation location, and clicking Finish. They must match each step to its purpose.

STUDENTS Students examine the installation wizard screenshots and discuss what each step does. They realize that installation is a guided process with clear steps.

Explain - Modelling & Concepts (10 min)

TEACHER Define installation as the process of copying software files to the computer so it can be run. Steps: find the software from a trusted source, download the installer file, run the installer, follow the installation wizard (choose language, agree to license, select location), and complete the installation. Explain safe sources: official websites, school-approved repositories, and trusted app stores. Warn about unsafe sources: random websites, pop-up ads, and unknown email attachments. Define uninstalling as removing software to free up space.

STUDENTS Students take notes on the installation process and create a flowchart showing the safe installation path versus the unsafe path.

Elaborate - Independent Application (10 min)

TEACHER Students must create a Software Installation Guide for a new student: write step-by-step instructions for installing LibreOffice on a school computer, include screenshots or drawings for each step, list safe sources for downloading, and explain what to do if the installation fails.

STUDENTS Students write clear, detailed installation guides with numbered steps and helpful tips for each stage of the process.

Evaluate - Check & Close (5 min)

TEACHER Students exchange their installation guides and follow them on a practice computer. The tester rates the guide on clarity, completeness, and accuracy. Teacher verifies that the guides would actually work. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students test each other guides and provide feedback. They revise their guides based on the testing results.

Cross-Curricular Connections

Mathematics: Science: Installation processes are like transplanting a plant - moving something from one place to another so it can grow
Science: Social Science: Software distribution connects to understanding supply chains and digital commerce
Language: Language: Writing installation guides practices procedural and technical writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Follow a pre-made installation guide with picture steps and checkboxes
On Level: Write a five-step installation guide with numbered instructions and brief explanations
Advanced: Create a comprehensive guide with troubleshooting tips and alternative installation methods
SEND: Click through an installation wizard with teacher reading each step aloud

Formative Assessment

Method: Installation guide tested by peer with rating on clarity, completeness, and accuracy
CT Skill Assessed: Algorithmic thinking: creating a step-by-step procedure that successfully installs software

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about the last time you installed an app on a phone. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.11 *File Management*

40 minutes **DEBUG**

Success Criteria - I can…

- Organize files and folders using a logical directory structure
- Rename, move, copy, and delete files safely
- Create a personal file organization system for schoolwork

Resources Needed:
- Phone update notification screenshots
- Virus attack news articles
- Update types matching cards
- Poster materials
- Update schedule examples

Key Vocabulary

Folder: Container holding many related files together like a box
Path: Address showing where file lives like folder inside folder chain
Extension: Suffix after dot indicating file type like .txt or .mp4
Backup: Extra copy kept safe in case original is lost or damaged accidentally

Engage - Retrieval & Hook (5 min)

TEACHER Show a messy desk covered with papers, books, and folders. Ask: Can you find the math homework from last week? Now show a neatly organized desk with labeled folders. Ask again. Which desk would you rather work at? Computer files need organizing just like physical papers.

STUDENTS Students easily see that organization saves time and reduces stress. They connect physical organization to digital file management.

Explore - Guided Discovery (10 min)

TEACHER Give students a computer with 20 disorganized files of different types (documents, images, music, spreadsheets) in a single folder. They must sort them into a logical folder structure within 5 minutes. Compare the different organizational systems students create.

STUDENTS Students hurriedly create folders and move files. Some organize by type, others by subject, others by date. Teacher highlights the different approaches and asks which is most useful.

Explain - Modelling & Concepts (10 min)

TEACHER Teach file management concepts: folders are containers that organize files, the root folder is the top level, subfolders create hierarchy, file extensions indicate file type (.odt for documents, .ods for spreadsheets, .jpg for images), and paths show the route to a file (Documents/School/Math/assignment1.odt). Teach file operations: rename (give a file a new name), move (transfer to a different folder), copy (make a duplicate), and delete (remove permanently or send to trash).

STUDENTS Students practice creating a folder structure: School, with subfolders for each subject (Kannada, English, Math, Science, Social), and within each, subfolders for Homework, Projects, and Notes.

Elaborate - Independent Application (10 min)

TEACHER Students must create a complete file organization system: create a root folder called My Schoolwork, create subfolders for each subject, move the 20 provided files into the correct folders, rename files with consistent naming conventions (subject_date_description), and create a README file explaining their organization system.

STUDENTS Students build their folder structures, carefully moving and renaming files. They write a README explaining their logic so someone else could find files in their system.

Evaluate - Check & Close (5 min)

TEACHER Students demonstrate their file systems. Teacher plays a game: find the file that matches a description within 30 seconds. The organized systems find files quickly. Students reflect on their naming conventions and folder structure. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students demonstrate finding files in their systems. Teacher times each search. Students discuss what makes a file system easy to navigate.

Cross-Curricular Connections

Mathematics: Math: Folder hierarchy connects to tree structures and organizational diagrams
Science: Library Science: File management follows the same principles as library cataloging
Language: Language: Writing file naming conventions practices systematic and consistent naming

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Sort files into two folders (Documents and Pictures) with picture labels
On Level: Create a three-level folder structure following a diagram and move files into it
Advanced: Design a complete organizational system with consistent naming and a README file
SEND: Create folders with teacher dictating the names and dragging files together

Formative Assessment

Method: File organization system evaluated on structure logic, naming consistency, and ease of navigation
CT Skill Assessed: Decomposition: breaking a large collection of files into a logical hierarchical structure

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If you had to organize all your books at home into folders, how would you group them?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.12 *Choosing Software* 40 minutes DEBUG

Success Criteria - I can...

- Compare different software options for the same task based on features, cost, and compatibility
- Make informed decisions about which software to use for specific educational tasks
- Explain the benefits of free and open-source software for education

Resources Needed:
- Software comparison matrix
- Free software list for schools
- Decision flowchart template
- Recommendation report rubric
- our school installed software list

Resources Needed:
- Computer with disorganized files
- File management step-by-step guide
- Naming convention reference
- Folder structure examples
- README template

Resources Needed:
- LibreOffice installer files
- Installation wizard screenshots
- Safe source list
- Installation guide template
- Practice computers

Key Vocabulary

Software: Set of programs that tells hardware what to do each time
Application: Program designed for specific task like writing or drawing pictures
System Software: Programs managing computer hardware and basic operations silently behind scenes
License: Permission describing how you may legally use a program without breaking rules

Engage - Retrieval & Hook (5 min)

TEACHER Present a scenario: the school needs software for three tasks - writing documents, creating presentations, and editing photos. Show two options for each task: a paid option and a free option. Ask: Which should the school choose and why?

STUDENTS Students debate the pros and cons of paid versus free software. They consider factors like cost, features, and ease of use.

Explore - Guided Discovery (10 min)

TEACHER Give students a Software Comparison Matrix with five tasks across the top (write, calculate, present, draw, browse) and two software options for each task. They must rate each option on three criteria: features (1-5), cost (free or paid), and ease of use (1-5).

STUDENTS Students fill in the comparison matrix, researching each option. They discover that free software often has comparable features to paid alternatives.

Explain - Modelling & Concepts (10 min)

TEACHER Teach software selection criteria: features (does it do what you need?), cost (can the school afford it?), compatibility (does it work on our computers?), support (is there help available?), and license (can we share it with other schools?). Explain free and open-source software (FOSS): free to use, free to share, free to modify, and supported by a community. Give our school examples: Ubuntu (operating system), LibreOffice (office suite), GIMP (image editing), and Audacity (audio editing) are all free.

STUDENTS Students create a decision-making flowchart: What task do you need? What features are required? Is free software available? Does it work on our computers? Choose the best option.

Elaborate - Independent Application (10 min)

TEACHER Students must create a Software Recommendation Report for the school: recommend software for five school needs (word processing, spreadsheets, presentations, image editing, audio editing), justify each recommendation with reasons related to features, cost, and compatibility, and present their findings to the class.

STUDENTS Students research and compare options for each task, building arguments for their recommendations. They consider the school budget and technical requirements.

Evaluate - Check & Close (5 min)

TEACHER Students present their recommendations. Class votes on the best software for each task. Teacher summarizes the key decision-making factors and explains why our school chose the software currently installed on school computers. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

STUDENTS Students present and defend their recommendations. Class discusses the real-world implications of software choices for schools.

Cross-Curricular Connections

Mathematics: Economics: Software cost-benefit analysis connects to financial decision-making
Science: Social Science: Open-source software movement is about community collaboration and sharing
Language: Language: Writing recommendation reports practices persuasive and analytical writing

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

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Differentiation

Approaching: Match software names to their functions using a picture-based matching sheet
On Level: Complete a comparison matrix for two software options with guided ratings
Advanced: Write a full recommendation report comparing paid and free options with detailed justification
SEND: Identify which software to use for a specific task with teacher explaining the alternatives

Formative Assessment

Method: Software Recommendation Report evaluated on comparison accuracy, justification quality, and presentation clarity

CT Skill Assessed: Evaluation: making informed software choices based on multiple criteria and justifying the decision

Dormitory Task - Interleaved Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: If your family could buy only one computer program, what would it be and why?. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Colophon

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Version: August 2026 - ICT Levels 1 (Class 6), 2 (Class 7), 3 (Classes 8-10) - 216 lessons
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